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(54) **SUGGESTING EXECUTABLE ACTIONS IN
RESPONSE TO DETECTING EVENTS**

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(58) **Field of Classification Search**

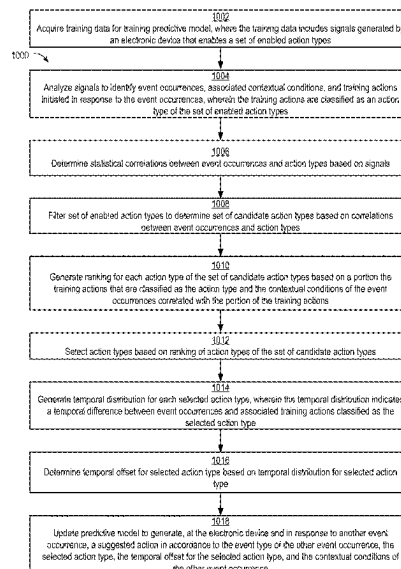
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(57) **ABSTRACT**

Systems and processes for providing, via an electronic device, suggested user actions. The suggested actions are provided in response to detecting an occurrence of a pre-defined event occurring in the user's day. The occurrence of the anchor is encoded in signals generated by the electronic device. The occurrence of the anchor is detectable via monitoring and analysis of electronic signals. Based on the user's previous interactions with the device, the occurrence of the anchor is indicative of user behavior and/or action taken in response to the anchor. Machine learning (ML) is employed to train an anchor model to associate actions taken in response to anchor occurrences. The trained anchor model is employed to detect anchors and provide suggested actions in response to the detected anchor occurrence. The suggested action is based on a type of anchor occurrence and contextual conditions of the anchor occurrences.

33 Claims, 17 Drawing Sheets



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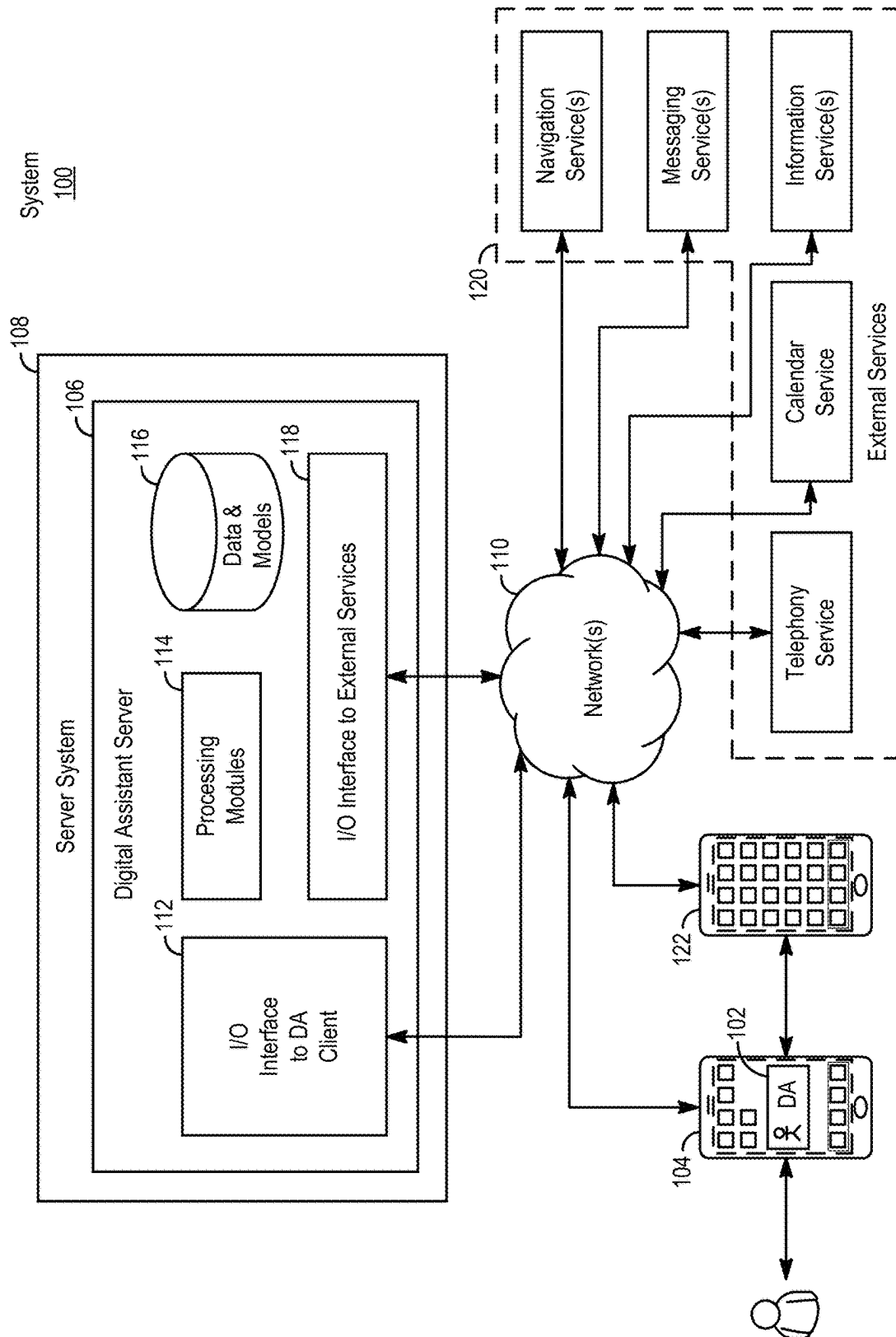


FIG. 1

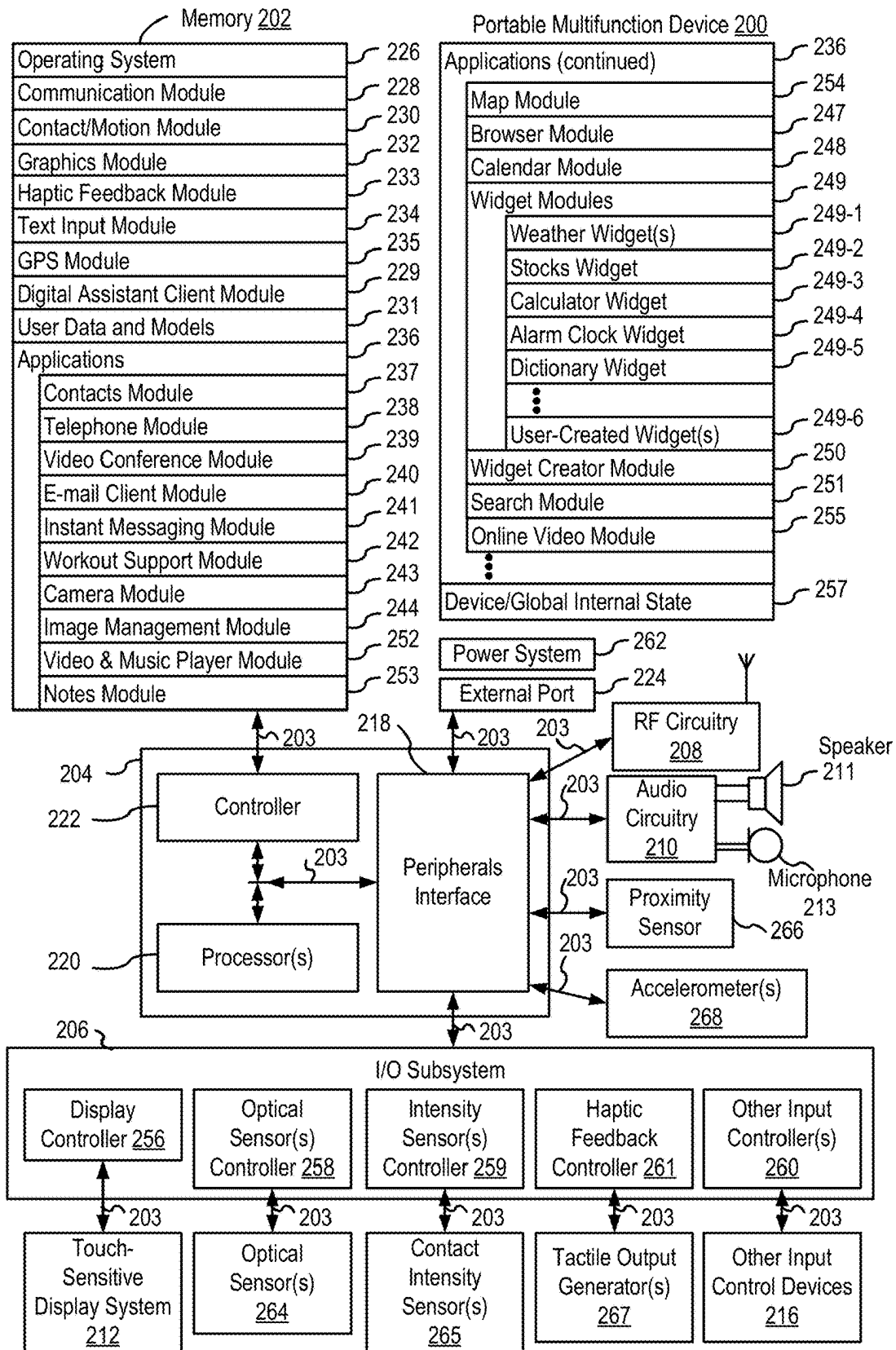


FIG. 2A

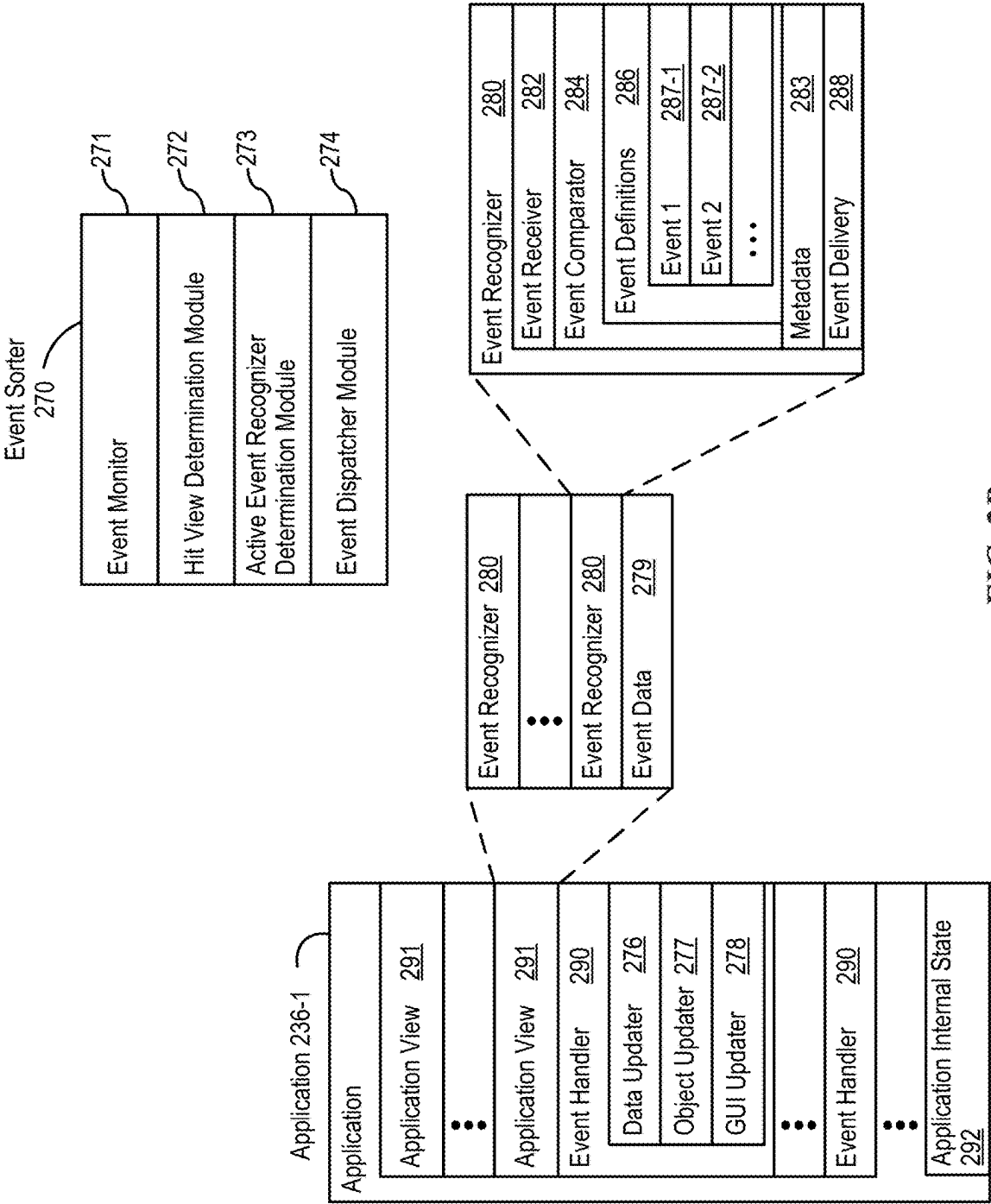


FIG. 2B

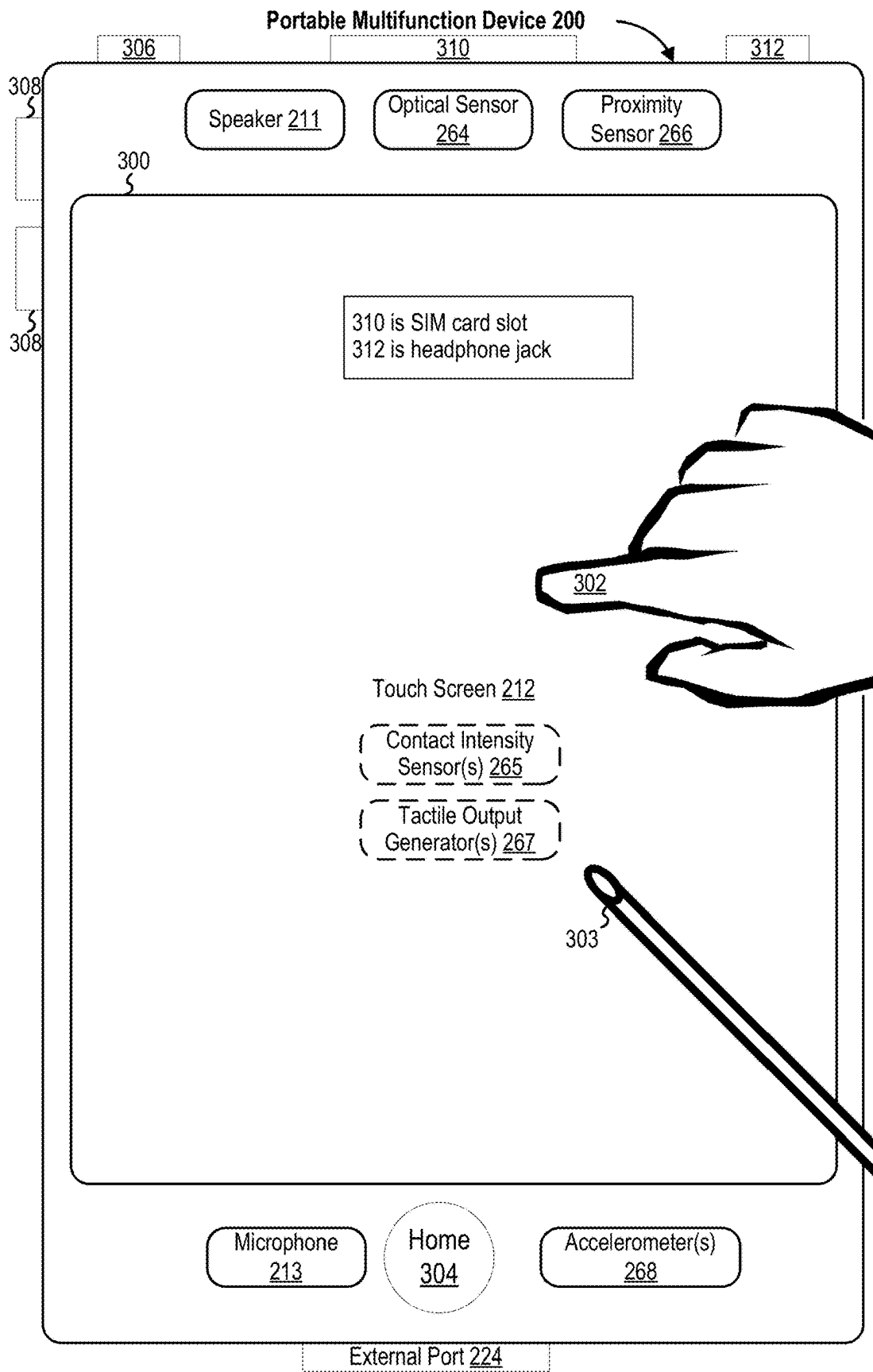


FIG. 3

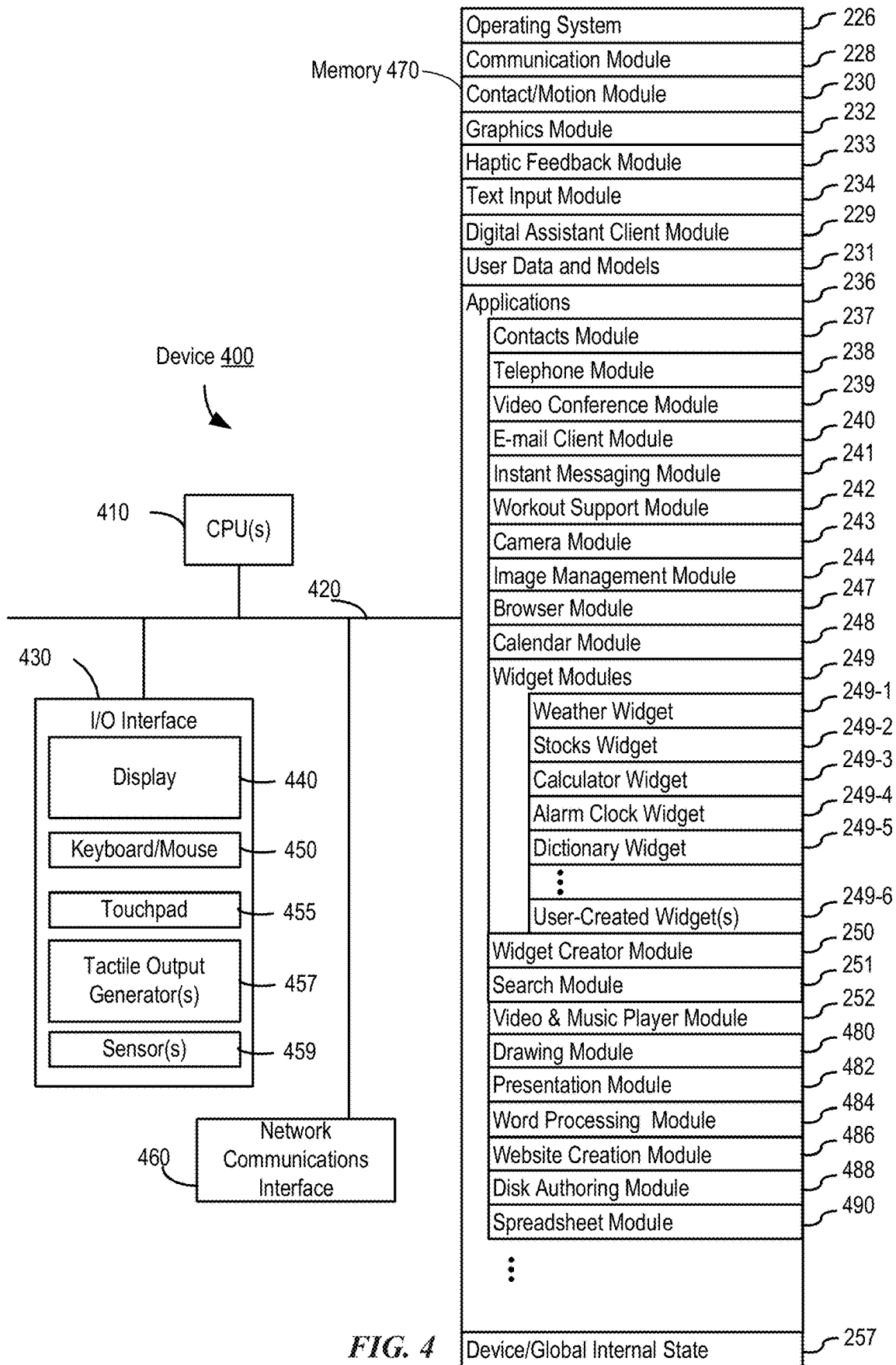


FIG. 4

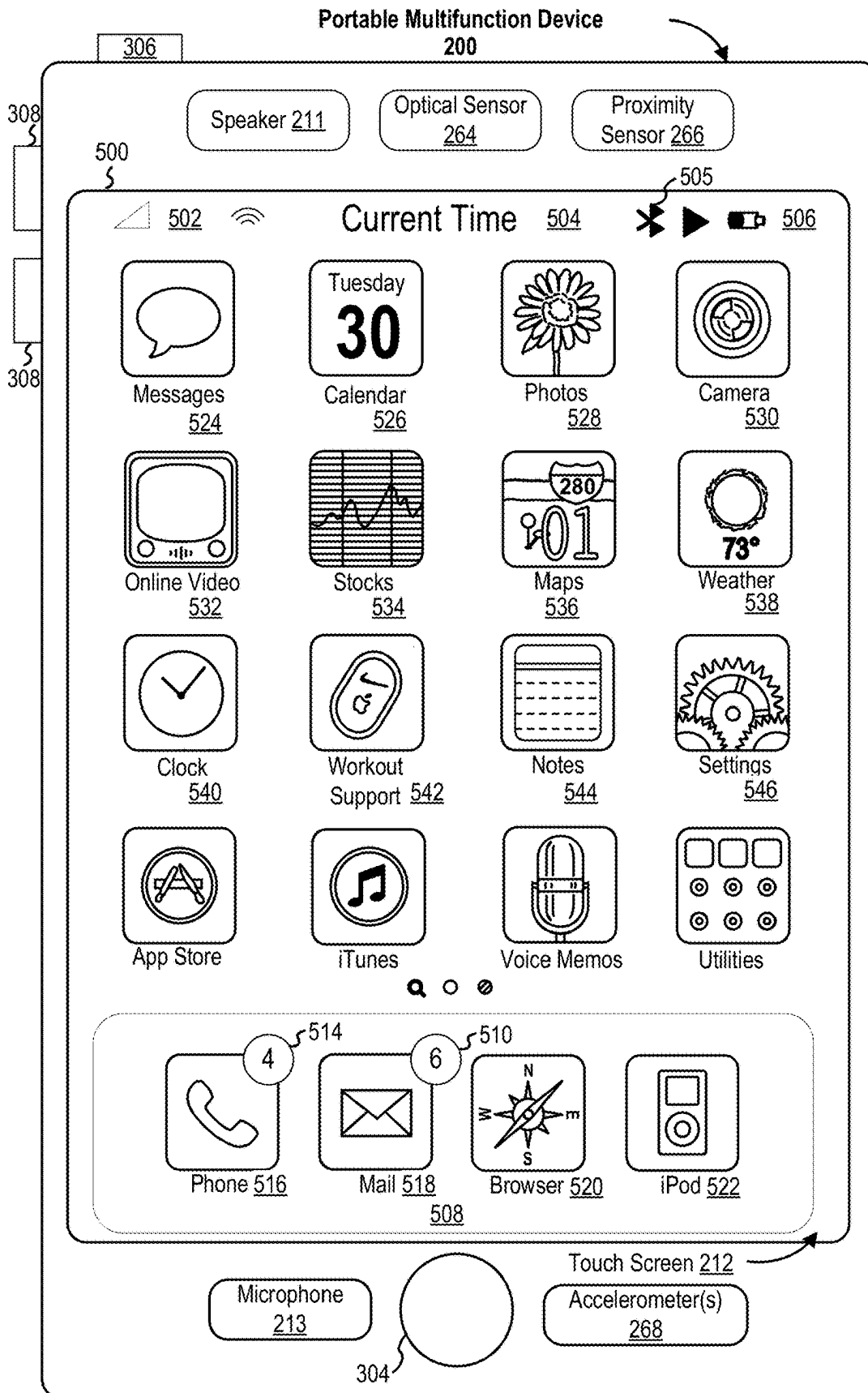


FIG. 5A

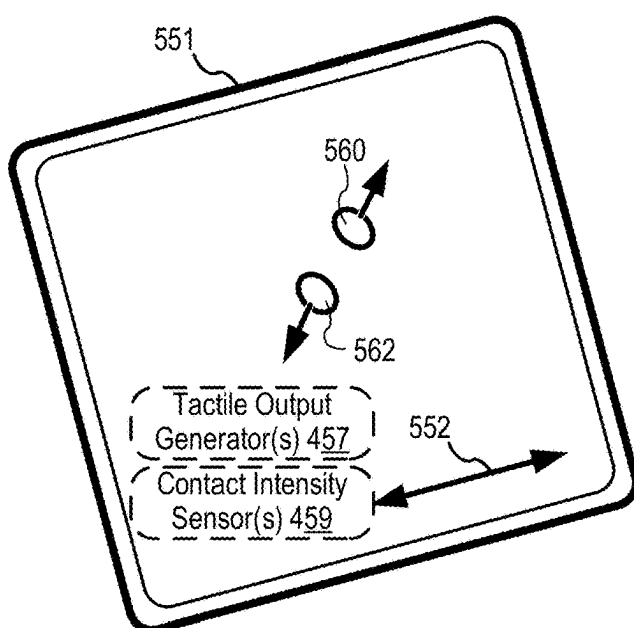
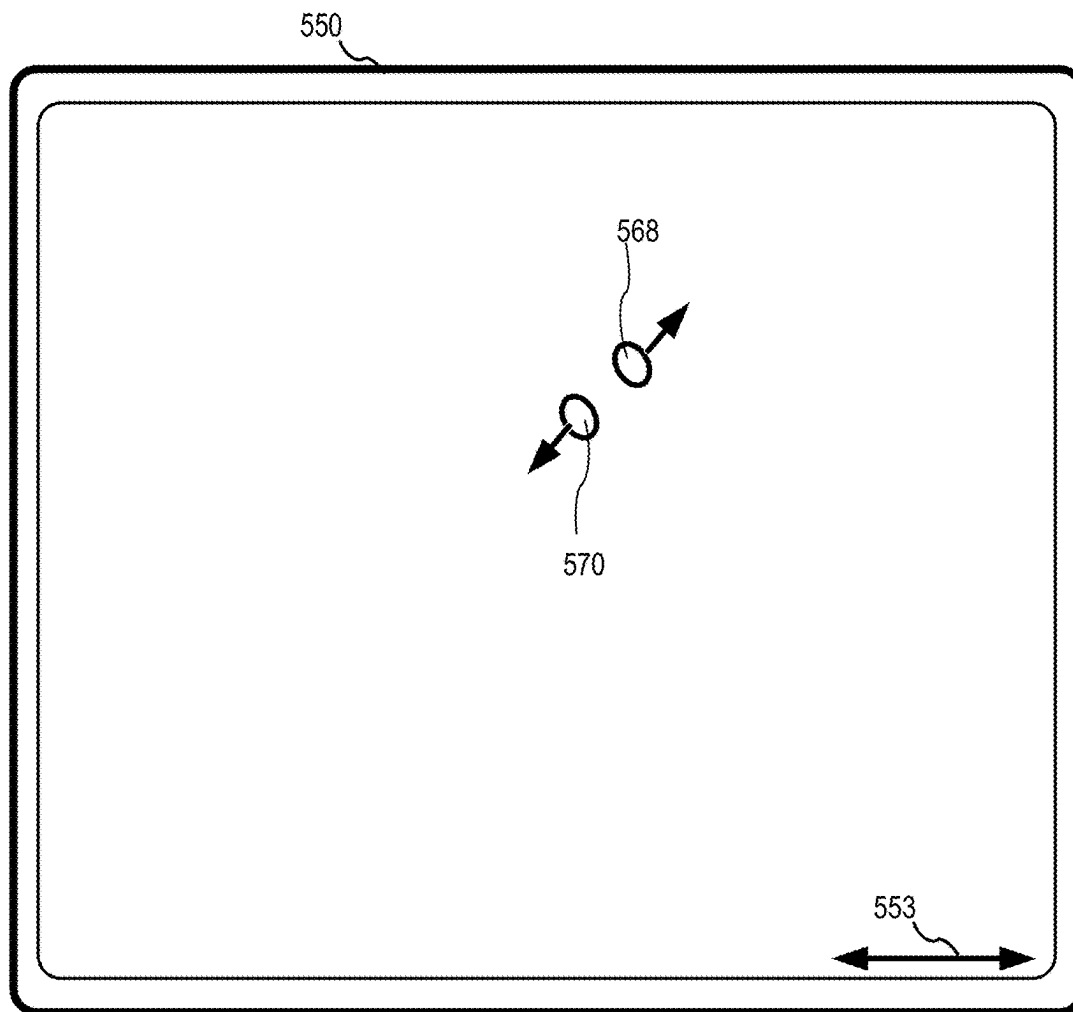
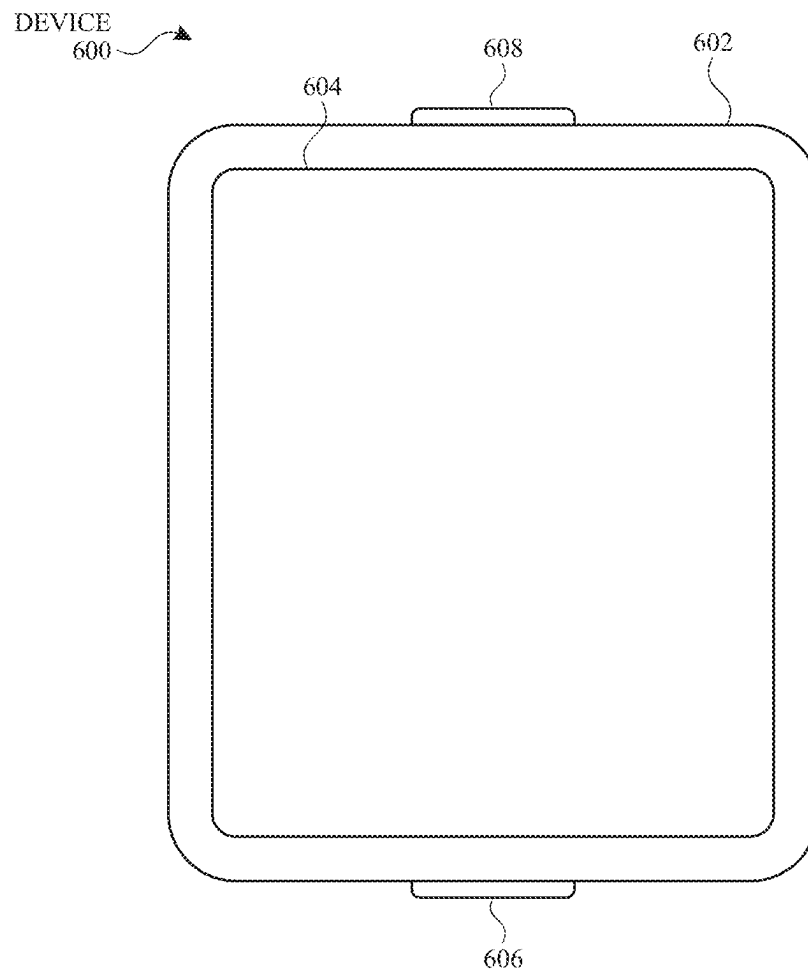


FIG. 5B

**FIG. 6A**

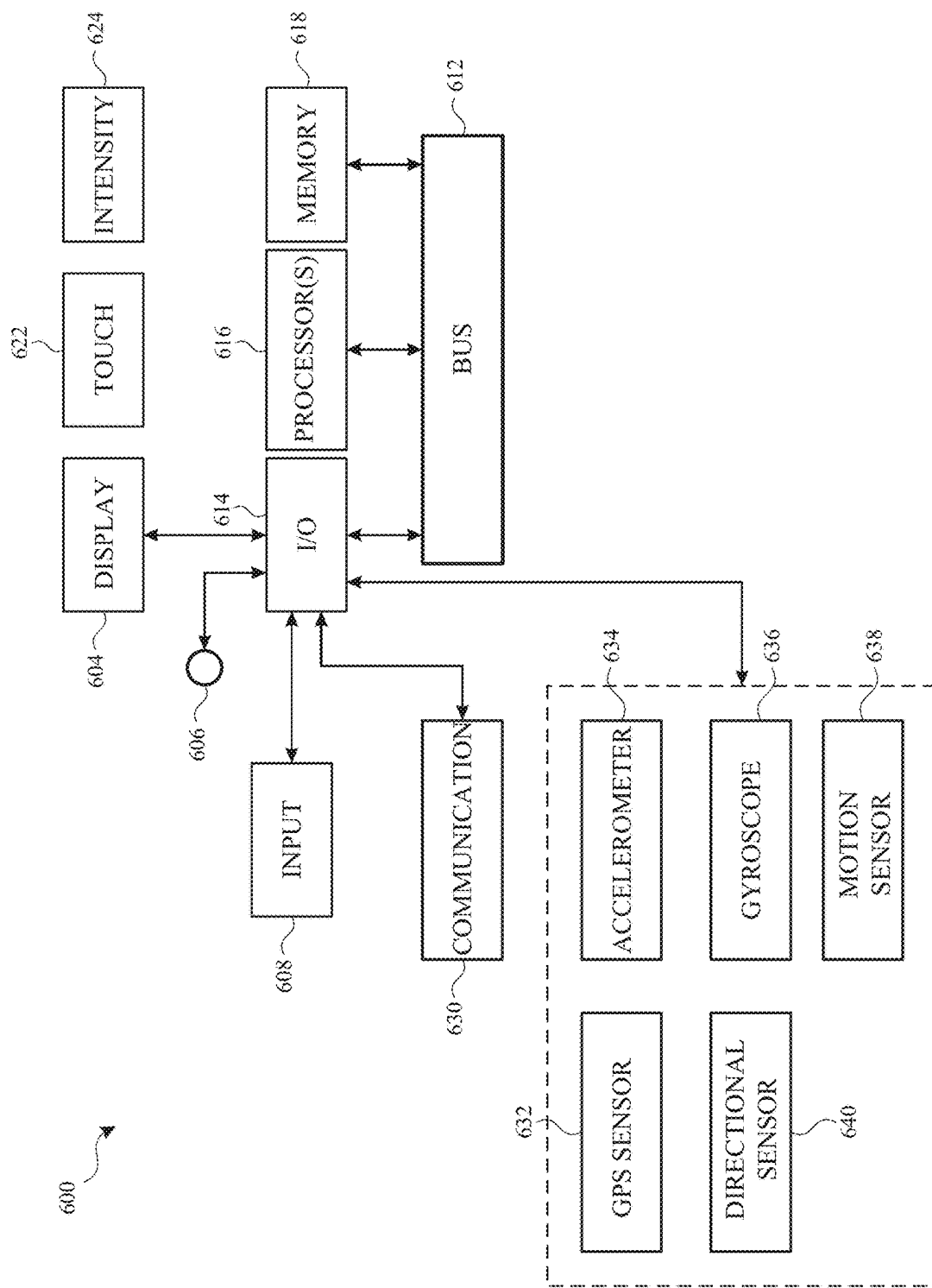


FIG. 6B

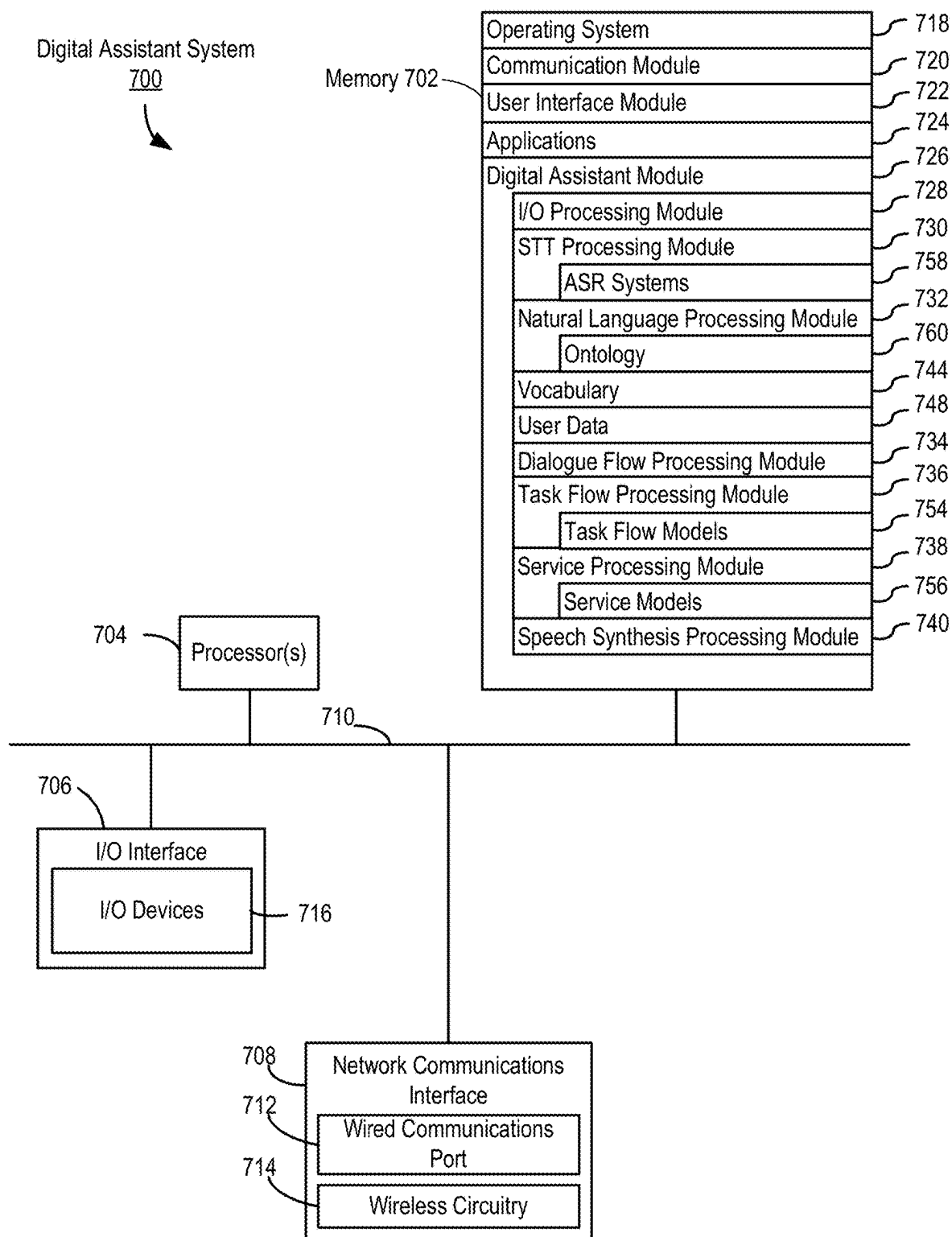


FIG. 7A

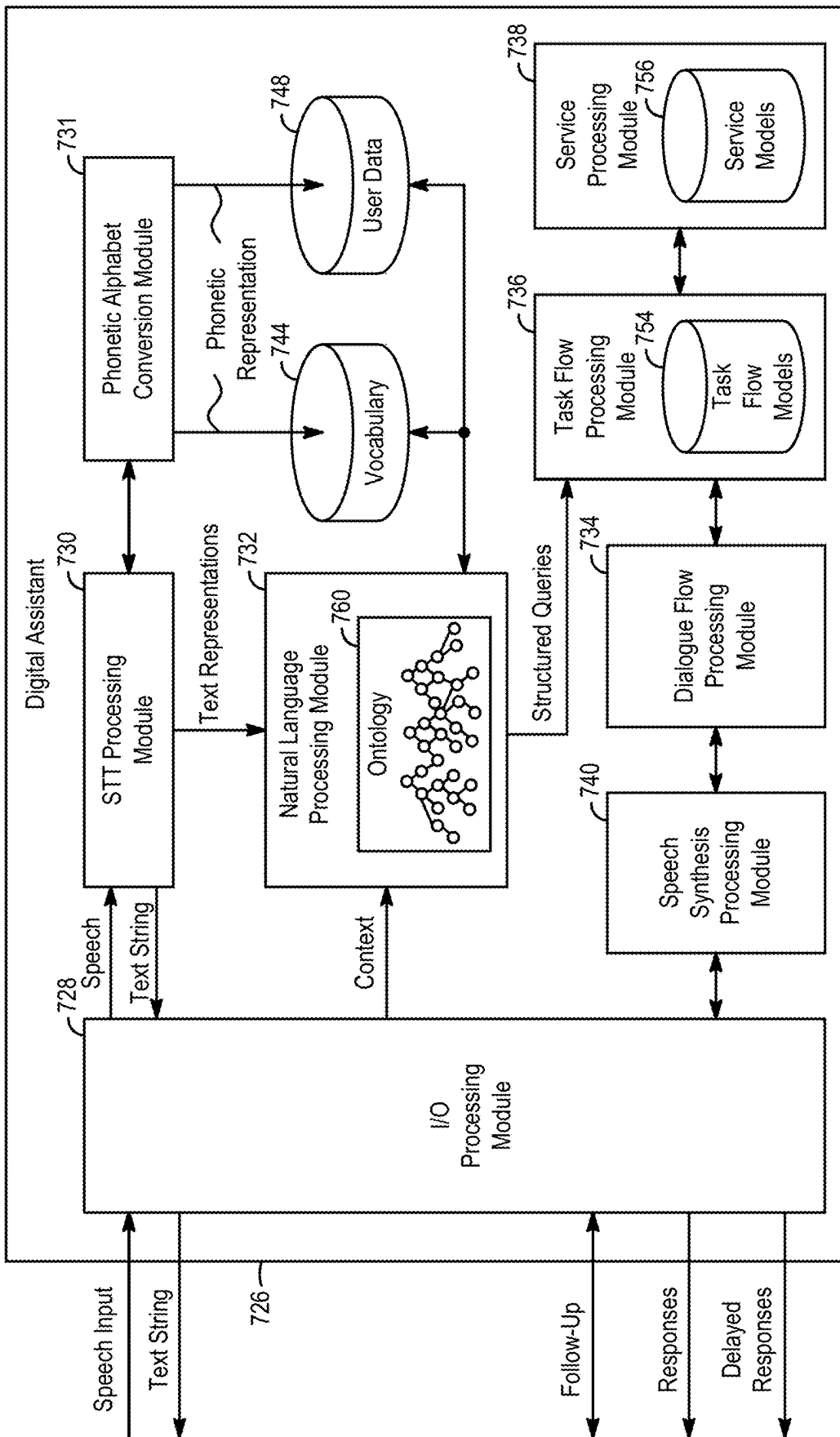


FIG. 7B

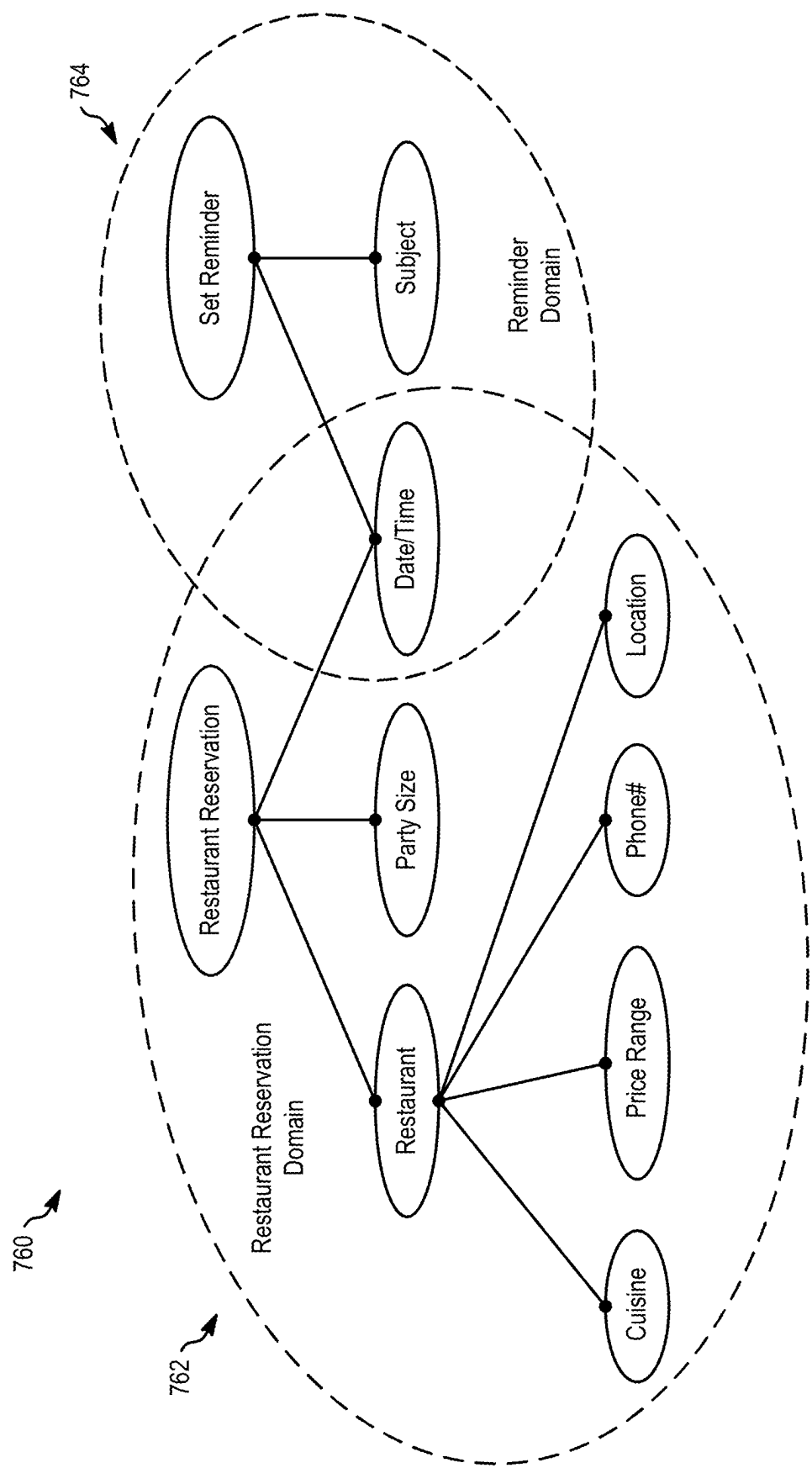


FIG. 7C

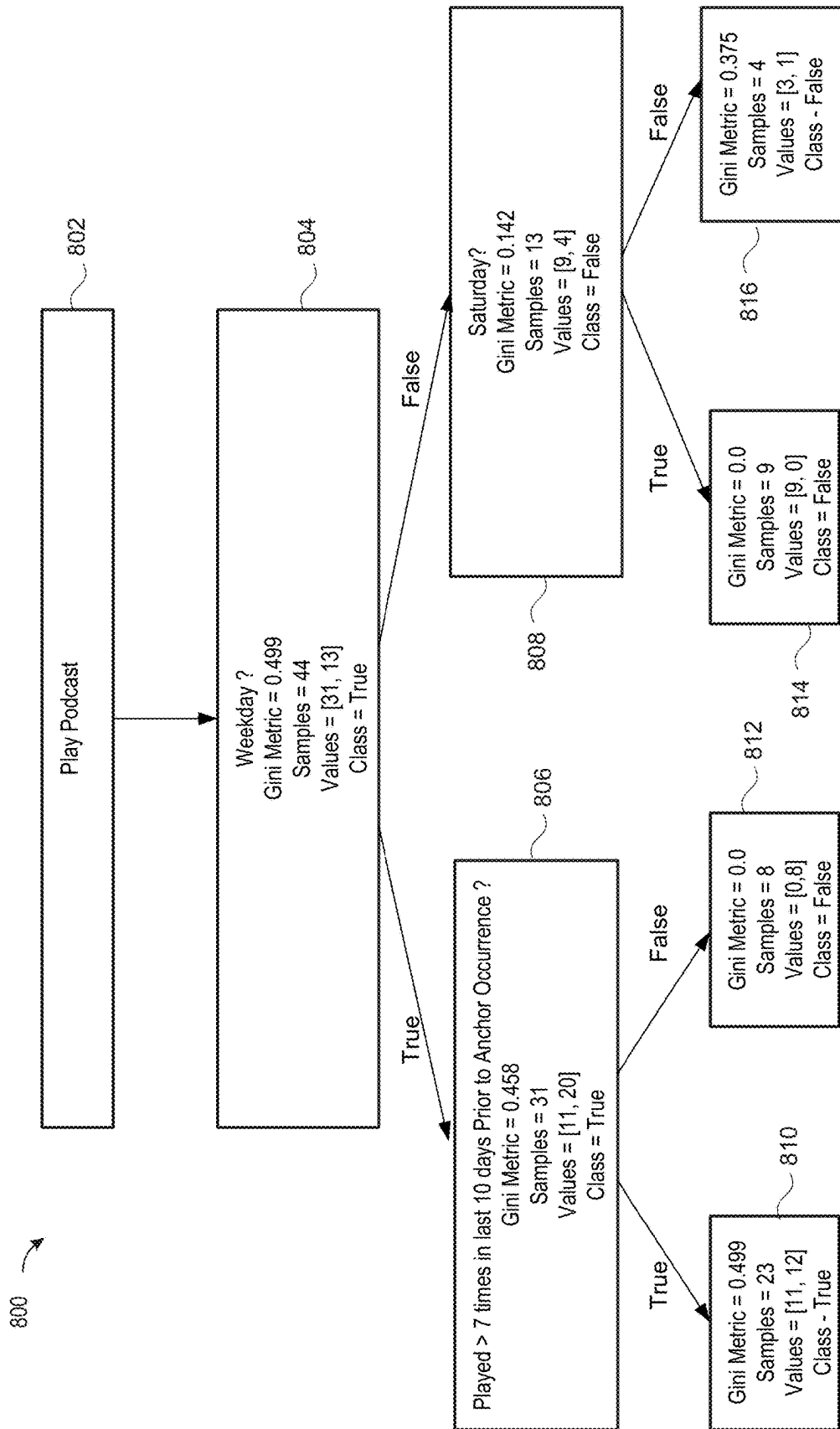


FIG. 8A

820

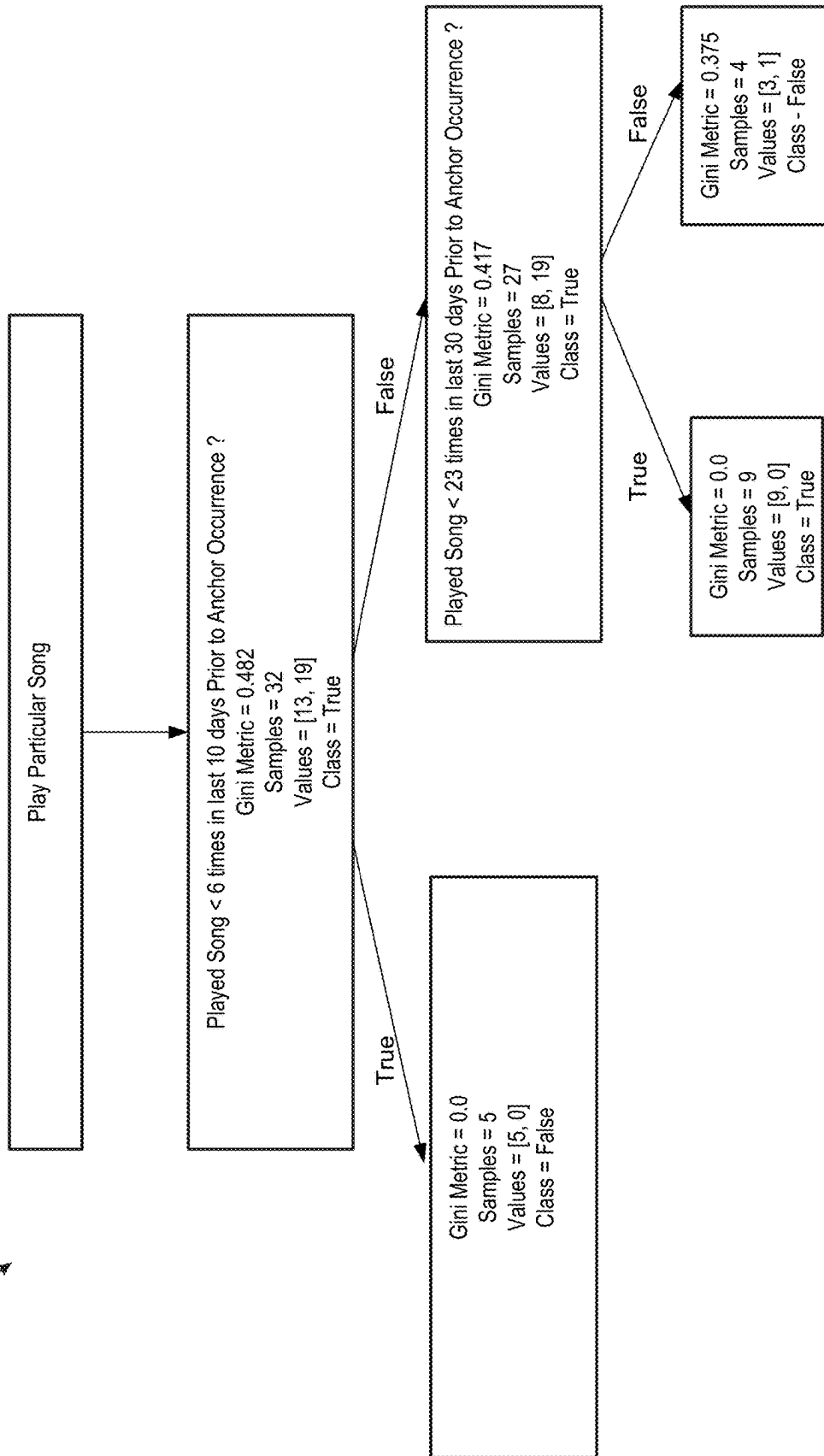


FIG. 8B

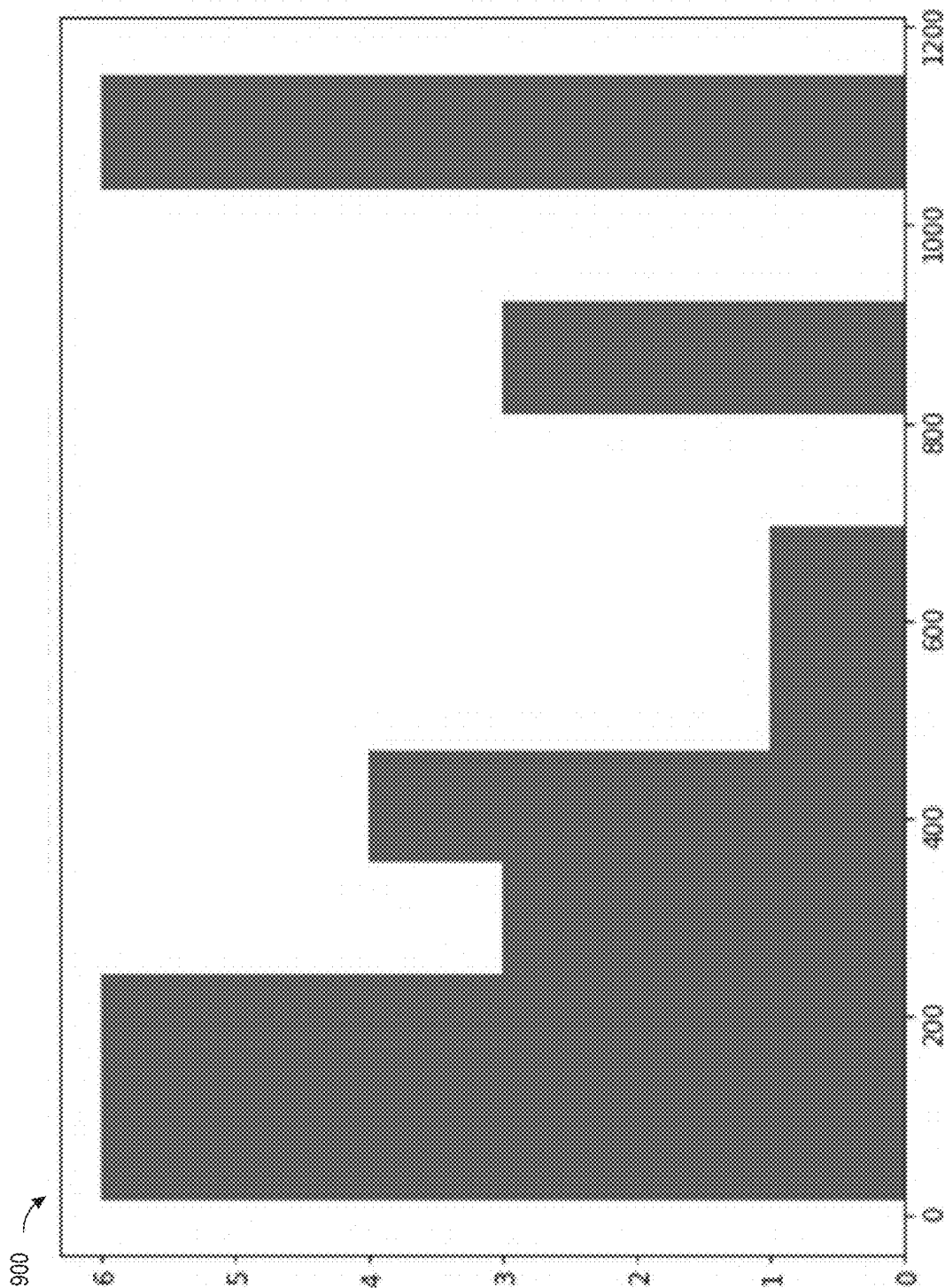


FIG. 9

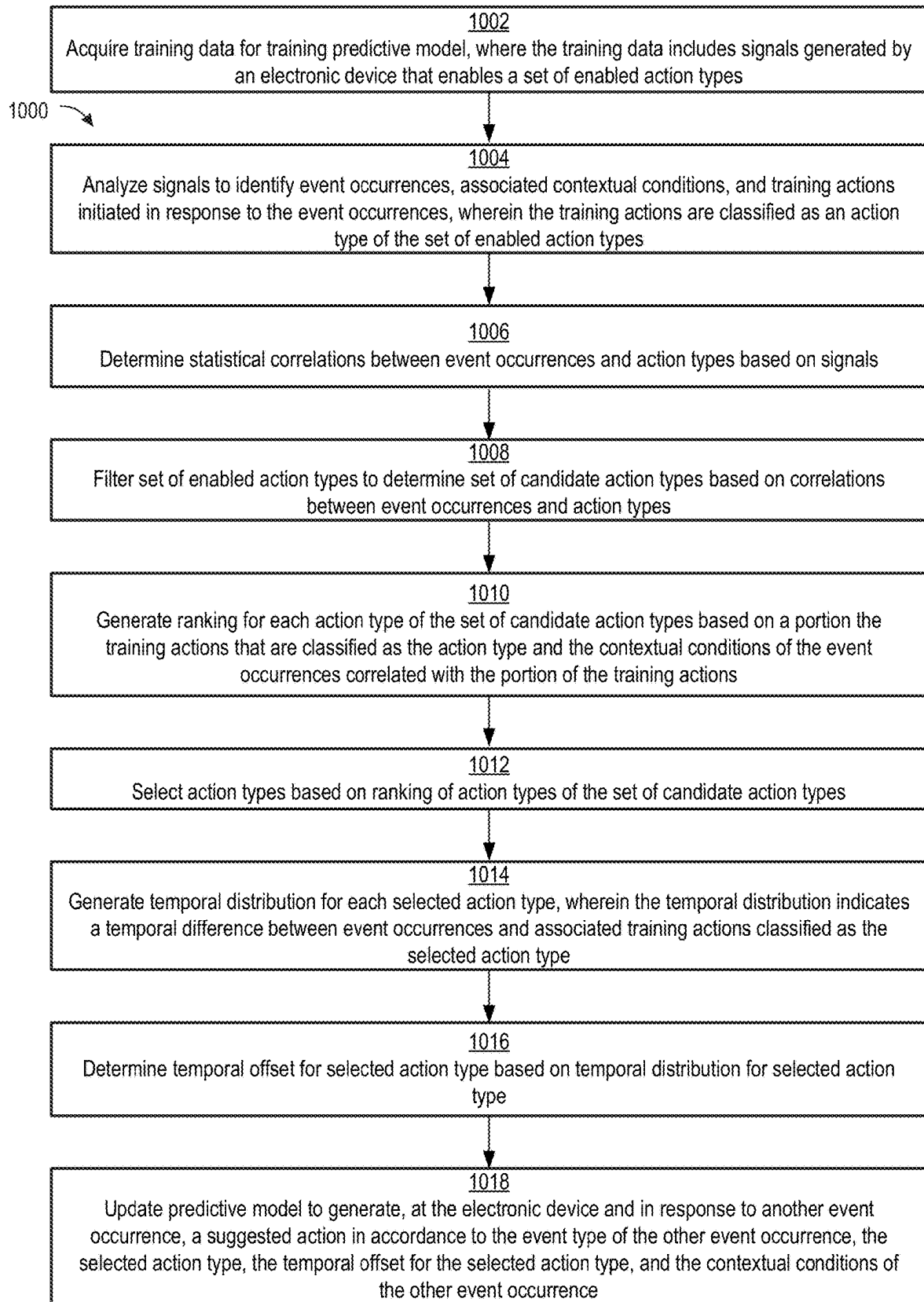
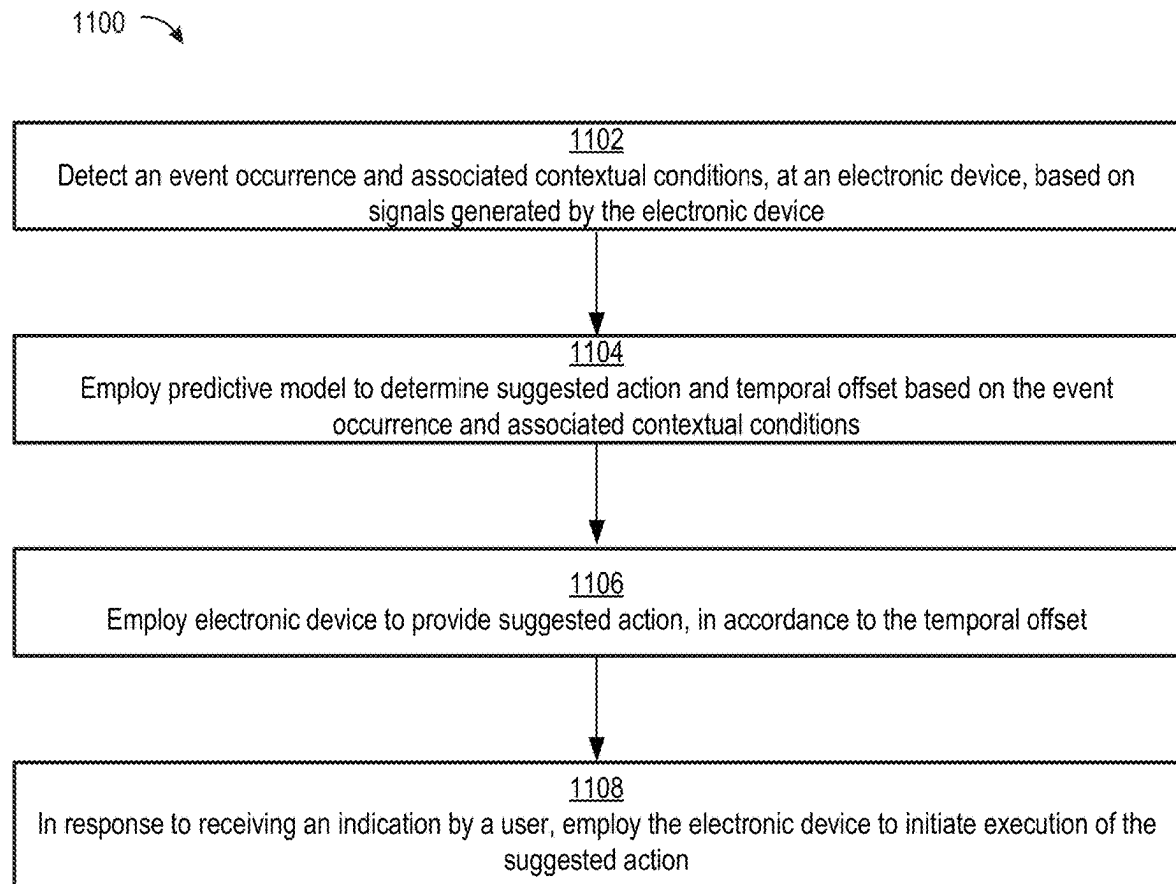


FIG. 10

**FIG. 11**

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SUGGESTING EXECUTABLE ACTIONS IN RESPONSE TO DETECTING EVENTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. non-Provisional application Ser. No. 17/336,159, filed Jun. 1, 2021, entitled “SUGGESTING EXECUTABLE ACTIONS IN RESPONSE TO DETECTING EVENTS” which claims priority to U.S. Provisional Application No. 63/033,110, filed Jun. 1, 2020, entitled “SUGGESTING EXECUTABLE ACTIONS IN RESPONSE TO DETECTING EVENTS.” The entire contents of each of these applications are hereby incorporated by reference.

FIELD

This relates generally to intelligent automated assistants and, more specifically, to enable digital assistants in intelligently suggesting executable actions, in response to detecting events that indicate user behavior, based on previous user interactions with the electronic device implementing the digital assistant.

BACKGROUND

Intelligent automated assistants (or digital assistants) can provide a beneficial interface between human users and electronic devices. Such assistants can allow users to interact with devices or systems using natural language in spoken and/or text forms. For example, a user can provide a speech input containing a user request to a digital assistant operating on an electronic device. The digital assistant can interpret the user's intent from the speech input and operationalize the user's intent into tasks. The tasks can then be performed by executing one or more services of the electronic device, and a relevant output responsive to the user request can be returned to the user.

SUMMARY

Example methods are disclosed herein. One example method includes training a predictive model (e.g., an anchor model) for a detected event (e.g., an anchor) that indicates a behavior of a user of an electronic device enabled to execute each action type of a set of enabled action types. The electronic device may have one or more processors and a memory. The electronic device may perform one or more actions and or operations. The method may include, at the electronic device and based on a plurality of signals generated by the electronic device, detecting a plurality of training actions and a plurality of event occurrences of the event. Each of the plurality of event occurrences may be associated with metadata indicating a contextual condition of the event occurrence. Each of the plurality of training actions may be initiated via the user interacting with the electronic device and is classified as an action type of the set of enabled action types. A set of candidate action types may be determined. Determining the set of candidate action types may be based on a plurality of correlations between each of the plurality of training actions and each of the plurality of event occurrences. The set of candidate action types may be a subset of the set of enabled action types. A ranking for each action type of the set of candidate action types may be determined. Determining the ranking for an action type may be based on a portion of the plurality of training actions that are classified

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as the action type. Determining the ranking of the action type may be further based on the one or more contextual conditions indicated by the metadata associated with a portion of the plurality of events occurrences that are correlated, via the plurality of correlations, with the portion of the plurality of training actions. In some embodiments, a first action type of the set of candidate action types may be selected based on the determined ranking for each action type of the set of candidate action types. A first portion of the plurality of training actions and a first portion of the plurality of event occurrences may be selected. Each of the first portion of the plurality of training actions may be classified as the first action type. Each of the first portion of the plurality of event occurrences may be correlated, via the plurality of correlations, with at least one of the first portion of the plurality of training actions. A temporal offset for the first action type may be determined based on a temporal distribution of the first of the plurality of training actions, with respect to the first portion of the plurality of event occurrences. In some embodiments, the predictive model may be updated to generate, in response to another occurrence of the event, a suggested action. The provided suggested action is in accordance with the first action type and the temporal offset of the first action type.

Example non-transitory computer-readable media are disclosed herein. An example non-transitory computer-readable storage medium stores one or more programs. The one or more programs comprise instructions, which when executed by one or more processors of an electronic device, cause the electronic device to perform actions for training a predictive model for an event that indicates a behavior user of an electronic device enabled to execute each action type of a set of enabled action types. The actions may include, at the electronic device and based on a plurality of signals generated by the electronic device, detecting a plurality of training actions and a plurality of event occurrences of the event. Each of the plurality of event occurrences may be associated with metadata indicating a contextual condition of the event occurrence. Each of the plurality of training actions may be initiated via the user interacting with the electronic device and is classified as an action type of the set of enabled action types. A set of candidate action types may be determined. Determining the set of candidate action types may be based on a plurality of correlations between each of the plurality of training actions and each of the plurality of event occurrences. The set of candidate action types may be a subset of the set of enabled action types. A ranking for each action type of the set of candidate action types may be determined. Determining the ranking for an action type may be based on a portion of the plurality of training actions that are classified as the action type. Determining the ranking of the action type may be further based on the one or more contextual conditions indicated by the metadata associated with a portion of the plurality of events occurrences that are correlated, via the plurality of correlations, with the portion of the plurality of training actions. In some embodiments, a first action type of the set of candidate action types may be selected based on the determined ranking for each action type of the set of candidate action types. A first portion of the plurality of training actions and a first portion of the plurality of event occurrences may be selected. Each of the first portion of the plurality of training actions may be classified as the first action type. Each of the first portion of the plurality of event occurrences may be correlated, via the plurality of correlations, with at least one of the first portion of the plurality of training actions. A temporal offset for the first action type may be determined based on a temporal

distribution of the first of the plurality of training actions, with respect to the first portion of the plurality of event occurrences. In some embodiments, the predictive model may be updated to generate, in response to another occurrence of the event, a suggested action. The provided suggested action is in accordance with the first action type and the temporal offset of the first action type.

Example electronic devices are disclosed herein. An example electronic device comprises one or more processors; a memory; and one or more programs, where the one or more programs are stored in the memory and configured to be executed by the one or more processors, the one or more programs including instructions for performing operations for training a predictive model for an event that indicates a behavior user of an electronic device enabled to execute each action type of a set of enabled action types. The operations may include, at the electronic device and based on a plurality of signals generated by the electronic device, detecting a plurality of training actions and a plurality of event occurrences of the event. Each of the plurality of event occurrences may be associated with metadata indicating a contextual condition of the event occurrence. Each of the plurality of training actions may be initiated via the user interacting with the electronic device and is classified as an action type of the set of enabled action types. A set of candidate action types may be determined. Determining the set of candidate action types may be based on a plurality of correlations between each of the plurality of training actions and each of the plurality of event occurrences. The set of candidate action types may be a subset of the set of enabled action types. A ranking for each action type of the set of candidate action types may be determined. Determining the ranking for an action type may be based on a portion of the plurality of training actions that are classified as the action type. Determining the ranking of the action type may be further based on the one or more contextual conditions indicated by the metadata associated with a portion of the plurality of events occurrences that are correlated, via the plurality of correlations, with the portion of the plurality of training actions. In some embodiments, a first action type of the set of candidate action types may be selected based on the determined ranking for each action type of the set of candidate action types. A first portion of the plurality of training actions and a first portion of the plurality of event occurrences may be selected. Each of the first portion of the plurality of training actions may be classified as the first action type. Each of the first portion of the plurality of event occurrences may be correlated, via the plurality of correlations, with at least one of the first portion of the plurality of training actions. A temporal offset for the first action type may be determined based on a temporal distribution of the first of the plurality of training actions, with respect to the first portion of the plurality of event occurrences. In some embodiments, the predictive model may be updated to generate, in response to another occurrence of the event, a suggested action. The provided suggested action is in accordance with the first action type and the temporal offset of the first action type.

Another example method may be for employing a predictive model for an event that indicates a behavior of a user of an electronic device. The other method may include, based on one or more signals generated by the electronic device, detecting an event occurrence of the event. The event occurrence may be associated with metadata indicating a contextual condition of the event occurrence. In accordance with the event occurrence, a suggested action and a temporal offset may be received from the predictive model. The

suggested action may be provided to the user within the temporal offset from the event occurrence.

Another example non-transitory computer-readable storage medium stores one or more other programs. The one or more other programs comprise instructions, which when executed by one or more processors of an electronic device, cause the electronic device to perform other operations for employing a predictive model for an event that indicates a behavior of a user of the electronic device. The other operations may include based on one or more signals generated by the electronic device, detecting an event occurrence of the event. The event occurrence may be associated with metadata indicating a contextual condition of the event occurrence. In accordance with the event occurrence, a suggested action and a temporal offset may be received from the predictive model. The suggested action may be provided to the user within the temporal offset from the event occurrence.

Another example electronic device comprises one or more processors; a memory; and one or more other programs, where the one or more other programs are stored in the memory and configured to be executed by the one or more processors, the one or more programs including instructions for performing other operations for employing a predictive model for an event that indicates a behavior of a user of the electronic device. The other operations may include, based on one or more signals generated by the electronic device, detecting an event occurrence of the event. The event occurrence may be associated with metadata indicating a contextual condition of the event occurrence. In accordance with the event occurrence, a suggested action and a temporal offset may be received from the predictive model. The suggested action may be provided to the user within the temporal offset from the event occurrence.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating a system and environment for implementing a digital assistant, according to various examples.

FIG. 2A is a block diagram illustrating a portable multifunction device implementing the client-side portion of a digital assistant, according to various examples.

FIG. 2B is a block diagram illustrating exemplary components for event handling, according to various examples.

FIG. 3 illustrates a portable multifunction device implementing the client-side portion of a digital assistant, according to various examples.

FIG. 4 is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface, according to various examples.

FIG. 5A illustrates an exemplary user interface for a menu of applications on a portable multifunction device, according to various examples.

FIG. 5B illustrates an exemplary user interface for a multifunction device with a touch-sensitive surface that is separate from the display, according to various examples.

FIG. 6A illustrates a personal electronic device, according to various examples.

FIG. 6B is a block diagram illustrating a personal electronic device, according to various examples.

FIG. 7A is a block diagram illustrating a digital assistant system or a server portion thereof, according to various examples.

FIG. 7B illustrates the functions of the digital assistant shown in FIG. 7A, according to various examples.

FIG. 7C illustrates a portion of an ontology, according to various examples.

FIG. 8A illustrates a classification decision tree for the action type of playing a particular podcast and the anchor type of the user waking up, according to various examples.

FIG. 8B illustrates a classification decision tree for the action type of playing a particular song and the anchor type of the user waking up, according to various examples.

FIG. 9 illustrates a histogram for the temporal distribution of the action type of playing a particular podcast, in conjunction with a waking up anchor, according to various examples.

FIG. 10 illustrates process for training a predictive model that enables providing suggested actions in response to an occurrence of an anchor event, according to various examples.

FIG. 11 illustrates process 1100 for deploying a trained predictive model at an electronic device, according to various examples.

DETAILED DESCRIPTION

In the following description of examples, reference is made to the accompanying drawings in which are shown by way of illustration specific examples that can be practiced. It is to be understood that other examples can be used and structural changes can be made without departing from the scope of the various examples.

The embodiments are directed to providing, via an electronic device, suggested user actions. The suggested actions may be provided in response to detecting an occurrence of an anchor event. An anchor event (or simply an anchor) may be a predefined event occurring in the user's day. For example, upon waking up in the morning (e.g., a user waking up may be an example of an anchor), an electronic device may provide a notification that includes a suggestion for playing the newest episode of a particular podcast (i.e., a suggested user action), where the user frequently listens to the particular podcast upon waking up in the morning. The occurrence of the anchor (or event) may be detected via one or more signals generated by the electronic device (e.g., the device's alarm clock function executing, the device being transitioned from a "Do Not Disturb Mode" to an "Active Mode," or the like). An occurrence of the anchor may be detectable via the monitoring and analysis of electronic signals generated by the electronic device. Based on the user's previous interactions with the device (e.g., encoded in training data), the occurrence of the anchor may be indicative of user behavior and/or action taken in response to the event in the user's day. By way of non-limiting examples, an anchor may include the user entering a location of interest (LOI), such as but not limited to the user's home, office, a fitness center, airport, shopping center, or the like. Because an anchor may be an event that indicates user behavior, an anchor may be referred throughout as an anchor event, or simply an event. The occurrence of an anchor may be referred to as an anchor occurrence.

Other non-limiting examples of an occurrence of an anchor include the user beginning to use an electronic device after an extended period of device idle time, the user finishing a workout, the user waking up, the user going to bed, the user pairing a Bluetooth-enabled device to the electronic device, the user beginning or completing a calendar event, and the like. Still other examples of an anchor include the user launching a specific application and/or employing a specific functionality of the device. Any event

detectable via signatures encoded in signals generated by the electronic device may be an anchor. A signature (e.g., a pattern within the electronic signals generated by the device) that indicates an occurrence of a specific anchor may be pre-determined, pre-computed, and/or pre-learned (e.g., via supervised or unsupervised machine learning (ML) methods). Thus, the electronic device may monitor its various signals and detect an occurrence of the specific anchor by identifying and/or detecting one or more signatures within the signals that indicates the specific anchor. For example, an occurrence of a waking up anchor may be detected via signals of the electronic device that indicates that the user has terminated, or otherwise ended, an "idle" state of the electronic device (or that the device's alarm went off). Upon detecting an occurrence of the anchor, the various embodiments may suggest one or more user actions that correlate (via previously generated training data) with the occurrence of the anchor.

A user action (i.e., an action) may include an invocation, execution, or otherwise launching a specific application, capability, functionality, or command that is enabled via the electronic device. By way of non-limiting examples, a user action may include but is not limited to playing a specific audio/video content (e.g., a podcast, a music playlist, an audio book, a lecture, a television series, a movie, or the like), launching an application installed on the device (e.g., a workout application, a meditation application, a ride-share application, a food delivery application, a social network application, or the like), sending an electronic communication (e.g., email, SMS, tweet, or the like) to another user, user group, or social network, turning on/off functionality of the device (e.g., turning off/on an Airplane mode of the device), updating various settings and/or configurations of the device, creating a calendar event, or the like.

When providing a suggested action, the various embodiments may provide a notification (e.g., a "pop-up" notification) that indicates the suggested action. The notification may be interactive, in that the user may initiate the action via an interactive selection of the notification. Upon selecting the notification, the suggested action may be executed and/or the execution of the action may be initiated, by the device.

In the various embodiments, one or more machine learning (ML) models may be trained to learn various statistical correlations (or associations) between occurrences of a specific anchor and actions (e.g., innovations of device functionalities or capabilities) initiated by the user. Any of the various ML models employed by the various embodiments may be herein collectively referred to as "anchor models," because the models are trained to detect occurrences of various anchors and suggest one or more actions that the user is statistically likely to be initiate in response to the occurrence (or impending occurrence) of the anchor. Thus, the various embodiments include the training of anchor models, as well as employing the trained anchor models to enhance the user's experience (UX) of employing the device. Supervised or unsupervised ML may be employed to train the models to determine the context (e.g., contextual features) of an anchor occurrence, and base, tailor, or selectively target the suggested actions on the context of the anchor.

Although the following description uses terms "first," "second," etc. to describe various elements, these elements should not be limited by the terms. These terms are only used to distinguish one element from another. For example, a first input could be termed a second input, and, similarly, a second input could be termed a first input, without depart-

ing from the scope of the various described examples. The first input and the second input are both inputs and, in some cases, are separate and different inputs.

The terminology used in the description of the various described examples herein is for the purpose of describing particular examples only and is not intended to be limiting. As used in the description of the various described examples and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

The term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” may be construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

1. System and Environment

FIG. 1 illustrates a block diagram of system 100 according to various examples. In some examples, system 100 implements a digital assistant. The terms “digital assistant,” “virtual assistant,” “intelligent automated assistant,” or “automatic digital assistant” refer to any information processing system that interprets natural language input in spoken and/or textual form to infer user intent, and performs actions based on the inferred user intent. For example, to act on an inferred user intent, the system performs one or more of the following: identifying a task flow with steps and parameters designed to accomplish the inferred user intent, inputting specific requirements from the inferred user intent into the task flow; executing the task flow by invoking programs, methods, services, APIs, or the like; and generating output responses to the user in an audible (e.g., speech) and/or visual form.

Specifically, a digital assistant is capable of accepting a user request at least partially in the form of a natural language command, request, statement, narrative, and/or inquiry. Typically, the user request seeks either an informational answer or performance of a task by the digital assistant. A satisfactory response to the user request includes a provision of the requested informational answer, a performance of the requested task, or a combination of the two. For example, a user asks the digital assistant a question, such as “Where am I right now?” Based on the user’s current location, the digital assistant answers, “You are in Central Park near the west gate.” The user also requests the performance of a task, for example, “Please invite my friends to my girlfriend’s birthday party next week.” In response, the digital assistant can acknowledge the request by saying “Yes, right away,” and then send a suitable calendar invite on behalf of the user to each of the user’s friends listed in the user’s electronic address book. During performance of a requested task, the digital assistant sometimes interacts with the user in a continuous dialogue involving multiple

exchanges of information over an extended period of time. There are numerous other ways of interacting with a digital assistant to request information or performance of various tasks. In addition to providing verbal responses and taking programmed actions, the digital assistant also provides responses in other visual or audio forms, e.g., as text, alerts, music, videos, animations, etc.

As shown in FIG. 1, in some examples, a digital assistant is implemented according to a client-server model. The digital assistant includes client-side portion 102 (hereafter “DA client 102”) executed on user device 104 and server-side portion 106 (hereafter “DA server 106”) executed on server system 108. DA client 102 communicates with DA server 106 through one or more networks 110. DA client 102 provides client-side functionalities such as user-facing input and output processing and communication with DA server 106. DA server 106 provides server-side functionalities for any number of DA clients 102 each residing on a respective user device 104.

In some examples, DA server 106 includes client-facing I/O interface 112, one or more processing modules 114, data and models 116, and I/O interface to external services 118. The client-facing I/O interface 112 facilitates the client-facing input and output processing for DA server 106. One or more processing modules 114 utilize data and models 116 to process speech input and determine the user’s intent based on natural language input. Further, one or more processing modules 114 perform task execution based on inferred user intent. In some examples, DA server 106 communicates with external services 120 through network(s) 110 for task completion or information acquisition. I/O interface to external services 118 facilitates such communications.

User device 104 can be any suitable electronic device. In some examples, user device 104 is a portable multifunctional device (e.g., device 200, described below with reference to FIG. 2A), a multifunctional device (e.g., device 400, described below with reference to FIG. 4), or a personal electronic device (e.g., device 600, described below with reference to FIGS. 6A-6B.) A portable multifunctional device is, for example, a mobile telephone that also contains other functions, such as PDA and/or music player functions. Specific examples of portable multifunction devices include the Apple Watch®, iPhone®, iPod Touch®, and iPad® devices from Apple Inc. of Cupertino, California. Other examples of portable multifunction devices include, without limitation, earphones/headphones, speakers, and laptop or tablet computers. Further, in some examples, user device 104 is a non-portable multifunctional device. In particular, user device 104 is a desktop computer, a game console, a speaker, a television, or a television set-top box. In some examples, user device 104 includes a touch-sensitive surface (e.g., touch screen displays and/or touchpads). Further, user device 104 optionally includes one or more other physical user-interface devices, such as a physical keyboard, a mouse, and/or a joystick. Various examples of electronic devices, such as multifunctional devices, are described below in greater detail.

Examples of communication network(s) 110 include local area networks (LAN) and wide area networks (WAN), e.g., the Internet. Communication network(s) 110 is implemented using any known network protocol, including various wired or wireless protocols, such as, for example, Ethernet, Universal Serial Bus (USB), FIREWIRE, Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), code division multiple access (CDMA), time division multiple access (TDMA), Blu-

etooth, Wi-Fi, voice over Internet Protocol (VOIP), Wi-MAX, or any other suitable communication protocol.

Server system 108 is implemented on one or more standalone data processing apparatus or a distributed network of computers. In some examples, server system 108 also employs various virtual devices and/or services of third-party service providers (e.g., third-party cloud service providers) to provide the underlying computing resources and/or infrastructure resources of server system 108.

In some examples, user device 104 communicates with DA server 106 via second user device 122. Second user device 122 is similar or identical to user device 104. For example, second user device 122 is similar to devices 200, 400, or 600 described below with reference to FIGS. 2A, 4, and 6A-6B. User device 104 is configured to communicatively couple to second user device 122 via a direct communication connection, such as Bluetooth, NFC, BTLE, or the like, or via a wired or wireless network, such as a local Wi-Fi network. In some examples, second user device 122 is configured to act as a proxy between user device 104 and DA server 106. For example, DA client 102 of user device 104 is configured to transmit information (e.g., a user request received at user device 104) to DA server 106 via second user device 122. DA server 106 processes the information and returns relevant data (e.g., data content responsive to the user request) to user device 104 via second user device 122.

In some examples, user device 104 is configured to communicate abbreviated requests for data to second user device 122 to reduce the amount of information transmitted from user device 104. Second user device 122 is configured to determine supplemental information to add to the abbreviated request to generate a complete request to transmit to DA server 106. This system architecture can advantageously allow user device 104 having limited communication capabilities and/or limited battery power (e.g., a watch or a similar compact electronic device) to access services provided by DA server 106 by using second user device 122, having greater communication capabilities and/or battery power (e.g., a mobile phone, laptop computer, tablet computer, or the like), as a proxy to DA server 106. While only two user devices 104 and 122 are shown in FIG. 1, it should be appreciated that system 100, in some examples, includes any number and type of user devices configured in this proxy configuration to communicate with DA server system 106.

Although the digital assistant shown in FIG. 1 includes both a client-side portion (e.g., DA client 102) and a server-side portion (e.g., DA server 106), in some examples, the functions of a digital assistant are implemented as a standalone application installed on a user device. In addition, the divisions of functionalities between the client and server portions of the digital assistant can vary in different implementations. For instance, in some examples, the DA client is a thin-client that provides only user-facing input and output processing functions, and delegates all other functionalities of the digital assistant to a backend server.

2. Electronic Devices

Attention is now directed toward embodiments of electronic devices for implementing the client-side portion of a digital assistant. FIG. 2A is a block diagram illustrating portable multifunction device 200 with touch-sensitive display system 212 in accordance with some embodiments. Touch-sensitive display 212 is sometimes called a “touch screen” for convenience and is sometimes known as or called a “touch-sensitive display system.” Device 200

includes memory 202 (which optionally includes one or more computer-readable storage mediums), memory controller 222, one or more processing units (CPUs) 220, peripherals interface 218, RF circuitry 208, audio circuitry 210, speaker 211, microphone 213, input/output (I/O) subsystem 206, other input control devices 216, and external port 224. Device 200 optionally includes one or more optical sensors 264. Device 200 optionally includes one or more contact intensity sensors 265 for detecting intensity of contacts on device 200 (e.g., a touch-sensitive surface such as touch-sensitive display system 212 of device 200). Device 200 optionally includes one or more tactile output generators 267 for generating tactile outputs on device 200 (e.g., generating tactile outputs on a touch-sensitive surface such as touch-sensitive display system 212 of device 200 or touchpad 455 of device 400). These components optionally communicate over one or more communication buses or signal lines 203.

As used in the specification and claims, the term “intensity” of a contact on a touch-sensitive surface refers to the force or pressure (force per unit area) of a contact (e.g., a finger contact) on the touch-sensitive surface, or to a substitute (proxy) for the force or pressure of a contact on the touch-sensitive surface. The intensity of a contact has a range of values that includes at least four distinct values and more typically includes hundreds of distinct values (e.g., at least 256). Intensity of a contact is, optionally, determined (or measured) using various approaches and various sensors or combinations of sensors. For example, one or more force sensors underneath or adjacent to the touch-sensitive surface are, optionally, used to measure force at various points on the touch-sensitive surface. In some implementations, force measurements from multiple force sensors are combined (e.g., a weighted average) to determine an estimated force of a contact. Similarly, a pressure-sensitive tip of a stylus is, optionally, used to determine a pressure of the stylus on the touch-sensitive surface. Alternatively, the size of the contact area detected on the touch-sensitive surface and/or changes thereto, the capacitance of the touch-sensitive surface proximate to the contact and/or changes thereto, and/or the resistance of the touch-sensitive surface proximate to the contact and/or changes thereto are, optionally, used as a substitute for the force or pressure of the contact on the touch-sensitive surface. In some implementations, the substitute measurements for contact force or pressure are used directly to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is described in units corresponding to the substitute measurements). In some implementations, the substitute measurements for contact force or pressure are converted to an estimated force or pressure, and the estimated force or pressure is used to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is a pressure threshold measured in units of pressure). Using the intensity of a contact as an attribute of a user input allows for user access to additional device functionality that may otherwise not be accessible by the user on a reduced-size device with limited real estate for displaying affordances (e.g., on a touch-sensitive display) and/or receiving user input (e.g., via a touch-sensitive display, a touch-sensitive surface, or a physical/mechanical control such as a knob or a button).

As used in the specification and claims, the term “tactile output” refers to physical displacement of a device relative to a previous position of the device, physical displacement of a component (e.g., a touch-sensitive surface) of a device relative to another component (e.g., housing) of the device, or displacement of the component relative to a center of

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mass of the device that will be detected by a user with the user's sense of touch. For example, in situations where the device or the component of the device is in contact with a surface of a user that is sensitive to touch (e.g., a finger, palm, or other part of a user's hand), the tactile output generated by the physical displacement will be interpreted by the user as a tactile sensation corresponding to a perceived change in physical characteristics of the device or the component of the device. For example, movement of a touch-sensitive surface (e.g., a touch-sensitive display or trackpad) is, optionally, interpreted by the user as a "down click" or "up click" of a physical actuator button. In some cases, a user will feel a tactile sensation such as an "down click" or "up click" even when there is no movement of a physical actuator button associated with the touch-sensitive surface that is physically pressed (e.g., displaced) by the user's movements. As another example, movement of the touch-sensitive surface is, optionally, interpreted or sensed by the user as "roughness" of the touch-sensitive surface, even when there is no change in smoothness of the touch-sensitive surface. While such interpretations of touch by a user will be subject to the individualized sensory perceptions of the user, there are many sensory perceptions of touch that are common to a large majority of users. Thus, when a tactile output is described as corresponding to a particular sensory perception of a user (e.g., an "up click," a "down click," "roughness"), unless otherwise stated, the generated tactile output corresponds to physical displacement of the device or a component thereof that will generate the described sensory perception for a typical (or average) user.

It should be appreciated that device **200** is only one example of a portable multifunction device, and that device **200** optionally has more or fewer components than shown, optionally combines two or more components, or optionally has a different configuration or arrangement of the components. The various components shown in FIG. 2A are implemented in hardware, software, or a combination of both hardware and software, including one or more signal processing and/or application-specific integrated circuits.

Memory **202** includes one or more computer-readable storage mediums. The computer-readable storage mediums are, for example, tangible and non-transitory. Memory **202** includes high-speed random access memory and also includes non-volatile memory, such as one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Memory controller **222** controls access to memory **202** by other components of device **200**.

In some examples, a non-transitory computer-readable storage medium of memory **202** is used to store instructions (e.g., for performing aspects of processes described below) for use by or in connection with an instruction execution system, apparatus, or device, such as a computer-based system, processor-containing system, or other system that can fetch the instructions from the instruction execution system, apparatus, or device and execute the instructions. In other examples, the instructions (e.g., for performing aspects of the processes described below) are stored on a non-transitory computer-readable storage medium (not shown) of the server system **108** or are divided between the non-transitory computer-readable storage medium of memory **202** and the non-transitory computer-readable storage medium of server system **108**.

Peripherals interface **218** is used to couple input and output peripherals of the device to CPU **220** and memory **202**. The one or more processors **220** run or execute various software programs and/or sets of instructions stored in

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memory **202** to perform various functions for device **200** and to process data. In some embodiments, peripherals interface **218**, CPU **220**, and memory controller **222** are implemented on a single chip, such as chip **204**. In some other embodiments, they are implemented on separate chips.

RF (radio frequency) circuitry **208** receives and sends RF signals, also called electromagnetic signals. RF circuitry **208** converts electrical signals to/from electromagnetic signals and communicates with communications networks and other communications devices via the electromagnetic signals. RF circuitry **208** optionally includes well-known circuitry for performing these functions, including but not limited to an antenna system, an RF transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a CODEC chipset, a subscriber identity module (SIM) card, memory, and so forth. RF circuitry **208** optionally communicates with networks, such as the Internet, also referred to as the World Wide Web (WWW), an intranet and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN) and/or a metropolitan area network (MAN), and other devices by wireless communication. The RF circuitry **208** optionally includes well-known circuitry for detecting near field communication (NFC) fields, such as by a short-range communication radio. The wireless communication optionally uses any of a plurality of communications standards, protocols, and technologies, including but not limited to Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), high-speed downlink packet access (HSDPA), high-speed uplink packet access (HSUPA), Evolution, Data-Only (EV-DO), HSPA, HSPA+, Dual-Cell HSPA (DC-HSPA), long term evolution (LTE), near field communication (NFC), wideband code division multiple access (W-CDMA), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Bluetooth Low Energy (BTLE), Wireless Fidelity (Wi-Fi) (e.g., IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, IEEE 802.11n, and/or IEEE 802.11ac), voice over Internet Protocol (VOIP), Wi-MAX, a protocol for e mail (e.g., Internet message access protocol (IMAP) and/or post office protocol (POP)), instant messaging (e.g., extensible messaging and presence protocol (XMPP), Session Initiation Protocol for Instant Messaging and Presence Leveraging Extensions (SIMPLE), Instant Messaging and Presence Service (IMPS)), and/or Short Message Service (SMS), or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

Audio circuitry **210**, speaker **211**, and microphone **213** provide an audio interface between a user and device **200**. Audio circuitry **210** receives audio data from peripherals interface **218**, converts the audio data to an electrical signal, and transmits the electrical signal to speaker **211**. Speaker **211** converts the electrical signal to human-audible sound waves. Audio circuitry **210** also receives electrical signals converted by microphone **213** from sound waves. Audio circuitry **210** converts the electrical signal to audio data and transmits the audio data to peripherals interface **218** for processing. Audio data are retrieved from and/or transmitted to memory **202** and/or RF circuitry **208** by peripherals interface **218**. In some embodiments, audio circuitry **210** also includes a headset jack (e.g., **312**, FIG. 3). The headset jack provides an interface between audio circuitry **210** and removable audio input/output peripherals, such as output-only headphones or a headset with both output (e.g., a headphone for one or both ears) and input (e.g., a microphone).

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I/O subsystem **206** couples input/output peripherals on device **200**, such as touch screen **212** and other input control devices **216**, to peripherals interface **218**. I/O subsystem **206** optionally includes display controller **256**, optical sensor controller **258**, intensity sensor controller **259**, haptic feedback controller **261**, and one or more input controllers **260** for other input or control devices. The one or more input controllers **260** receive/send electrical signals from/to other input control devices **216**. The other input control devices **216** optionally include physical buttons (e.g., push buttons, rocker buttons, etc.), dials, slider switches, joysticks, click wheels, and so forth. In some alternate embodiments, input controller(s) **260** are, optionally, coupled to any (or none) of the following: a keyboard, an infrared port, a USB port, and a pointer device such as a mouse. The one or more buttons (e.g., **308**, FIG. 3) optionally include an up/down button for volume control of speaker **211** and/or microphone **213**. The one or more buttons optionally include a push button (e.g., **306**, FIG. 3).

A quick press of the push button disengages a lock of touch screen **212** or begin a process that uses gestures on the touch screen to unlock the device, as described in U.S. patent application Ser. No. 11/322,549, "Unlocking a Device by Performing Gestures on an Unlock Image," filed Dec. 23, 2005, U.S. Pat. No. 7,657,849, which is hereby incorporated by reference in its entirety. A longer press of the push button (e.g., **306**) turns power to device **200** on or off. The user is able to customize a functionality of one or more of the buttons. Touch screen **212** is used to implement virtual or soft buttons and one or more soft keyboards.

Touch-sensitive display **212** provides an input interface and an output interface between the device and a user. Display controller **256** receives and/or sends electrical signals from/to touch screen **212**. Touch screen **212** displays visual output to the user. The visual output includes graphics, text, icons, video, and any combination thereof (collectively termed "graphics"). In some embodiments, some or all of the visual output correspond to user-interface objects.

Touch screen **212** has a touch-sensitive surface, sensor, or set of sensors that accepts input from the user based on haptic and/or tactile contact. Touch screen **212** and display controller **256** (along with any associated modules and/or sets of instructions in memory **202**) detect contact (and any movement or breaking of the contact) on touch screen **212** and convert the detected contact into interaction with user-interface objects (e.g., one or more soft keys, icons, web pages, or images) that are displayed on touch screen **212**. In an exemplary embodiment, a point of contact between touch screen **212** and the user corresponds to a finger of the user.

Touch screen **212** uses LCD (liquid crystal display) technology, LPD (light emitting polymer display) technology, or LED (light emitting diode) technology, although other display technologies may be used in other embodiments. Touch screen **212** and display controller **256** detect contact and any movement or breaking thereof using any of a plurality of touch sensing technologies now known or later developed, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with touch screen **212**. In an exemplary embodiment, projected mutual capacitance sensing technology is used, such as that found in the iPhone® and iPod Touch® from Apple Inc. of Cupertino, California.

A touch-sensitive display in some embodiments of touch screen **212** is analogous to the multi-touch sensitive touchpads described in the following U.S. Pat. No. 6,323,846

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(Westerman et al.), U.S. Pat. No. 6,570,557 (Westerman et al.), and/or U.S. Pat. No. 6,677,932 (Westerman), and/or U.S. Patent Publication 2002/0015024A1, each of which is hereby incorporated by reference in its entirety. However, touch screen **212** displays visual output from device **200**, whereas touch-sensitive touchpads do not provide visual output.

A touch-sensitive display in some embodiments of touch screen **212** is as described in the following applications: (1) U.S. patent application Ser. No. 11/381,313, "Multipoint Touch Surface Controller," filed May 2, 2006; (2) U.S. patent application Ser. No. 10/840,862, "Multipoint Touchscreen," filed May 6, 2004; (3) U.S. patent application Ser. No. 10/903,964, "Gestures For Touch Sensitive Input Devices," filed Jul. 30, 2004; (4) U.S. patent application Ser. No. 11/048,264, "Gestures For Touch Sensitive Input Devices," filed Jan. 31, 2005; (5) U.S. patent application Ser. No. 11/038,590, "Mode-Based Graphical User Interfaces For Touch Sensitive Input Devices," filed Jan. 18, 2005; (6) U.S. patent application Ser. No. 11/228,758, "Virtual Input Device Placement On A Touch Screen User Interface," filed Sep. 16, 2005; (7) U.S. patent application Ser. No. 11/228,700, "Operation Of A Computer With A Touch Screen Interface," filed Sep. 16, 2005; (8) U.S. patent application Ser. No. 11/228,737, "Activating Virtual Keys Of A Touch-Screen Virtual Keyboard," filed Sep. 16, 2005; and (9) U.S. patent application Ser. No. 11/367,749, "Multi-Functional Hand-Held Device," filed Mar. 3, 2006. All of these applications are incorporated by reference herein in their entirety.

Touch screen **212** has, for example, a video resolution in excess of 100 dpi. In some embodiments, the touch screen has a video resolution of approximately 160 dpi. The user makes contact with touch screen **212** using any suitable object or appendage, such as a stylus, a finger, and so forth. In some embodiments, the user interface is designed to work primarily with finger-based contacts and gestures, which can be less precise than stylus-based input due to the larger area of contact of a finger on the touch screen. In some embodiments, the device translates the rough finger-based input into a precise pointer/cursor position or command for performing the actions desired by the user.

In some embodiments, in addition to the touch screen, device **200** includes a touchpad (not shown) for activating or deactivating particular functions. In some embodiments, the touchpad is a touch-sensitive area of the device that, unlike the touch screen, does not display visual output. The touchpad is a touch-sensitive surface that is separate from touch screen **212** or an extension of the touch-sensitive surface formed by the touch screen.

Device **200** also includes power system **262** for powering the various components. Power system **262** includes a power management system, one or more power sources (e.g., battery, alternating current (AC)), a recharging system, a power failure detection circuit, a power converter or inverter, a power status indicator (e.g., a light-emitting diode (LED)) and any other components associated with the generation, management and distribution of power in portable devices.

Device **200** also includes one or more optical sensors **264**. FIG. 2A shows an optical sensor coupled to optical sensor controller **258** in I/O subsystem **206**. Optical sensor **264** includes charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS) phototransistors. Optical sensor **264** receives light from the environment, projected through one or more lenses, and converts the light to data representing an image. In conjunction with imaging module **243** (also called a camera module), optical sensor

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264 captures still images or video. In some embodiments, an optical sensor is located on the back of device **200**, opposite touch screen display **212** on the front of the device so that the touch screen display is used as a viewfinder for still and/or video image acquisition. In some embodiments, an optical sensor is located on the front of the device so that the user's image is obtained for video conferencing while the user views the other video conference participants on the touch screen display. In some embodiments, the position of optical sensor **264** can be changed by the user (e.g., by rotating the lens and the sensor in the device housing) so that a single optical sensor **264** is used along with the touch screen display for both video conferencing and still and/or video image acquisition.

Device **200** optionally also includes one or more contact intensity sensors **265**. FIG. 2A shows a contact intensity sensor coupled to intensity sensor controller **259** in I/O subsystem **206**. Contact intensity sensor **265** optionally includes one or more piezoresistive strain gauges, capacitive force sensors, electric force sensors, piezoelectric force sensors, optical force sensors, capacitive touch-sensitive surfaces, or other intensity sensors (e.g., sensors used to measure the force (or pressure) of a contact on a touch-sensitive surface). Contact intensity sensor **265** receives contact intensity information (e.g., pressure information or a proxy for pressure information) from the environment. In some embodiments, at least one contact intensity sensor is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system **212**). In some embodiments, at least one contact intensity sensor is located on the back of device **200**, opposite touch screen display **212**, which is located on the front of device **200**.

Device **200** also includes one or more proximity sensors **266**. FIG. 2A shows proximity sensor **266** coupled to peripherals interface **218**. Alternately, proximity sensor **266** is coupled to input controller **260** in I/O subsystem **206**. Proximity sensor **266** is performed as described in U.S. patent application Ser. No. 11/241,839, "Proximity Detector In Handheld Device"; Ser. No. 11/240,788, "Proximity Detector In Handheld Device"; Ser. No. 11/620,702, "Using Ambient Light Sensor To Augment Proximity Sensor Output"; Ser. No. 11/586,862, "Automated Response To And Sensing Of User Activity In Portable Devices"; and Ser. No. 11/638,251, "Methods And Systems For Automatic Configuration Of Peripherals," which are hereby incorporated by reference in their entirety. In some embodiments, the proximity sensor turns off and disables touch screen **212** when the multifunction device is placed near the user's ear (e.g., when the user is making a phone call).

Device **200** optionally also includes one or more tactile output generators **267**. FIG. 2A shows a tactile output generator coupled to haptic feedback controller **261** in I/O subsystem **206**. Tactile output generator **267** optionally includes one or more electroacoustic devices such as speakers or other audio components and/or electromechanical devices that convert energy into linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating component (e.g., a component that converts electrical signals into tactile outputs on the device). Contact intensity sensor **265** receives tactile feedback generation instructions from haptic feedback module **233** and generates tactile outputs on device **200** that are capable of being sensed by a user of device **200**. In some embodiments, at least one tactile output generator is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system **212**) and, optionally, generates a tactile output by moving

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the touch-sensitive surface vertically (e.g., in/out of a surface of device **200**) or laterally (e.g., back and forth in the same plane as a surface of device **200**). In some embodiments, at least one tactile output generator sensor is located on the back of device **200**, opposite touch screen display **212**, which is located on the front of device **200**.

Device **200** also includes one or more accelerometers **268**. FIG. 2A shows accelerometer **268** coupled to peripherals interface **218**. Alternately, accelerometer **268** is coupled to an input controller **260** in I/O subsystem **206**. Accelerometer **268** performs, for example, as described in U.S. Patent Publication No. 20050190059, "Acceleration-based Theft Detection System for Portable Electronic Devices," and U.S. Patent Publication No. 20060017692, "Methods And Apparatuses For Operating A Portable Device Based On An Accelerometer," both of which are incorporated by reference herein in their entirety. In some embodiments, information is displayed on the touch screen display in a portrait view or a landscape view based on an analysis of data received from the one or more accelerometers. Device **200** optionally includes, in addition to accelerometer(s) **268**, a magnetometer (not shown) and a GPS (or GLONASS or other global navigation system) receiver (not shown) for obtaining information concerning the location and orientation (e.g., portrait or landscape) of device **200**.

In some embodiments, the software components stored in memory **202** include operating system **226**, communication module (or set of instructions) **228**, contact/motion module (or set of instructions) **230**, graphics module (or set of instructions) **232**, text input module (or set of instructions) **234**, Global Positioning System (GPS) module (or set of instructions) **235**, Digital Assistant Client Module **229**, and applications (or sets of instructions) **236**. Further, memory **202** stores data and models, such as user data and models **231**. Furthermore, in some embodiments, memory **202** (FIG. 2A) or **470** (FIG. 4) stores device/global internal state **257**, as shown in FIGS. 2A and 4. Device/global internal state **257** includes one or more of: active application state, indicating which applications, if any, are currently active; display state, indicating what applications, views or other information occupy various regions of touch screen display **212**; sensor state, including information obtained from the device's various sensors and input control devices **216**; and location information concerning the device's location and/or attitude.

Operating system **226** (e.g., Darwin, RTXC, LINUX, UNIX, OS X, IOS, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communication between various hardware and software components.

Communication module **228** facilitates communication with other devices over one or more external ports **224** and also includes various software components for handling data received by RF circuitry **208** and/or external port **224**. External port **224** (e.g., Universal Serial Bus (USB), FIRE-WIRE, etc.) is adapted for coupling directly to other devices or indirectly over a network (e.g., the Internet, wireless LAN, etc.). In some embodiments, the external port is a multi-pin (e.g., 30-pin) connector that is the same as, or similar to and/or compatible with, the 30-pin connector used on iPod® (trademark of Apple Inc.) devices.

Contact/motion module **230** optionally detects contact with touch screen **212** (in conjunction with display controller **256**) and other touch-sensitive devices (e.g., a touchpad or physical click wheel). Contact/motion module **230** includes various software components for performing vari-

ous operations related to detection of contact, such as determining if contact has occurred (e.g., detecting a finger-down event), determining an intensity of the contact (e.g., the force or pressure of the contact or a substitute for the force or pressure of the contact), determining if there is movement of the contact and tracking the movement across the touch-sensitive surface (e.g., detecting one or more finger-dragging events), and determining if the contact has ceased (e.g., detecting a finger-up event or a break in contact). Contact/motion module **230** receives contact data from the touch-sensitive surface. Determining movement of the point of contact, which is represented by a series of contact data, optionally includes determining speed (magnitude), velocity (magnitude and direction), and/or an acceleration (a change in magnitude and/or direction) of the point of contact. These operations are, optionally, applied to single contacts (e.g., one finger contacts) or to multiple simultaneous contacts (e.g., “multitouch”/multiple finger contacts). In some embodiments, contact/motion module **230** and display controller **256** detect contact on a touchpad.

In some embodiments, contact/motion module **230** uses a set of one or more intensity thresholds to determine whether an operation has been performed by a user (e.g., to determine whether a user has “clicked” on an icon). In some embodiments, at least a subset of the intensity thresholds are determined in accordance with software parameters (e.g., the intensity thresholds are not determined by the activation thresholds of particular physical actuators and can be adjusted without changing the physical hardware of device **200**). For example, a mouse “click” threshold of a trackpad or touch screen display can be set to any of a large range of predefined threshold values without changing the trackpad or touch screen display hardware. Additionally, in some implementations, a user of the device is provided with software settings for adjusting one or more of the set of intensity thresholds (e.g., by adjusting individual intensity thresholds and/or by adjusting a plurality of intensity thresholds at once with a system-level click “intensity” parameter).

Contact/motion module **230** optionally detects a gesture input by a user. Different gestures on the touch-sensitive surface have different contact patterns (e.g., different motions, timings, and/or intensities of detected contacts). Thus, a gesture is, optionally, detected by detecting a particular contact pattern. For example, detecting a finger tap gesture includes detecting a finger-down event followed by detecting a finger-up (liftoff) event at the same position (or substantially the same position) as the finger-down event (e.g., at the position of an icon). As another example, detecting a finger swipe gesture on the touch-sensitive surface includes detecting a finger-down event followed by detecting one or more finger-dragging events, and subsequently followed by detecting a finger-up (liftoff) event.

Graphics module **232** includes various known software components for rendering and displaying graphics on touch screen **212** or other display, including components for changing the visual impact (e.g., brightness, transparency, saturation, contrast, or other visual property) of graphics that are displayed. As used herein, the term “graphics” includes any object that can be displayed to a user, including, without limitation, text, web pages, icons (such as user-interface objects including soft keys), digital images, videos, animations, and the like.

In some embodiments, graphics module **232** stores data representing graphics to be used. Each graphic is, optionally, assigned a corresponding code. Graphics module **232** receives, from applications etc., one or more codes specifying graphics to be displayed along with, if necessary,

coordinate data and other graphic property data, and then generates screen image data to output to display controller **256**.

Haptic feedback module **233** includes various software components for generating instructions used by tactile output generator(s) **267** to produce tactile outputs at one or more locations on device **200** in response to user interactions with device **200**.

Text input module **234**, which is, in some examples, a component of graphics module **232**, provides soft keyboards for entering text in various applications (e.g., contacts **237**, e-mail client module **240**, IM **241**, browser **247**, and any other application that needs text input).

GPS module **235** determines the location of the device and provides this information for use in various applications (e.g., to telephone module **238** for use in location-based dialing; to camera module **243** as picture/video metadata; and to applications that provide location-based services such as weather widgets, local yellow page widgets, and map/navigation widgets).

Digital assistant client module **229** includes various client-side digital assistant instructions to provide the client-side functionalities of the digital assistant. For example, digital assistant client module **229** is capable of accepting voice input (e.g., speech input), text input, touch input, and/or gestural input through various user interfaces (e.g., microphone **213**, accelerometer(s) **268**, touch-sensitive display system **212**, optical sensor(s) **264**, other input control devices **216**, etc.) of portable multifunction device **200**. Digital assistant client module **229** is also capable of providing output in audio (e.g., speech output), visual, and/or tactile forms through various output interfaces (e.g., speaker **211**, touch-sensitive display system **212**, tactile output generator(s) **267**, etc.) of portable multifunction device **200**. For example, output is provided as voice, sound, alerts, text messages, menus, graphics, videos, animations, vibrations, and/or combinations of two or more of the above. During operation, digital assistant client module **229** communicates with DA server **106** using RF circuitry **208**.

User data and models **231** include various data associated with the user (e.g., user-specific vocabulary data, user preference data, user-specified name pronunciations, data from the user’s electronic address book, to-do lists, shopping lists, etc.) to provide the client-side functionalities of the digital assistant. Further, user data and models **231** include various models (e.g., speech recognition models, statistical language models, natural language processing models, ontology, task flow models, service models, etc.) for processing user input and determining user intent.

In some examples, digital assistant client module **229** utilizes the various sensors, subsystems, and peripheral devices of portable multifunction device **200** to gather additional information from the surrounding environment of the portable multifunction device **200** to establish a context associated with a user, the current user interaction, and/or the current user input. In some examples, digital assistant client module **229** provides the contextual information or a subset thereof with the user input to DA server **106** to help infer the user’s intent. In some examples, the digital assistant also uses the contextual information to determine how to prepare and deliver outputs to the user. Contextual information is referred to as context data.

In some examples, the contextual information that accompanies the user input includes sensor information, e.g., lighting, ambient noise, ambient temperature, images or videos of the surrounding environment, etc. In some examples, the contextual information can also include the

physical state of the device, e.g., device orientation, device location, device temperature, power level, speed, acceleration, motion patterns, cellular signals strength, etc. In some examples, information related to the software state of DA server 106, e.g., running processes, installed programs, past and present network activities, background services, error logs, resources usage, etc., and of portable multifunction device 200 is provided to DA server 106 as contextual information associated with a user input.

In some examples, the digital assistant client module 229 selectively provides information (e.g., user data 231) stored on the portable multifunction device 200 in response to requests from DA server 106. In some examples, digital assistant client module 229 also elicits additional input from the user via a natural language dialogue or other user interfaces upon request by DA server 106. Digital assistant client module 229 passes the additional input to DA server 106 to help DA server 106 in intent deduction and/or fulfillment of the user's intent expressed in the user request.

A more detailed description of a digital assistant is described below with reference to FIGS. 7A-7C. It should be recognized that digital assistant client module 229 can include any number of the sub-modules of digital assistant module 726 described below.

Applications 236 include the following modules (or sets of instructions), or a subset or superset thereof:

Contacts module 237 (sometimes called an address book or contact list);

Telephone module 238;

Video conference module 239;

E-mail client module 240;

Instant messaging (IM) module 241;

Workout support module 242;

Camera module 243 for still and/or video images;

Image management module 244;

Video player module;

Music player module;

Browser module 247;

Calendar module 248;

Widget modules 249, which includes, in some examples, one or more of: weather widget 249-1, stocks widget 249-2, calculator widget 249-3, alarm clock widget 249-4, dictionary widget 249-5, and other widgets obtained by the user, as well as user-created widgets 249-6;

Widget creator module 250 for making user-created widgets 249-6;

Search module 251;

Video and music player module 252, which merges video player module and music player module;

Notes module 253;

Map module 254; and/or

Online video module 255.

Examples of other applications 236 that are stored in memory 202 include other word processing applications, other image editing applications, drawing applications, presentation applications, JAVA-enabled applications, encryption, digital rights management, voice recognition, and voice replication.

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, contacts module 237 are used to manage an address book or contact list (e.g., stored in application internal state 292 of contacts module 237 in memory 202 or memory 470), including: adding name(s) to the address book; deleting name(s) from the address book; associating telephone number(s), e-mail address(es), physi-

cal address(es) or other information with a name; associating an image with a name; categorizing and sorting names; providing telephone numbers or e-mail addresses to initiate and/or facilitate communications by telephone 238, video conference module 239, e-mail 240, or IM 241; and so forth.

In conjunction with RF circuitry 208, audio circuitry 210, speaker 211, microphone 213, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, telephone module 238 are used to enter a sequence of characters corresponding to a telephone number, access one or more telephone numbers in contacts module 237, modify a telephone number that has been entered, dial a respective telephone number, conduct a conversation, and disconnect or hang up when the conversation is completed. As noted above, the wireless communication uses any of a plurality of communications standards, protocols, and technologies.

In conjunction with RF circuitry 208, audio circuitry 210, speaker 211, microphone 213, touch screen 212, display controller 256, optical sensor 264, optical sensor controller 258, contact/motion module 230, graphics module 232, text input module 234, contacts module 237, and telephone module 238, video conference module 239 includes executable instructions to initiate, conduct, and terminate a video conference between a user and one or more other participants in accordance with user instructions.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, e-mail client module 240 includes executable instructions to create, send, receive, and manage e-mail in response to user instructions. In conjunction with image management module 244, e-mail client module 240 makes it very easy to create and send e-mails with still or video images taken with camera module 243.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, the instant messaging module 241 includes executable instructions to enter a sequence of characters corresponding to an instant message, to modify previously entered characters, to transmit a respective instant message (for example, using a Short Message Service (SMS) or Multimedia Message Service (MMS) protocol for telephony-based instant messages or using XMPP, SIMPLE, or IMPS for Internet-based instant messages), to receive instant messages, and to view received instant messages. In some embodiments, transmitted and/or received instant messages include graphics, photos, audio files, video files and/or other attachments as are supported in an MMS and/or an Enhanced Messaging Service (EMS). As used herein, "instant messaging" refers to both telephony-based messages (e.g., messages sent using SMS or MMS) and Internet-based messages (e.g., messages sent using XMPP, SIMPLE, or IMPS).

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, GPS module 235, map module 254, and music player module, workout support module 242 includes executable instructions to create workouts (e.g., with time, distance, and/or calorie burning goals); communicate with workout sensors (sports devices); receive workout sensor data; calibrate sensors used to monitor a workout; select and play music for a workout; and display, store, and transmit workout data.

In conjunction with touch screen 212, display controller 256, optical sensor(s) 264, optical sensor controller 258, contact/motion module 230, graphics module 232, and

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image management module **244**, camera module **243** includes executable instructions to capture still images or video (including a video stream) and store them into memory **202**, modify characteristics of a still image or video, or delete a still image or video from memory **202**.

In conjunction with touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, text input module **234**, and camera module **243**, image management module **244** includes executable instructions to arrange, modify (e.g., edit), or otherwise manipulate, label, delete, present (e.g., in a digital slide show or album), and store still and/or video images.

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, and text input module **234**, browser module **247** includes executable instructions to browse the Internet in accordance with user instructions, including searching, linking to, receiving, and displaying web pages or portions thereof, as well as attachments and other files linked to web pages.

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, text input module **234**, e-mail client module **240**, and browser module **247**, calendar module **248** includes executable instructions to create, display, modify, and store calendars and data associated with calendars (e.g., calendar entries, to-do lists, etc.) in accordance with user instructions.

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, text input module **234**, and browser module **247**, widget modules **249** are mini-applications that can be downloaded and used by a user (e.g., weather widget **249-1**, stocks widget **249-2**, calculator widget **249-3**, alarm clock widget **249-4**, and dictionary widget **249-5**) or created by the user (e.g., user-created widget **249-6**). In some embodiments, a widget includes an HTML (Hypertext Markup Language) file, a CSS (Cascading Style Sheets) file, and a JavaScript file. In some embodiments, a widget includes an XML (Extensible Markup Language) file and a JavaScript file (e.g., Yahoo! Widgets).

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, text input module **234**, and browser module **247**, the widget creator module **250** are used by a user to create widgets (e.g., turning a user-specified portion of a web page into a widget).

In conjunction with touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, and text input module **234**, search module **251** includes executable instructions to search for text, music, sound, image, video, and/or other files in memory **202** that match one or more search criteria (e.g., one or more user-specified search terms) in accordance with user instructions.

In conjunction with touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, audio circuitry **210**, speaker **211**, RF circuitry **208**, and browser module **247**, video and music player module **252** includes executable instructions that allow the user to download and play back recorded music and other sound files stored in one or more file formats, such as MP3 or AAC files, and executable instructions to display, present, or otherwise play back videos (e.g., on touch screen **212** or on an external, connected display via external port **224**). In some embodiments, device **200** optionally includes the functionality of an MP3 player, such as an iPod (trademark of Apple Inc.).

In conjunction with touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, and

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text input module **234**, notes module **253** includes executable instructions to create and manage notes, to-do lists, and the like in accordance with user instructions.

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, text input module **234**, GPS module **235**, and browser module **247**, map module **254** are used to receive, display, modify, and store maps and data associated with maps (e.g., driving directions, data on stores and other points of interest at or near a particular location, and other location-based data) in accordance with user instructions.

In conjunction with touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, audio circuitry **210**, speaker **211**, RF circuitry **208**, text input module **234**, e-mail client module **240**, and browser module **247**, online video module **255** includes instructions that allow the user to access, browse, receive (e.g., by streaming and/or download), play back (e.g., on the touch screen or on an external, connected display via external port **224**), send an e-mail with a link to a particular online video, and otherwise manage online videos in one or more file formats, such as H.264. In some embodiments, instant messaging module **241**, rather than e-mail client module **240**, is used to send a link to a particular online video. Additional description of the online video application can be found in U.S. Provisional Patent Application No. 60/936,562, "Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos," filed Jun. 20, 2007, and U.S. patent application Ser. No. 11/968,067, "Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos," filed Dec. 31, 2007, the contents of which are hereby incorporated by reference in their entirety.

Each of the above-identified modules and applications corresponds to a set of executable instructions for performing one or more functions described above and the methods described in this application (e.g., the computer-implemented methods and other information processing methods described herein). These modules (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules can be combined or otherwise rearranged in various embodiments. For example, video player module can be combined with music player module into a single module (e.g., video and music player module **252**, FIG. 2A). In some embodiments, memory **202** stores a subset of the modules and data structures identified above. Furthermore, memory **202** stores additional modules and data structures not described above.

In some embodiments, device **200** is a device where operation of a predefined set of functions on the device is performed exclusively through a touch screen and/or a touchpad. By using a touch screen and/or a touchpad as the primary input control device for operation of device **200**, the number of physical input control devices (such as push buttons, dials, and the like) on device **200** is reduced.

The predefined set of functions that are performed exclusively through a touch screen and/or a touchpad optionally include navigation between user interfaces. In some embodiments, the touchpad, when touched by the user, navigates device **200** to a main, home, or root menu from any user interface that is displayed on device **200**. In such embodiments, a "menu button" is implemented using a touchpad. In some other embodiments, the menu button is a physical push button or other physical input control device instead of a touchpad.

FIG. 2B is a block diagram illustrating exemplary components for event handling in accordance with some

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embodiments. In some embodiments, memory 202 (FIG. 2A) or 470 (FIG. 4) includes event sorter 270 (e.g., in operating system 226) and a respective application 236-1 (e.g., any of the aforementioned applications 237-251, 255, 480-490).

Event sorter 270 receives event information and determines the application 236-1 and application view 291 of application 236-1 to which to deliver the event information. Event sorter 270 includes event monitor 271 and event dispatcher module 274. In some embodiments, application 236-1 includes application internal state 292, which indicates the current application view(s) displayed on touch-sensitive display 212 when the application is active or executing. In some embodiments, device/global internal state 257 is used by event sorter 270 to determine which application(s) is (are) currently active, and application internal state 292 is used by event sorter 270 to determine application views 291 to which to deliver event information.

In some embodiments, application internal state 292 includes additional information, such as one or more of: resume information to be used when application 236-1 resumes execution, user interface state information that indicates information being displayed or that is ready for display by application 236-1, a state queue for enabling the user to go back to a prior state or view of application 236-1, and a redo/undo queue of previous actions taken by the user.

Event monitor 271 receives event information from peripherals interface 218. Event information includes information about a sub-event (e.g., a user touch on touch-sensitive display 212, as part of a multi-touch gesture). Peripherals interface 218 transmits information it receives from I/O subsystem 206 or a sensor, such as proximity sensor 266, accelerometer(s) 268, and/or microphone 213 (through audio circuitry 210). Information that peripherals interface 218 receives from I/O subsystem 206 includes information from touch-sensitive display 212 or a touch-sensitive surface.

In some embodiments, event monitor 271 sends requests to the peripherals interface 218 at predetermined intervals. In response, peripherals interface 218 transmits event information. In other embodiments, peripherals interface 218 transmits event information only when there is a significant event (e.g., receiving an input above a predetermined noise threshold and/or for more than a predetermined duration).

In some embodiments, event sorter 270 also includes a hit view determination module 272 and/or an active event recognizer determination module 273.

Hit view determination module 272 provides software procedures for determining where a sub-event has taken place within one or more views when touch-sensitive display 212 displays more than one view. Views are made up of controls and other elements that a user can see on the display.

Another aspect of the user interface associated with an application is a set of views, sometimes herein called application views or user interface windows, in which information is displayed and touch-based gestures occur. The application views (of a respective application) in which a touch is detected correspond to programmatic levels within a programmatic or view hierarchy of the application. For example, the lowest level view in which a touch is detected is called the hit view, and the set of events that are recognized as proper inputs is determined based, at least in part, on the hit view of the initial touch that begins a touch-based gesture.

Hit view determination module 272 receives information related to sub events of a touch-based gesture. When an

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application has multiple views organized in a hierarchy, hit view determination module 272 identifies a hit view as the lowest view in the hierarchy which should handle the sub-event. In most circumstances, the hit view is the lowest level view in which an initiating sub-event occurs (e.g., the first sub-event in the sequence of sub-events that form an event or potential event). Once the hit view is identified by the hit view determination module 272, the hit view typically receives all sub-events related to the same touch or input source for which it was identified as the hit view.

Active event recognizer determination module 273 determines which view or views within a view hierarchy should receive a particular sequence of sub-events. In some embodiments, active event recognizer determination module 273 determines that only the hit view should receive a particular sequence of sub-events. In other embodiments, active event recognizer determination module 273 determines that all views that include the physical location of a sub-event are actively involved views, and therefore determines that all actively involved views should receive a particular sequence of sub-events. In other embodiments, even if touch sub-events were entirely confined to the area associated with one particular view, views higher in the hierarchy would still remain as actively involved views.

Event dispatcher module 274 dispatches the event information to an event recognizer (e.g., event recognizer 280). In embodiments including active event recognizer determination module 273, event dispatcher module 274 delivers the event information to an event recognizer determined by active event recognizer determination module 273. In some embodiments, event dispatcher module 274 stores in an event queue the event information, which is retrieved by a respective event receiver 282.

In some embodiments, operating system 226 includes event sorter 270. Alternatively, application 236-1 includes event sorter 270. In yet other embodiments, event sorter 270 is a stand-alone module, or a part of another module stored in memory 202, such as contact/motion module 230.

In some embodiments, application 236-1 includes a plurality of event handlers 290 and one or more application views 291, each of which includes instructions for handling touch events that occur within a respective view of the application's user interface. Each application view 291 of the application 236-1 includes one or more event recognizers 280. Typically, a respective application view 291 includes a plurality of event recognizers 280. In other embodiments, one or more of event recognizers 280 are part of a separate module, such as a user interface kit (not shown) or a higher level object from which application 236-1 inherits methods and other properties. In some embodiments, a respective event handler 290 includes one or more of: data updater 276, object updater 277, GUI updater 278, and/or event data 279 received from event sorter 270. Event handler 290 utilizes or calls data updater 276, object updater 277, or GUI updater 278 to update the application internal state 292. Alternatively, one or more of the application views 291 include one or more respective event handlers 290. Also, in some embodiments, one or more of data updater 276, object updater 277, and GUI updater 278 are included in a respective application view 291.

A respective event recognizer 280 receives event information (e.g., event data 279) from event sorter 270 and identifies an event from the event information. Event recognizer 280 includes event receiver 282 and event comparator 284. In some embodiments, event recognizer 280 also

includes at least a subset of: metadata **283**, and event delivery instructions **288** (which include sub-event delivery instructions).

Event receiver **282** receives event information from event sorter **270**. The event information includes information about a sub-event, for example, a touch or a touch movement. Depending on the sub-event, the event information also includes additional information, such as location of the sub-event. When the sub-event concerns motion of a touch, the event information also includes speed and direction of the sub-event. In some embodiments, events include rotation of the device from one orientation to another (e.g., from a portrait orientation to a landscape orientation, or vice versa), and the event information includes corresponding information about the current orientation (also called device attitude) of the device.

Event comparator **284** compares the event information to predefined event or sub-event definitions and, based on the comparison, determines an event or sub event, or determines or updates the state of an event or sub-event. In some embodiments, event comparator **284** includes event definitions **286**. Event definitions **286** contain definitions of events (e.g., predefined sequences of sub-events), for example, event **1** (**287-1**), event **2** (**287-2**), and others. In some embodiments, sub-events in an event (**287**) include, for example, touch begin, touch end, touch movement, touch cancellation, and multiple touching. In one example, the definition for event **1** (**287-1**) is a double tap on a displayed object. The double tap, for example, comprises a first touch (touch begin) on the displayed object for a predetermined phase, a first liftoff (touch end) for a predetermined phase, a second touch (touch begin) on the displayed object for a predetermined phase, and a second liftoff (touch end) for a predetermined phase. In another example, the definition for event **2** (**287-2**) is a dragging on a displayed object. The dragging, for example, comprises a touch (or contact) on the displayed object for a predetermined phase, a movement of the touch across touch-sensitive display **212**, and liftoff of the touch (touch end). In some embodiments, the event also includes information for one or more associated event handlers **290**.

In some embodiments, event definition **287** includes a definition of an event for a respective user-interface object. In some embodiments, event comparator **284** performs a hit test to determine which user-interface object is associated with a sub-event. For example, in an application view in which three user-interface objects are displayed on touch-sensitive display **212**, when a touch is detected on touch-sensitive display **212**, event comparator **284** performs a hit test to determine which of the three user-interface objects is associated with the touch (sub-event). If each displayed object is associated with a respective event handler **290**, the event comparator uses the result of the hit test to determine which event handler **290** should be activated. For example, event comparator **284** selects an event handler associated with the sub-event and the object triggering the hit test.

In some embodiments, the definition for a respective event (**287**) also includes delayed actions that delay delivery of the event information until after it has been determined whether the sequence of sub-events does or does not correspond to the event recognizer's event type.

When a respective event recognizer **280** determines that the series of sub-events do not match any of the events in event definitions **286**, the respective event recognizer **280** enters an event impossible, event failed, or event ended state, after which it disregards subsequent sub-events of the touch-based gesture. In this situation, other event recogniz-

ers, if any, that remain active for the hit view continue to track and process sub-events of an ongoing touch-based gesture.

In some embodiments, a respective event recognizer **280** includes metadata **283** with configurable properties, flags, and/or lists that indicate how the event delivery system should perform sub-event delivery to actively involved event recognizers. In some embodiments, metadata **283** includes configurable properties, flags, and/or lists that indicate how event recognizers interact, or are enabled to interact, with one another. In some embodiments, metadata **283** includes configurable properties, flags, and/or lists that indicate whether sub-events are delivered to varying levels in the view or programmatic hierarchy.

In some embodiments, a respective event recognizer **280** activates event handler **290** associated with an event when one or more particular sub-events of an event are recognized. In some embodiments, a respective event recognizer **280** delivers event information associated with the event to event handler **290**. Activating an event handler **290** is distinct from sending (and deferred sending) sub-events to a respective hit view. In some embodiments, event recognizer **280** throws a flag associated with the recognized event, and event handler **290** associated with the flag catches the flag and performs a predefined process.

In some embodiments, event delivery instructions **288** include sub-event delivery instructions that deliver event information about a sub-event without activating an event handler. Instead, the sub-event delivery instructions deliver event information to event handlers associated with the series of sub-events or to actively involved views. Event handlers associated with the series of sub-events or with actively involved views receive the event information and perform a predetermined process.

In some embodiments, data updater **276** creates and updates data used in application **236-1**. For example, data updater **276** updates the telephone number used in contacts module **237**, or stores a video file used in video player module. In some embodiments, object updater **277** creates and updates objects used in application **236-1**. For example, object updater **277** creates a new user-interface object or updates the position of a user-interface object. GUI updater **278** updates the GUI. For example, GUI updater **278** prepares display information and sends it to graphics module **232** for display on a touch-sensitive display.

In some embodiments, event handler(s) **290** includes or has access to data updater **276**, object updater **277**, and GUI updater **278**. In some embodiments, data updater **276**, object updater **277**, and GUI updater **278** are included in a single module of a respective application **236-1** or application view **291**. In other embodiments, they are included in two or more software modules.

It shall be understood that the foregoing discussion regarding event handling of user touches on touch-sensitive displays also applies to other forms of user inputs to operate multifunction devices **200** with input devices, not all of which are initiated on touch screens. For example, mouse movement and mouse button presses, optionally coordinated with single or multiple keyboard presses or holds; contact movements such as taps, drags, scrolls, etc. on touchpads; pen stylus inputs; movement of the device; oral instructions; detected eye movements; biometric inputs; and/or any combination thereof are optionally utilized as inputs corresponding to sub-events which define an event to be recognized.

FIG. 3 illustrates a portable multifunction device **200** having a touch screen **212** in accordance with some embodiments. The touch screen optionally displays one or more

graphics within user interface (UI) **300**. In this embodiment, as well as others described below, a user is enabled to select one or more of the graphics by making a gesture on the graphics, for example, with one or more fingers **302** (not drawn to scale in the figure) or one or more styluses **303** (not drawn to scale in the figure). In some embodiments, selection of one or more graphics occurs when the user breaks contact with the one or more graphics. In some embodiments, the gesture optionally includes one or more taps, one or more swipes (from left to right, right to left, upward and/or downward), and/or a rolling of a finger (from right to left, left to right, upward and/or downward) that has made contact with device **200**. In some implementations or circumstances, inadvertent contact with a graphic does not select the graphic. For example, a swipe gesture that sweeps over an application icon optionally does not select the corresponding application when the gesture corresponding to selection is a tap.

Device **200** also includes one or more physical buttons, such as “home” or menu button **304**. As described previously, menu button **304** is used to navigate to any application **236** in a set of applications that is executed on device **200**. Alternatively, in some embodiments, the menu button is implemented as a soft key in a GUI displayed on touch screen **212**.

In one embodiment, device **200** includes touch screen **212**, menu button **304**, push button **306** for powering the device on/off and locking the device, volume adjustment button(s) **308**, subscriber identity module (SIM) card slot **310**, headset jack **312**, and docking/charging external port **224**. Push button **306** is, optionally, used to turn the power on/off on the device by depressing the button and holding the button in the depressed state for a predefined time interval; to lock the device by depressing the button and releasing the button before the predefined time interval has elapsed; and/or to unlock the device or initiate an unlock process. In an alternative embodiment, device **200** also accepts verbal input for activation or deactivation of some functions through microphone **213**. Device **200** also, optionally, includes one or more contact intensity sensors **265** for detecting intensity of contacts on touch screen **212** and/or one or more tactile output generators **267** for generating tactile outputs for a user of device **200**.

FIG. 4 is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface in accordance with some embodiments. Device **400** need not be portable. In some embodiments, device **400** is a laptop computer, a desktop computer, a tablet computer, a multimedia player device, a navigation device, an educational device (such as a child’s learning toy), a gaming system, or a control device (e.g., a home or industrial controller). Device **400** typically includes one or more processing units (CPUs) **410**, one or more network or other communications interfaces **460**, memory **470**, and one or more communication buses **420** for interconnecting these components. Communication buses **420** optionally include circuitry (sometimes called a chipset) that interconnects and controls communications between system components. Device **400** includes input/output (I/O) interface **430** comprising display **440**, which is typically a touch screen display. I/O interface **430** also optionally includes a keyboard and/or mouse (or other pointing device) **450** and touchpad **455**, tactile output generator **457** for generating tactile outputs on device **400** (e.g., similar to tactile output generator(s) **267** described above with reference to FIG. 2A), sensors **459** (e.g., optical, acceleration, proximity, touch-sensitive, and/or contact intensity sensors similar to contact intensity sensor(s) **265**

described above with reference to FIG. 2A). Memory **470** includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices; and optionally includes non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. Memory **470** optionally includes one or more storage devices remotely located from CPU(s) **410**. In some embodiments, memory **470** stores programs, modules, and data structures analogous to the programs, modules, and data structures stored in memory **202** of portable multifunction device **200** (FIG. 2A), or a subset thereof. Furthermore, memory **470** optionally stores additional programs, modules, and data structures not present in memory **202** of portable multifunction device **200**. For example, memory **470** of device **400** optionally stores drawing module **480**, presentation module **482**, word processing module **484**, website creation module **486**, disk authoring module **488**, and/or spreadsheet module **490**, while memory **202** of portable multifunction device **200** (FIG. 2A) optionally does not store these modules.

Each of the above-identified elements in FIG. 4 is, in some examples, stored in one or more of the previously mentioned memory devices. Each of the above-identified modules corresponds to a set of instructions for performing a function described above. The above-identified modules or programs (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules are combined or otherwise rearranged in various embodiments. In some embodiments, memory **470** stores a subset of the modules and data structures identified above. Furthermore, memory **470** stores additional modules and data structures not described above.

Attention is now directed towards embodiments of user interfaces that can be implemented on, for example, portable multifunction device **200**.

FIG. 5A illustrates an exemplary user interface for a menu of applications on portable multifunction device **200** in accordance with some embodiments. Similar user interfaces are implemented on device **400**. In some embodiments, user interface **500** includes the following elements, or a subset or superset thereof:

Signal strength indicator(s) **502** for wireless communication(s), such as cellular and Wi-Fi signals;

Time **504**;

Bluetooth indicator **505**;

Battery status indicator **506**;

Tray **508** with icons for frequently used applications, such as:

Icon **516** for telephone module **238**, labeled “Phone,” which optionally includes an indicator **514** of the number of missed calls or voicemail messages;

Icon **518** for e-mail client module **240**, labeled “Mail,” which optionally includes an indicator **510** of the number of unread e-mails;

Icon **520** for browser module **247**, labeled “Browser;” and

Icon **522** for video and music player module **252**, also referred to as iPod (trademark of Apple Inc.) module **252**, labeled “iPod;” and

Icons for other applications, such as:

Icon **524** for IM module **241**, labeled “Messages;”

Icon **526** for calendar module **248**, labeled “Calendar;”

Icon **528** for image management module **244**, labeled “Photos;”

Icon **530** for camera module **243**, labeled “Camera;”

Icon **532** for online video module **255**, labeled “Online Video;”

Icon **534** for stocks widget **249-2**, labeled “Stocks;”

Icon **536** for map module **254**, labeled “Maps;”

Icon **538** for weather widget **249-1**, labeled “Weather;”

Icon **540** for alarm clock widget **249-4**, labeled “Clock;”

Icon **542** for workout support module **242**, labeled “Workout Support;”

Icon **544** for notes module **253**, labeled “Notes;” and

Icon **546** for a settings application or module, labeled “Settings;” which provides access to settings for device **200** and its various applications **236**.

It should be noted that the icon labels illustrated in FIG. **5A** are merely exemplary. For example, icon **522** for video and music player module **252** is optionally labeled “Music” or “Music Player.” Other labels are, optionally, used for various application icons. In some embodiments, a label for a respective application icon includes a name of an application corresponding to the respective application icon. In some embodiments, a label for a particular application icon is distinct from a name of an application corresponding to the particular application icon.

FIG. **5B** illustrates an exemplary user interface on a device (e.g., device **400**, FIG. **4**) with a touch-sensitive surface **551** (e.g., a tablet or touchpad **455**, FIG. **4**) that is separate from the display **550** (e.g., touch screen display **212**). Device **400** also, optionally, includes one or more contact intensity sensors (e.g., one or more of sensors **457**) for detecting intensity of contacts on touch-sensitive surface **551** and/or one or more tactile output generators **459** for generating tactile outputs for a user of device **400**.

Although some of the examples which follow will be given with reference to inputs on touch screen display **212** (where the touch-sensitive surface and the display are combined), in some embodiments, the device detects inputs on a touch-sensitive surface that is separate from the display, as shown in FIG. **5B**. In some embodiments, the touch-sensitive surface (e.g., **551** in FIG. **5B**) has a primary axis (e.g., **552** in FIG. **5B**) that corresponds to a primary axis (e.g., **553** in FIG. **5B**) on the display (e.g., **550**). In accordance with these embodiments, the device detects contacts (e.g., **560** and **562** in FIG. **5B**) with the touch-sensitive surface **551** at locations that correspond to respective locations on the display (e.g., in FIG. **5B**, **560** corresponds to **568** and **562** corresponds to **570**). In this way, user inputs (e.g., contacts **560** and **562**, and movements thereof) detected by the device on the touch-sensitive surface (e.g., **551** in FIG. **5B**) are used by the device to manipulate the user interface on the display (e.g., **550** in FIG. **5B**) of the multifunction device when the touch-sensitive surface is separate from the display. It should be understood that similar methods are, optionally, used for other user interfaces described herein.

Additionally, while the following examples are given primarily with reference to finger inputs (e.g., finger contacts, finger tap gestures, finger swipe gestures), it should be understood that, in some embodiments, one or more of the finger inputs are replaced with input from another input device (e.g., a mouse-based input or stylus input). For example, a swipe gesture is, optionally, replaced with a mouse click (e.g., instead of a contact) followed by movement of the cursor along the path of the swipe (e.g., instead of movement of the contact). As another example, a tap gesture is, optionally, replaced with a mouse click while the cursor is located over the location of the tap gesture (e.g., instead of detection of the contact followed by ceasing to detect the contact). Similarly, when multiple user inputs are

simultaneously detected, it should be understood that multiple computer mice are, optionally, used simultaneously, or a mouse and finger contacts are, optionally, used simultaneously.

FIG. **6A** illustrates exemplary personal electronic device **600**. Device **600** includes body **602**. In some embodiments, device **600** includes some or all of the features described with respect to devices **200** and **400** (e.g., FIGS. **2A-4**). In some embodiments, device **600** has touch-sensitive display screen **604**, hereafter touch screen **604**. Alternatively, or in addition to touch screen **604**, device **600** has a display and a touch-sensitive surface. As with devices **200** and **400**, in some embodiments, touch screen **604** (or the touch-sensitive surface) has one or more intensity sensors for detecting intensity of contacts (e.g., touches) being applied. The one or more intensity sensors of touch screen **604** (or the touch-sensitive surface) provide output data that represents the intensity of touches. The user interface of device **600** responds to touches based on their intensity, meaning that touches of different intensities can invoke different user interface operations on device **600**.

Techniques for detecting and processing touch intensity are found, for example, in related applications: International Patent Application Serial No. PCT/US2013/040061, titled “Device, Method, and Graphical User Interface for Displaying User Interface Objects Corresponding to an Application,” filed May 8, 2013, and International Patent Application Serial No. PCT/US2013/069483, titled “Device, Method, and Graphical User Interface for Transitioning Between Touch Input to Display Output Relationships,” filed Nov. 11, 2013, each of which is hereby incorporated by reference in their entirety.

In some embodiments, device **600** has one or more input mechanisms **606** and **608**. Input mechanisms **606** and **608**, if included, are physical. Examples of physical input mechanisms include push buttons and rotatable mechanisms. In some embodiments, device **600** has one or more attachment mechanisms. Such attachment mechanisms, if included, can permit attachment of device **600** with, for example, hats, eyewear, earrings, necklaces, shirts, jackets, bracelets, watch straps, chains, trousers, belts, shoes, purses, backpacks, and so forth. These attachment mechanisms permit device **600** to be worn by a user.

FIG. **6B** depicts exemplary personal electronic device **600**. In some embodiments, device **600** includes some or all of the components described with respect to FIGS. **2A**, **2B**, and **4**. Device **600** has bus **612** that operatively couples I/O section **614** with one or more computer processors **616** and memory **618**. I/O section **614** is connected to display **604**, which can have touch-sensitive component **622** and, optionally, touch-intensity sensitive component **624**. In addition, I/O section **614** is connected with communication unit **630** for receiving application and operating system data, using Wi-Fi, Bluetooth, near field communication (NFC), cellular, and/or other wireless communication techniques. Device **600** includes input mechanisms **606** and/or **608**. Input mechanism **606** is a rotatable input device or a depressible and rotatable input device, for example. Input mechanism **608** is a button, in some examples.

Input mechanism **608** is a microphone, in some examples. Personal electronic device **600** includes, for example, various sensors, such as GPS sensor **632**, accelerometer **634**, directional sensor **640** (e.g., compass), gyroscope **636**, motion sensor **638**, and/or a combination thereof, all of which are operatively connected to I/O section **614**.

Memory **618** of personal electronic device **600** is a non-transitory computer-readable storage medium, for stor-

ing computer-executable instructions, which, when executed by one or more computer processors 616, for example, cause the computer processors to perform the techniques and processes described below. The computer-executable instructions, for example, are also stored and/or transported within any non-transitory computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device, such as a computer-based system, processor-containing system, or other system that can fetch the instructions from the instruction execution system, apparatus, or device and execute the instructions. Personal electronic device 600 is not limited to the components and configuration of FIG. 6B, but can include other or additional components in multiple configurations.

As used here, the term “affordance” refers to a user-interactive graphical user interface object that is, for example, displayed on the display screen of devices 200, 400, and/or 600 (FIGS. 2A, 4, and 6A-6B). For example, an image (e.g., icon), a button, and text (e.g., hyperlink) each constitutes an affordance.

As used herein, the term “focus selector” refers to an input element that indicates a current part of a user interface with which a user is interacting. In some implementations that include a cursor or other location marker, the cursor acts as a “focus selector” so that when an input (e.g., a press input) is detected on a touch-sensitive surface (e.g., touchpad 455 in FIG. 4 or touch-sensitive surface 551 in FIG. 5B) while the cursor is over a particular user interface element (e.g., a button, window, slider or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations that include a touch screen display (e.g., touch-sensitive display system 212 in FIG. 2A or touch screen 212 in FIG. 5A) that enables direct interaction with user interface elements on the touch screen display, a detected contact on the touch screen acts as a “focus selector” so that when an input (e.g., a press input by the contact) is detected on the touch screen display at a location of a particular user interface element (e.g., a button, window, slider, or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations, focus is moved from one region of a user interface to another region of the user interface without corresponding movement of a cursor or movement of a contact on a touch screen display (e.g., by using a tab key or arrow keys to move focus from one button to another button); in these implementations, the focus selector moves in accordance with movement of focus between different regions of the user interface. Without regard to the specific form taken by the focus selector, the focus selector is generally the user interface element (or contact on a touch screen display) that is controlled by the user so as to communicate the user’s intended interaction with the user interface (e.g., by indicating, to the device, the element of the user interface with which the user is intending to interact). For example, the location of a focus selector (e.g., a cursor, a contact, or a selection box) over a respective button while a press input is detected on the touch-sensitive surface (e.g., a touchpad or touch screen) will indicate that the user is intending to activate the respective button (as opposed to other user interface elements shown on a display of the device).

As used in the specification and claims, the term “characteristic intensity” of a contact refers to a characteristic of the contact based on one or more intensities of the contact. In some embodiments, the characteristic intensity is based on multiple intensity samples. The characteristic intensity is, optionally, based on a predefined number of intensity

samples, or a set of intensity samples collected during a predetermined time period (e.g., 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10 seconds) relative to a predefined event (e.g., after detecting the contact, prior to detecting liftoff of the contact, before or after detecting a start of movement of the contact, prior to detecting an end of the contact, before or after detecting an increase in intensity of the contact, and/or before or after detecting a decrease in intensity of the contact). A characteristic intensity of a contact is, optionally based on one or more of: a maximum value of the intensities of the contact, a mean value of the intensities of the contact, an average value of the intensities of the contact, a top 10 percentile value of the intensities of the contact, a value at the half maximum of the intensities of the contact, a value at the 90 percent maximum of the intensities of the contact, or the like. In some embodiments, the duration of the contact is used in determining the characteristic intensity (e.g., when the characteristic intensity is an average of the intensity of the contact over time). In some embodiments, the characteristic intensity is compared to a set of one or more intensity thresholds to determine whether an operation has been performed by a user. For example, the set of one or more intensity thresholds includes a first intensity threshold and a second intensity threshold. In this example, a contact with a characteristic intensity that does not exceed the first threshold results in a first operation, a contact with a characteristic intensity that exceeds the first intensity threshold and does not exceed the second intensity threshold results in a second operation, and a contact with a characteristic intensity that exceeds the second threshold results in a third operation. In some embodiments, a comparison between the characteristic intensity and one or more thresholds is used to determine whether or not to perform one or more operations (e.g., whether to perform a respective operation or forgo performing the respective operation) rather than being used to determine whether to perform a first operation or a second operation.

In some embodiments, a portion of a gesture is identified for purposes of determining a characteristic intensity. For example, a touch-sensitive surface receives a continuous swipe contact transitioning from a start location and reaching an end location, at which point the intensity of the contact increases. In this example, the characteristic intensity of the contact at the end location is based on only a portion of the continuous swipe contact, and not the entire swipe contact (e.g., only the portion of the swipe contact at the end location). In some embodiments, a smoothing algorithm is applied to the intensities of the swipe contact prior to determining the characteristic intensity of the contact. For example, the smoothing algorithm optionally includes one or more of: an unweighted sliding-average smoothing algorithm, a triangular smoothing algorithm, a median filter smoothing algorithm, and/or an exponential smoothing algorithm. In some circumstances, these smoothing algorithms eliminate narrow spikes or dips in the intensities of the swipe contact for purposes of determining a characteristic intensity.

The intensity of a contact on the touch-sensitive surface is characterized relative to one or more intensity thresholds, such as a contact-detection intensity threshold, a light press intensity threshold, a deep press intensity threshold, and/or one or more other intensity thresholds. In some embodiments, the light press intensity threshold corresponds to an intensity at which the device will perform operations typically associated with clicking a button of a physical mouse or a trackpad. In some embodiments, the deep press intensity threshold corresponds to an intensity at which the device

will perform operations that are different from operations typically associated with clicking a button of a physical mouse or a trackpad. In some embodiments, when a contact is detected with a characteristic intensity below the light press intensity threshold (e.g., and above a nominal contact-detection intensity threshold below which the contact is no longer detected), the device will move a focus selector in accordance with movement of the contact on the touch-sensitive surface without performing an operation associated with the light press intensity threshold or the deep press intensity threshold. Generally, unless otherwise stated, these intensity thresholds are consistent between different sets of user interface figures.

An increase of characteristic intensity of the contact from an intensity below the light press intensity threshold to an intensity between the light press intensity threshold and the deep press intensity threshold is sometimes referred to as a “light press” input. An increase of characteristic intensity of the contact from an intensity below the deep press intensity threshold to an intensity above the deep press intensity threshold is sometimes referred to as a “deep press” input. An increase of characteristic intensity of the contact from an intensity below the contact-detection intensity threshold to an intensity between the contact-detection intensity threshold and the light press intensity threshold is sometimes referred to as detecting the contact on the touch-surface. A decrease of characteristic intensity of the contact from an intensity above the contact-detection intensity threshold to an intensity below the contact-detection intensity threshold is sometimes referred to as detecting liftoff of the contact from the touch-surface. In some embodiments, the contact-detection intensity threshold is zero. In some embodiments, the contact-detection intensity threshold is greater than zero.

In some embodiments described herein, one or more operations are performed in response to detecting a gesture that includes a respective press input or in response to detecting the respective press input performed with a respective contact (or a plurality of contacts), where the respective press input is detected based at least in part on detecting an increase in intensity of the contact (or plurality of contacts) above a press-input intensity threshold. In some embodiments, the respective operation is performed in response to detecting the increase in intensity of the respective contact above the press-input intensity threshold (e.g., a “down stroke” of the respective press input). In some embodiments, the press input includes an increase in intensity of the respective contact above the press-input intensity threshold and a subsequent decrease in intensity of the contact below the press-input intensity threshold, and the respective operation is performed in response to detecting the subsequent decrease in intensity of the respective contact below the press-input threshold (e.g., an “up stroke” of the respective press input).

In some embodiments, the device employs intensity hysteresis to avoid accidental inputs sometimes termed “jitter,” where the device defines or selects a hysteresis intensity threshold with a predefined relationship to the press-input intensity threshold (e.g., the hysteresis intensity threshold is X intensity units lower than the press-input intensity threshold or the hysteresis intensity threshold is 75%, 90%, or some reasonable proportion of the press-input intensity threshold). Thus, in some embodiments, the press input includes an increase in intensity of the respective contact above the press-input intensity threshold and a subsequent decrease in intensity of the contact below the hysteresis intensity threshold that corresponds to the press-input intensity threshold, and the respective operation is performed in

response to detecting the subsequent decrease in intensity of the respective contact below the hysteresis intensity threshold (e.g., an “up stroke” of the respective press input). Similarly, in some embodiments, the press input is detected only when the device detects an increase in intensity of the contact from an intensity at or below the hysteresis intensity threshold to an intensity at or above the press-input intensity threshold and, optionally, a subsequent decrease in intensity of the contact to an intensity at or below the hysteresis intensity, and the respective operation is performed in response to detecting the press input (e.g., the increase in intensity of the contact or the decrease in intensity of the contact, depending on the circumstances).

For ease of explanation, the descriptions of operations performed in response to a press input associated with a press-input intensity threshold or in response to a gesture including the press input are, optionally, triggered in response to detecting either: an increase in intensity of a contact above the press-input intensity threshold, an increase in intensity of a contact from an intensity below the hysteresis intensity threshold to an intensity above the press-input intensity threshold, a decrease in intensity of the contact below the press-input intensity threshold, and/or a decrease in intensity of the contact below the hysteresis intensity threshold corresponding to the press-input intensity threshold. Additionally, in examples where an operation is described as being performed in response to detecting a decrease in intensity of a contact below the press-input intensity threshold, the operation is, optionally, performed in response to detecting a decrease in intensity of the contact below a hysteresis intensity threshold corresponding to, and lower than, the press-input intensity threshold.

3. Digital Assistant System

FIG. 7A illustrates a block diagram of digital assistant system 700 in accordance with various examples. In some examples, digital assistant system 700 is implemented on a standalone computer system. In some examples, digital assistant system 700 is distributed across multiple computers. In some examples, some of the modules and functions of the digital assistant are divided into a server portion and a client portion, where the client portion resides on one or more user devices (e.g., devices 104, 122, 200, 400, or 600) and communicates with the server portion (e.g., server system 108) through one or more networks, e.g., as shown in FIG. 1. In some examples, digital assistant system 700 is an implementation of server system 108 (and/or DA server 106) shown in FIG. 1. It should be noted that digital assistant system 700 is only one example of a digital assistant system, and that digital assistant system 700 can have more or fewer components than shown, can combine two or more components, or can have a different configuration or arrangement of the components. The various components shown in FIG. 7A are implemented in hardware, software instructions for execution by one or more processors, firmware, including one or more signal processing and/or application specific integrated circuits, or a combination thereof.

Digital assistant system 700 includes memory 702, one or more processors 704, input/output (I/O) interface 706, and network communications interface 708. These components can communicate with one another over one or more communication buses or signal lines 710.

In some examples, memory 702 includes a non-transitory computer-readable medium, such as high-speed random access memory and/or a non-volatile computer-readable

storage medium (e.g., one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices).

In some examples, I/O interface **706** couples input/output devices **716** of digital assistant system **700**, such as displays, keyboards, touch screens, and microphones, to user interface module **722**. I/O interface **706**, in conjunction with user interface module **722**, receives user inputs (e.g., voice input, keyboard inputs, touch inputs, etc.) and processes them accordingly. In some examples, e.g., when the digital assistant is implemented on a standalone user device, digital assistant system **700** includes any of the components and I/O communication interfaces described with respect to devices **200**, **400**, or **600** in FIGS. **2A**, **4**, **6A-6B**, respectively. In some examples, digital assistant system **700** represents the server portion of a digital assistant implementation, and can interact with the user through a client-side portion residing on a user device (e.g., devices **104**, **200**, **400**, or **600**).

In some examples, the network communications interface **708** includes wired communication port(s) **712** and/or wireless transmission and reception circuitry **714**. The wired communication port(s) receives and send communication signals via one or more wired interfaces, e.g., Ethernet, Universal Serial Bus (USB), FIREWIRE, etc. The wireless circuitry **714** receives and sends RF signals and/or optical signals from/to communications networks and other communications devices. The wireless communications use any of a plurality of communications standards, protocols, and technologies, such as GSM, EDGE, CDMA, TDMA, Bluetooth, Wi-Fi, VOIP, Wi-MAX, or any other suitable communication protocol. Network communications interface **708** enables communication between digital assistant system **700** with networks, such as the Internet, an intranet, and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN), and/or a metropolitan area network (MAN), and other devices.

In some examples, memory **702**, or the computer-readable storage media of memory **702**, stores programs, modules, instructions, and data structures including all or a subset of: operating system **718**, communications module **720**, user interface module **722**, one or more applications **724**, and digital assistant module **726**. In particular, memory **702**, or the computer-readable storage media of memory **702**, stores instructions for performing the processes described below. One or more processors **704** execute these programs, modules, and instructions, and reads/writes from/to the data structures.

Operating system **718** (e.g., Darwin, RTXC, LINUX, UNIX, iOS, OS X, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communications between various hardware, firmware, and software components.

Communications module **720** facilitates communications between digital assistant system **700** with other devices over network communications interface **708**. For example, communications module **720** communicates with RF circuitry **208** of electronic devices such as devices **200**, **400**, and **600** shown in FIGS. **2A**, **4**, **6A-6B**, respectively. Communications module **720** also includes various components for handling data received by wireless circuitry **714** and/or wired communications port **712**.

User interface module **722** receives commands and/or inputs from a user via I/O interface **706** (e.g., from a keyboard, touch screen, pointing device, controller, and/or

microphone), and generate user interface objects on a display. User interface module **722** also prepares and delivers outputs (e.g., speech, sound, animation, text, icons, vibrations, haptic feedback, light, etc.) to the user via the I/O interface **706** (e.g., through displays, audio channels, speakers, touch-pads, etc.).

Applications **724** include programs and/or modules that are configured to be executed by one or more processors **704**. For example, if the digital assistant system is implemented on a standalone user device, applications **724** include user applications, such as games, a calendar application, a navigation application, or an email application. If digital assistant system **700** is implemented on a server, applications **724** include resource management applications, diagnostic applications, or scheduling applications, for example.

Memory **702** also stores digital assistant module **726** (or the server portion of a digital assistant). In some examples, digital assistant module **726** includes the following sub-modules, or a subset or superset thereof: input/output processing module **728**, speech-to-text (STT) processing module **730**, natural language processing module **732**, dialogue flow processing module **734**, task flow processing module **736**, service processing module **738**, and speech synthesis processing module **740**. Each of these modules has access to one or more of the following systems or data and models of the digital assistant module **726**, or a subset or superset thereof: ontology **760**, vocabulary index **744**, user data **748**, task flow models **754**, service models **756**, and ASR systems **758**.

In some examples, using the processing modules, data, and models implemented in digital assistant module **726**, the digital assistant can perform at least some of the following: converting speech input into text; identifying a user's intent expressed in a natural language input received from the user; actively eliciting and obtaining information needed to fully infer the user's intent (e.g., by disambiguating words, games, intentions, etc.); determining the task flow for fulfilling the inferred intent; and executing the task flow to fulfill the inferred intent.

In some examples, as shown in FIG. **7B**, I/O processing module **728** interacts with the user through I/O devices **716** in FIG. **7A** or with a user device (e.g., devices **104**, **200**, **400**, or **600**) through network communications interface **708** in FIG. **7A** to obtain user input (e.g., a speech input) and to provide responses (e.g., as speech outputs) to the user input. I/O processing module **728** optionally obtains contextual information associated with the user input from the user device, along with or shortly after the receipt of the user input. The contextual information includes user-specific data, vocabulary, and/or preferences relevant to the user input. In some examples, the contextual information also includes software and hardware states of the user device at the time the user request is received, and/or information related to the surrounding environment of the user at the time that the user request was received. In some examples, I/O processing module **728** also sends follow-up questions to, and receive answers from, the user regarding the user request. When a user request is received by I/O processing module **728** and the user request includes speech input, I/O processing module **728** forwards the speech input to STT processing module **730** (or speech recognizer) for speech-to-text conversions.

STT processing module **730** includes one or more ASR systems **758**. The one or more ASR systems **758** can process the speech input that is received through I/O processing module **728** to produce a recognition result. Each ASR

system 758 includes a front-end speech pre-processor. The front-end speech pre-processor extracts representative features from the speech input. For example, the front-end speech pre-processor performs a Fourier transform on the speech input to extract spectral features that characterize the speech input as a sequence of representative multi-dimensional vectors. Further, each ASR system 758 includes one or more speech recognition models (e.g., acoustic models and/or language models) and implements one or more speech recognition engines. Examples of speech recognition models include Hidden Markov Models, Gaussian-Mixture Models, Deep Neural Network Models, n-gram language models, and other statistical models. Examples of speech recognition engines include the dynamic time warping based engines and weighted finite-state transducers (WFST) based engines. The one or more speech recognition models and the one or more speech recognition engines are used to process the extracted representative features of the front-end speech pre-processor to produce intermediate recognition results (e.g., phonemes, phonemic strings, and sub-words), and ultimately, text recognition results (e.g., words, word strings, or sequence of tokens). In some examples, the speech input is processed at least partially by a third-party service or on the user's device (e.g., device 104, 200, 400, or 600) to produce the recognition result. Once STT processing module 730 produces recognition results containing a text string (e.g., words, or sequence of words, or sequence of tokens), the recognition result is passed to natural language processing module 732 for intent deduction. In some examples, STT processing module 730 produces multiple candidate text representations of the speech input. Each candidate text representation is a sequence of words or tokens corresponding to the speech input. In some examples, each candidate text representation is associated with a speech recognition confidence score. Based on the speech recognition confidence scores, STT processing module 730 ranks the candidate text representations and provides the n-best (e.g., n highest ranked) candidate text representation(s) to natural language processing module 732 for intent deduction, where n is a predetermined integer greater than zero. For example, in one example, only the highest ranked (n=1) candidate text representation is passed to natural language processing module 732 for intent deduction. In another example, the five highest ranked (n=5) candidate text representations are passed to natural language processing module 732 for intent deduction.

More details on the speech-to-text processing are described in U.S. Utility application Ser. No. 13/236,942 for "Consolidating Speech Recognition Results," filed on Sep. 20, 2011, the entire disclosure of which is incorporated herein by reference.

In some examples, STT processing module 730 includes and/or accesses a vocabulary of recognizable words via phonetic alphabet conversion module 731. Each vocabulary word is associated with one or more candidate pronunciations of the word represented in a speech recognition phonetic alphabet. In particular, the vocabulary of recognizable words includes a word that is associated with a plurality of candidate pronunciations. For example, the vocabulary includes the word "tomato" that is associated with the candidate pronunciations of /tə'meɪrəʊ/ and /tə'matəʊ/. Further, vocabulary words are associated with custom candidate pronunciations that are based on previous speech inputs from the user. Such custom candidate pronunciations are stored in STT processing module 730 and are associated with a particular user via the user's profile on the device. In some examples, the candidate pronunciations for words are

determined based on the spelling of the word and one or more linguistic and/or phonetic rules. In some examples, the candidate pronunciations are manually generated, e.g., based on known canonical pronunciations.

In some examples, the candidate pronunciations are ranked based on the commonness of the candidate pronunciation. For example, the candidate pronunciation /tə'meɪrəʊ/ is ranked higher than /tə'matəʊ/, because the former is a more commonly used pronunciation (e.g., among all users, for users in a particular geographical region, or for any other appropriate subset of users). In some examples, candidate pronunciations are ranked based on whether the candidate pronunciation is a custom candidate pronunciation associated with the user. For example, custom candidate pronunciations are ranked higher than canonical candidate pronunciations. This can be useful for recognizing proper nouns having a unique pronunciation that deviates from canonical pronunciation. In some examples, candidate pronunciations are associated with one or more speech characteristics, such as geographic origin, nationality, or ethnicity. For example, the candidate pronunciation /tə'meɪrəʊ/ is associated with the United States, whereas the candidate pronunciation /tə'matəʊ/ is associated with Great Britain. Further, the rank of the candidate pronunciation is based on one or more characteristics (e.g., geographic origin, nationality, ethnicity, etc.) of the user stored in the user's profile on the device. For example, it can be determined from the user's profile that the user is associated with the United States. Based on the user being associated with the United States, the candidate pronunciation /tə'meɪrəʊ/ (associated with the United States) is ranked higher than the candidate pronunciation /tə'matəʊ/ (associated with Great Britain). In some examples, one of the ranked candidate pronunciations is selected as a predicted pronunciation (e.g., the most likely pronunciation).

When a speech input is received, STT processing module 730 is used to determine the phonemes corresponding to the speech input (e.g., using an acoustic model), and then attempt to determine words that match the phonemes (e.g., using a language model). For example, if STT processing module 730 first identifies the sequence of phonemes /tə'meɪrəʊ/ corresponding to a portion of the speech input, it can then determine, based on vocabulary index 744, that this sequence corresponds to the word "tomato."

In some examples, STT processing module 730 uses approximate matching techniques to determine words in an utterance. Thus, for example, the STT processing module 730 determines that the sequence of phonemes /tə'meɪrəʊ/ corresponds to the word "tomato," even if that particular sequence of phonemes is not one of the candidate sequence of phonemes for that word.

Natural language processing module 732 ("natural language processor") of the digital assistant takes the n-best candidate text representation(s) ("word sequence(s)" or "token sequence(s)") generated by STT processing module 730, and attempts to associate each of the candidate text representations with one or more "actionable intents" recognized by the digital assistant. An "actionable intent" (or "user intent") represents a task that can be performed by the digital assistant, and can have an associated task flow implemented in task flow models 754. The associated task flow is a series of programmed actions and steps that the digital assistant takes in order to perform the task. The scope of a digital assistant's capabilities is dependent on the number and variety of task flows that have been implemented and stored in task flow models 754, or in other words, on the number and variety of "actionable intents" that

the digital assistant recognizes. The effectiveness of the digital assistant, however, also depends on the assistant's ability to infer the correct "actionable intent(s)" from the user request expressed in natural language.

In some examples, in addition to the sequence of words or tokens obtained from STT processing module 730, natural language processing module 732 also receives contextual information associated with the user request, e.g., from I/O processing module 728. The natural language processing module 732 optionally uses the contextual information to clarify, supplement, and/or further define the information contained in the candidate text representations received from STT processing module 730. The contextual information includes, for example, user preferences, hardware, and/or software states of the user device, sensor information collected before, during, or shortly after the user request, prior interactions (e.g., dialogue) between the digital assistant and the user, and the like. As described herein, contextual information is, in some examples, dynamic, and changes with time, location, content of the dialogue, and other factors.

In some examples, the natural language processing is based on, e.g., ontology 760. Ontology 760 is a hierarchical structure containing many nodes, each node representing either an "actionable intent" or a "property" relevant to one or more of the "actionable intents" or other "properties." As noted above, an "actionable intent" represents a task that the digital assistant is capable of performing, i.e., it is "actionable" or can be acted on. A "property" represents a parameter associated with an actionable intent or a sub-aspect of another property. A linkage between an actionable intent node and a property node in ontology 760 defines how a parameter represented by the property node pertains to the task represented by the actionable intent node.

In some examples, ontology 760 is made up of actionable intent nodes and property nodes. Within ontology 760, each actionable intent node is linked to one or more property nodes either directly or through one or more intermediate property nodes. Similarly, each property node is linked to one or more actionable intent nodes either directly or through one or more intermediate property nodes. For example, as shown in FIG. 7C, ontology 760 includes a "restaurant reservation" node (i.e., an actionable intent node). Property nodes "restaurant," "date/time" (for the reservation), and "party size" are each directly linked to the actionable intent node (i.e., the "restaurant reservation" node).

In addition, property nodes "cuisine," "price range," "phone number," and "location" are sub-nodes of the property node "restaurant," and are each linked to the "restaurant reservation" node (i.e., the actionable intent node) through the intermediate property node "restaurant." For another example, as shown in FIG. 7C, ontology 760 also includes a "set reminder" node (i.e., another actionable intent node). Property nodes "date/time" (for setting the reminder) and "subject" (for the reminder) are each linked to the "set reminder" node. Since the property "date/time" is relevant to both the task of making a restaurant reservation and the task of setting a reminder, the property node "date/time" is linked to both the "restaurant reservation" node and the "set reminder" node in ontology 760.

An actionable intent node, along with its linked property nodes, is described as a "domain." In the present discussion, each domain is associated with a respective actionable intent, and refers to the group of nodes (and the relationships there between) associated with the particular actionable intent. For example, ontology 760 shown in FIG. 7C

includes an example of restaurant reservation domain 762 and an example of reminder domain 764 within ontology 760. The restaurant reservation domain includes the actionable intent node "restaurant reservation," property nodes "restaurant," "date/time," and "party size," and sub-property nodes "cuisine," "price range," "phone number," and "location." Reminder domain 764 includes the actionable intent node "set reminder," and property nodes "subject" and "date/time." In some examples, ontology 760 is made up of many domains. Each domain shares one or more property nodes with one or more other domains. For example, the "date/time" property node is associated with many different domains (e.g., a scheduling domain, a travel reservation domain, a movie ticket domain, etc.), in addition to restaurant reservation domain 762 and reminder domain 764.

While FIG. 7C illustrates two example domains within ontology 760, other domains include, for example, "find a movie," "initiate a phone call," "find directions," "schedule a meeting," "send a message," and "provide an answer to a question," "read a list," "providing navigation instructions," "provide instructions for a task" and so on. A "send a message" domain is associated with a "send a message" actionable intent node, and further includes property nodes such as "recipient(s)," "message type," and "message body." The property node "recipient" is further defined, for example, by the sub-property nodes such as "recipient name" and "message address."

In some examples, ontology 760 includes all the domains (and hence actionable intents) that the digital assistant is capable of understanding and acting upon. In some examples, ontology 760 is modified, such as by adding or removing entire domains or nodes, or by modifying relationships between the nodes within the ontology 760.

In some examples, nodes associated with multiple related actionable intents are clustered under a "super domain" in ontology 760. For example, a "travel" super-domain includes a cluster of property nodes and actionable intent nodes related to travel. The actionable intent nodes related to travel includes "airline reservation," "hotel reservation," "car rental," "get directions," "find points of interest," and so on. The actionable intent nodes under the same super domain (e.g., the "travel" super domain) have many property nodes in common. For example, the actionable intent nodes for "airline reservation," "hotel reservation," "car rental," "get directions," and "find points of interest" share one or more of the property nodes "start location," "destination," "departure date/time," "arrival date/time," and "party size."

In some examples, each node in ontology 760 is associated with a set of words and/or phrases that are relevant to the property or actionable intent represented by the node. The respective set of words and/or phrases associated with each node are the so-called "vocabulary" associated with the node. The respective set of words and/or phrases associated with each node are stored in vocabulary index 744 in association with the property or actionable intent represented by the node. For example, returning to FIG. 7B, the vocabulary associated with the node for the property of "restaurant" includes words such as "food," "drinks," "cuisine," "hungry," "eat," "pizza," "fast food," "meal," and so on. For another example, the vocabulary associated with the node for the actionable intent of "initiate a phone call" includes words and phrases such as "call," "phone," "dial," "ring," "call this number," "make a call to," and so on. The vocabulary index 744 optionally includes words and phrases in different languages.

Natural language processing module 732 receives the candidate text representations (e.g., text string(s) or token sequence(s)) from STT processing module 730, and for each candidate representation, determines what nodes are implicated by the words in the candidate text representation. In some examples, if a word or phrase in the candidate text representation is found to be associated with one or more nodes in ontology 760 (via vocabulary index 744), the word or phrase “triggers” or “activates” those nodes. Based on the quantity and/or relative importance of the activated nodes, natural language processing module 732 selects one of the actionable intents as the task that the user intended the digital assistant to perform. In some examples, the domain that has the most “triggered” nodes is selected. In some examples, the domain having the highest confidence value (e.g., based on the relative importance of its various triggered nodes) is selected. In some examples, the domain is selected based on a combination of the number and the importance of the triggered nodes. In some examples, additional factors are considered in selecting the node as well, such as whether the digital assistant has previously correctly interpreted a similar request from a user.

User data 748 includes user-specific information, such as user-specific vocabulary, user preferences, user address, user’s default and secondary languages, user’s contact list, and other short-term or long-term information for each user. In some examples, natural language processing module 732 uses the user-specific information to supplement the information contained in the user input to further define the user intent. For example, for a user request “invite my friends to my birthday party,” natural language processing module 732 is able to access user data 748 to determine who the “friends” are and when and where the “birthday party” would be held, rather than requiring the user to provide such information explicitly in his/her request.

It should be recognized that in some examples, natural language processing module 732 is implemented using one or more machine learning mechanisms (e.g., neural networks). In particular, the one or more machine learning mechanisms are configured to receive a candidate text representation and contextual information associated with the candidate text representation. Based on the candidate text representation and the associated contextual information, the one or more machine learning mechanisms are configured to determine intent confidence scores over a set of candidate actionable intents. Natural language processing module 732 can select one or more candidate actionable intents from the set of candidate actionable intents based on the determined intent confidence scores. In some examples, an ontology (e.g., ontology 760) is also used to select the one or more candidate actionable intents from the set of candidate actionable intents.

Other details of searching an ontology based on a token string are described in U.S. Utility application Ser. No. 12/341,743 for “Method and Apparatus for Searching Using An Active Ontology,” filed Dec. 22, 2008, the entire disclosure of which is incorporated herein by reference.

In some examples, once natural language processing module 732 identifies an actionable intent (or domain) based on the user request, natural language processing module 732 generates a structured query to represent the identified actionable intent. In some examples, the structured query includes parameters for one or more nodes within the domain for the actionable intent, and at least some of the parameters are populated with the specific information and requirements specified in the user request. For example, the user says “Make me a dinner reservation at a sushi place at

7.” In this case, natural language processing module 732 is able to correctly identify the actionable intent to be “restaurant reservation” based on the user input. According to the ontology, a structured query for a “restaurant reservation” domain includes parameters such as {Cuisine}, {Time}, {Date}, {Party Size}, and the like. In some examples, based on the speech input and the text derived from the speech input using STT processing module 730, natural language processing module 732 generates a partial structured query for the restaurant reservation domain, where the partial structured query includes the parameters {Cuisine=“Sushi”} and {Time=“7 pm”}. However, in this example, the user’s utterance contains insufficient information to complete the structured query associated with the domain. Therefore, other necessary parameters such as {Party Size} and {Date} are not specified in the structured query based on the information currently available. In some examples, natural language processing module 732 populates some parameters of the structured query with received contextual information. For example, in some examples, if the user requested a sushi restaurant “near me,” natural language processing module 732 populates a {location} parameter in the structured query with GPS coordinates from the user device.

In some examples, natural language processing module 732 identifies multiple candidate actionable intents for each candidate text representation received from STT processing module 730. Further, in some examples, a respective structured query (partial or complete) is generated for each identified candidate actionable intent. Natural language processing module 732 determines an intent confidence score for each candidate actionable intent and ranks the candidate actionable intents based on the intent confidence scores. In some examples, natural language processing module 732 passes the generated structured query (or queries), including any completed parameters, to task flow processing module 736 (“task flow processor”). In some examples, the structured query (or queries) for the m-best (e.g., m highest ranked) candidate actionable intents are provided to task flow processing module 736, where m is a predetermined integer greater than zero. In some examples, the structured query (or queries) for the m-best candidate actionable intents are provided to task flow processing module 736 with the corresponding candidate text representation(s).

Other details of inferring a user intent based on multiple candidate actionable intents determined from multiple candidate text representations of a speech input are described in U.S. Utility application Ser. No. 14/298,725 for “System and Method for Inferring User Intent From Speech Inputs,” filed Jun. 6, 2014, the entire disclosure of which is incorporated herein by reference.

Task flow processing module 736 is configured to receive the structured query (or queries) from natural language processing module 732, complete the structured query, if necessary, and perform the actions required to “complete” the user’s ultimate request. In some examples, the various procedures necessary to complete these tasks are provided in task flow models 754. In some examples, task flow models 754 include procedures for obtaining additional information from the user and task flows for performing actions associated with the actionable intent.

As described above, in order to complete a structured query, task flow processing module 736 needs to initiate additional dialogue with the user in order to obtain additional information, and/or disambiguate potentially ambiguous utterances. When such interactions are necessary, task flow processing module 736 invokes dialogue flow processing module 734 to engage in a dialogue with the user. In

some examples, dialogue flow processing module 734 determines how (and/or when) to ask the user for the additional information and receives and processes the user responses. The questions are provided to and answers are received from the users through I/O processing module 728. In some examples, dialogue flow processing module 734 presents dialogue output to the user via audio and/or visual output, and receives input from the user via spoken or physical (e.g., clicking) responses. Continuing with the example above, when task flow processing module 736 invokes dialogue flow processing module 734 to determine the “party size” and “date” information for the structured query associated with the domain “restaurant reservation,” dialogue flow processing module 734 generates questions such as “For how many people?” and “On which day?” to pass to the user. Once answers are received from the user, dialogue flow processing module 734 then populates the structured query with the missing information, or pass the information to task flow processing module 736 to complete the missing information from the structured query.

Once task flow processing module 736 has completed the structured query for an actionable intent, task flow processing module 736 proceeds to perform the ultimate task associated with the actionable intent. Accordingly, task flow processing module 736 executes the steps and instructions in the task flow model according to the specific parameters contained in the structured query. For example, the task flow model for the actionable intent of “restaurant reservation” includes steps and instructions for contacting a restaurant and actually requesting a reservation for a particular party size at a particular time. For example, using a structured query such as: {restaurant reservation, restaurant=ABC Café, date=3/12/2012, time=7 pm, party size=5}, task flow processing module 736 performs the steps of: (1) logging onto a server of the ABC Café or a restaurant reservation system such as OPENTABLE®, (2) entering the date, time, and party size information in a form on the website, (3) submitting the form, and (4) making a calendar entry for the reservation in the user’s calendar.

In some examples, task flow processing module 736 employs the assistance of service processing module 738 (“service processing module”) to complete a task requested in the user input or to provide an informational answer requested in the user input. For example, service processing module 738 acts on behalf of task flow processing module 736 to make a phone call, set a calendar entry, invoke a map search, invoke or interact with other user applications installed on the user device, and invoke or interact with third-party services (e.g., a restaurant reservation portal, a social networking website, a banking portal, etc.). In some examples, the protocols and application programming interfaces (API) required by each service are specified by a respective service model among service models 756. Service processing module 738 accesses the appropriate service model for a service and generates requests for the service in accordance with the protocols and APIs required by the service according to the service model.

For example, if a restaurant has enabled an online reservation service, the restaurant submits a service model specifying the necessary parameters for making a reservation and the APIs for communicating the values of the necessary parameter to the online reservation service. When requested by task flow processing module 736, service processing module 738 establishes a network connection with the online reservation service using the web address stored in the service model, and sends the necessary parameters of the

reservation (e.g., time, date, party size) to the online reservation interface in a format according to the API of the online reservation service.

In some examples, natural language processing module 732, dialogue flow processing module 734, and task flow processing module 736 are used collectively and iteratively to infer and define the user’s intent, obtain information to further clarify and refine the user intent, and finally generate a response (i.e., an output to the user, or the completion of a task) to fulfill the user’s intent. The generated response is a dialogue response to the speech input that at least partially fulfills the user’s intent. Further, in some examples, the generated response is output as a speech output. In these examples, the generated response is sent to speech synthesis processing module 740 (e.g., speech synthesizer) where it can be processed to synthesize the dialogue response in speech form. In yet other examples, the generated response is data content relevant to satisfying a user request in the speech input.

In examples where task flow processing module 736 receives multiple structured queries from natural language processing module 732, task flow processing module 736 initially processes the first structured query of the received structured queries to attempt to complete the first structured query and/or execute one or more tasks or actions represented by the first structured query. In some examples, the first structured query corresponds to the highest ranked actionable intent. In other examples, the first structured query is selected from the received structured queries based on a combination of the corresponding speech recognition confidence scores and the corresponding intent confidence scores. In some examples, if task flow processing module 736 encounters an error during processing of the first structured query (e.g., due to an inability to determine a necessary parameter), the task flow processing module 736 can proceed to select and process a second structured query of the received structured queries that corresponds to a lower ranked actionable intent. The second structured query is selected, for example, based on the speech recognition confidence score of the corresponding candidate text representation, the intent confidence score of the corresponding candidate actionable intent, a missing necessary parameter in the first structured query, or any combination thereof.

Speech synthesis processing module 740 is configured to synthesize speech outputs for presentation to the user. Speech synthesis processing module 740 synthesizes speech outputs based on text provided by the digital assistant. For example, the generated dialogue response is in the form of a text string. Speech synthesis processing module 740 converts the text string to an audible speech output. Speech synthesis processing module 740 uses any appropriate speech synthesis technique in order to generate speech outputs from text, including, but not limited to, concatenative synthesis, unit selection synthesis, diphone synthesis, domain-specific synthesis, formant synthesis, articulatory synthesis, hidden Markov model (HMM) based synthesis, and sinewave synthesis. In some examples, speech synthesis processing module 740 is configured to synthesize individual words based on phonemic strings corresponding to the words. For example, a phonemic string is associated with a word in the generated dialogue response. The phonemic string is stored in metadata associated with the word. Speech synthesis processing module 740 is configured to directly process the phonemic string in the metadata to synthesize the word in speech form.

In some examples, instead of (or in addition to) using speech synthesis processing module 740, speech synthesis is

performed on a remote device (e.g., the server system 108), and the synthesized speech is sent to the user device for output to the user. For example, this can occur in some implementations where outputs for a digital assistant are generated at a server system. And because server systems generally have more processing power or resources than a user device, it is possible to obtain higher quality speech outputs than would be practical with client-side synthesis.

Additional details on digital assistants can be found in the U.S. Utility application Ser. No. 12/987,982, entitled “Intelligent Automated Assistant,” filed Jan. 10, 2011, and U.S. Utility application Ser. No. 13/251,088, entitled “Generating and Processing Task Items That Represent Tasks to Perform,” filed Sep. 30, 2011, the entire disclosures of which are incorporated herein by reference.

4. Process for Providing Suggested User Actions In Response to Detected Anchor

The embodiments are directed to providing, via an electronic device, suggested user actions. The suggested actions may be provided in response to detecting an occurrence of an anchor. An anchor may be a predefined event occurring in the user’s day. For example, upon waking up in the morning (e.g., a user waking up may be an example of an anchor), an electronic device may provide a notification that includes a suggestion for playing the newest episode of a particular podcast (i.e., a suggested user action), where the user frequently listens to the particular podcast upon waking up in the morning. The occurrence of the anchor may be detected via one or more signals generated by the electronic device (e.g., the device’s alarm clock function executing, the device being transitioned from a “Do Not Disturb Mode” to an “Active Mode,” or the like). As used herein, an anchor may be a marker or signal of an event in a user’s day. An occurrence of the anchor may be detectable via the monitoring and analysis of electronic signals generated by the electronic device. Based on the user’s previous interactions with the device, the occurrence of the anchor may be indicative of user behavior and/or action taken in response to the event in the user’s day. By way of non-limiting examples, an anchor may include the user entering a location of interest (LOI), such as but not limited to the user’s home, office, gym, airport, shopping center, or the like. Because an anchor may be an event that indicates user behavior, an anchor may be referred throughout as an anchor event, or simply an event. The occurrence of an anchor may be referred to as an anchor occurrence.

Other non-limiting examples of an occurrence of an anchor include the user beginning to use an electronic device after an extended period of device idle time, the user finishing a workout, the user waking up, the user going to bed, the user pairing a Bluetooth-enabled device to the electronic device, the user beginning or completing a calendar event, and the like. Still other examples of an anchor include the user launching a specific application and/or employing a specific functionality of the device. Any event detectable via signatures encoded in signals generated by the electronic device may be an anchor. A signature (e.g., a pattern within the electronic signals generated by the device) that indicates an occurrence of a specific anchor may be pre-determined, pre-computed, and/or pre-learned (e.g., via supervised or unsupervised machine learning (ML) methods). Thus, the electronic device may monitor its various signals and detect an occurrence of the specific anchor by identifying and/or detecting one or more signatures within the signals that indicates the specific anchor. For example,

an occurrence of a waking up anchor may be detected via signals of the electronic device that indicates that the user has terminated, or otherwise ended, an “idle” state of the electronic device (or that the device’s alarm went off). Upon detecting an occurrence of the anchor, the various embodiments may suggest one or more user actions that correlate (via previously generated training data) with the occurrence of the anchor.

A user action (i.e., an action) may include an invocation, execution, or otherwise launching a specific application, capability, functionality, or command that is enabled via the electronic device. By way of non-limiting examples, a user action may include but is not limited to playing a specific audio/video content (e.g., a podcast, a music playlist, an audio book, a lecture, a television series, a movie, or the like), launching an application installed on the device (e.g., a workout application, a meditation application, a ride-share application, a food delivery application, a social network application, or the like), sending an electronic communication (e.g., email, SMS, tweet, or the like) to another user, user group, or social network, turning on/off functionality of the device (e.g., turning off/on an Airplane mode of the device), updating various settings and/or configurations of the device, creating a calendar event, or the like.

When providing a suggested action, the various embodiments may provide a notification (e.g., a “pop-up” notification) that indicates the suggested action. The notification may be interactive, in that the user may initiate the action via an interactive selection of the notification. Upon selecting the notification, the suggested action may be executed and/or the execution of the action may be initiated, by the device. Such suggested action notifications include, but are not otherwise limited to pop-up notifications, toast notifications, passive pop-up notifications, snackbar notifications, bubble notifications, or any other such notification. In at least some embodiments, the suggested action may be automatically executed without the user’s interactions with the notification. For example, a user may employ a setting that enables the automatic execution of a suggested action. As noted throughout, the suggested action for any given anchor may not be unique. For example, multiple actions may be suggested to a user upon the user waking up. The context of the anchor may affect the determination of which actions to suggest. For example, the occurrence of a wake up anchor during a weekday may result suggesting the playing of a podcast, while the occurrence of a wake up anchor on a weekend day may result in suggesting the beginning of a workout, via a workout application installed on the device.

In the various embodiments, one or more machine learning (ML) models may be trained to learn various statistical correlations (or associations) between occurrences of a specific anchor and actions (e.g., innovations of device functionalities or capabilities) initiated by the user. Any of the various ML models employed by the various embodiments may be herein collectively referred to as “anchor models,” because the models are trained to detect occurrences of various anchors and suggest one or more actions that the user is statistically likely to be initiate in response to the occurrence (or impending occurrence) of the anchor. Thus, the various embodiments include the training of anchor models, as well as employing the trained anchor models to enhance the user’s experience (UX) of employing the device. That is, once deployed to the device, a trained anchor model significantly enhances the performance of the device because the anchor model is employed to anticipate, and suggest, one or more actions that the user is likely to take in response to various anchors (or events) that occur

throughout the user's day. Supervised or unsupervised ML may be employed to train the models to determine the context (e.g., contextual features) of an anchor occurrence, and base, tailor, and/or selectively target the suggested actions on the context of the anchor.

More specifically, the embodiments may detect, determine, and/or otherwise identify various contextual features of the anchor occurrence. The embodiments may employ the detected contextual features of the anchor occurrence, and tailor the one or more suggested actions to the context of the occurrence of the anchor. That is, in addition to the occurrence of the anchor, one or more signatures within the monitored signals may provide contextual information (e.g., contextual features) associated with a specific occurrence of the anchor. The one or more suggested actions that are provided may be based on the contextual information. Contextual features of an occurrence of an anchor (e.g., the user waking up) may include the calendar date, the day of the week, the location of the user, the time of day that the anchor occurred, and any other such contextualizing information. Various suggested actions may be provided based on, and tailored to, such contextual information. For instance, if the day of the week is a weekday, or otherwise is a workday for the user, the one or more suggested actions may include a suggestion to launch a navigation application that provides real-time road-traffic information and/or provides information related to commuting to work. In contrast, if the day of the week is a weekend or a non-workday for the user, the one or more suggested actions may include different and/or separate actions, such as but not limited to a suggestion to launch a workout application or play a podcast. Different suggested actions may be provided if the time of day that the user is waking up is in the morning, in contrast to the suggested actions provided if the user is waking up from a mid-afternoon nap. In such embodiments, the anchor models may be trained to recognize various patterns and/or correlations between the anchor occurrences, contextual features of the anchor occurrence, and the user's response to the anchor occurrences and the contextual features. For anchor types that are correlated with multiple actions, ML may be employed to train the models to discriminate between the multiple actions based on the context of the anchor occurrence. In some embodiments, even though more than one action type is positively correlated with a specific anchor type, when an occurrence of the anchor type is detected, only the most likely of the multiple actions are suggested, based on the context of the anchor occurrence. That is, even though multiple actions are correlated with an anchor, the anchor models may be trained to determine statistical correlations between the user's likely actions and the anchor occurrence, based on the specific contextual features of the specific anchor occurrence.

In some embodiments, the contextual information of an anchor occurrence may include contextual differences, changes, or variances that occur after (or before) the occurrence of the anchor. Such contextual information may indicate a temporal period (e.g., an amount of time) that has elapsed since the anchor occurrence, or a temporal period until an expected anchor occurrence. In such embodiments, the moment of time at which the electronic device provides the one or more suggested actions may be based on contextual information, such as but not limited to the amount of time that has passed since the occurrence of the anchor, or the amount of time till an expected anchor occurrence. Thus, the context of an anchor occurrence may be a context-based context. In some embodiments, the context of an anchor occurrence may be location-based context. For instance,

when a user enters a location of interest (LOI), the context may be location-based, e.g., is the LOI the user's home or the user's work office. In such embodiments that employ time-based context, an anchor model may be employed to determine a temporal offset, measured from the time of occurrence of the anchor (or the expected time of an expected anchor occurrence). The one or more suggested actions may be provided at a time based on the temporal offset. For example, the temporal offset for providing a suggested action may be 15 minutes after an occurrence of the anchor associated with the user waking up. The suggested action (e.g., playing a podcast) may be provided at the temporal offset (e.g., 15 minutes) after the anchor occurrence. That is, the suggestion of playing the podcast may be provided at the temporal offset, as measured after the detection of the anchor occurrence, e.g., 15 minutes after the user wakes up.

As noted above, one or more anchor models may be employed to detect anchor occurrences and provide the suggested actions. In various embodiments, an anchor model may be trained to learn, or otherwise predict, a likely temporal offset for providing suggested actions, as well as other contextual features of the anchors. The temporal offset may be positive (e.g., provide a suggested action after an occurrence of the anchor and after the tolling of the temporal offset) or negative (e.g., provide a suggested action before an expected occurrence of the anchor). The above example of providing a suggestion to play a podcast after 15 minutes upon a "wake-up" anchor is an example of a positive temporal offset. An example of a negative temporal offset includes providing a suggested action that includes hailing a ride, via a ride-share application, three hours before a scheduled flight departure for the user.

The detection of an anchor occurrence may trigger providing multiple suggested actions. The multiple suggested actions may be provided at separate temporal offsets. As a non-limiting example, a waking up anchor may be detected at 7:00 AM. The anchor may be detected via a signal generated when the user manually terminates (or the electronic device automatically terminates based on a timer setting of a "Do Not Disturb Mode") an extended period of "idle time" of the electronic device. At a first positive temporal offset of 5 minutes (e.g., at 7:05 AM), the electronic device provides the user a first suggested action of playing the most recent episode of one of the user's favorite podcast via a podcasting application installed on the device. At a second positive temporal offset of 60 minutes (e.g., at 8:00 AM), the electronic device provides the user a second suggested action of employing a weather application to check the local weather report. In some embodiments, whether to provide (or not provide) the second suggested action at the second temporal offset may be based on whether the user initiated the first suggested action. For example, the second suggested action may only be provided if and only if (iff) the users initiated the first suggested action. In other embodiments, the second suggested action may be provided at the second temporal offset, independent of whether or not the user initiated the first suggested action. In still other embodiments, the value of the second temporal offset may be based on whether or not the user initiated the first suggested action. For instance, if the user does not listen to the suggested episode of the podcast, the value of second temporal offset may be set to 60 minutes. If the user does listen to the podcast, the value of the second temporal offset may be set such that the second suggested action is provided when the episode of the podcast is finished, or the user otherwise stops listening to the podcast.

Continuing with this example regarding the user's day, the user may begin their commute to work by arriving at their bus station. The bus station may be a predetermined or a "learned" (via ML) LOI and the arrival at the bus station may be a second anchor in the user's day. The anchor occurrence may be detected via monitoring GPS signals. Thus, a LOI anchor may be detected when the user arrives at their bus station. The bus station may serve as a location-based context of the anchor occurrence. Upon arriving at the LOI (e.g., the bus station), and based on the location's context, a suggested action of checking when the user's bus is arriving via a bus schedule application may be provided to the user. As the user approaches their work office via the bus, another suggested action, such as but not limited to sending a message (e.g., SMS, email, or the like) to a co-worker may automatically be provided. Providing the suggested action may be location-based context dependent.

As still another example embodiment that suggest actions based on the detection of anchors, the user may have a flight scheduled for later in the day. The flight may serve as an anchor, or an expected anchor occurrence. Based on the expectation of the occurrence of the flight, one or more suggested actions may be provided, via a negative temporal offset, before the flight. For example, the flight may be scheduled for a 7:00 PM takeoff. Based on a -1:30 temporal offset (that was "learned" during the training of an anchor model), the user may be provided with a suggested action, at 5:30 PM. The suggested action may include requesting a rideshare service for transport to the airport, via a rideshare application installed on the user's device. As the user approaches the airport, a location-based context suggest action (e.g., using another application) of checking into their flight may be provided to the user. Another time-based context suggest action, with another negative temporal offset (e.g., providing a family member a message that the flight is about to take off) may be provided at 6:55 PM.

The anchor models may be trained, during a training period, via training data generated from the user interacting with the electronic device. At least a portion of the anchor-signifying signatures within the device's signals may be predetermined and/or pre-computed. At least a portion of the signatures may be indicative of contextual features of an anchor occurrence. Such contextual signals may also be predetermined and/or pre-computed. In other embodiments, the anchor-signifying signals and/or the contextual feature-signifying signals may be learned and/or determined, via training the model. For example, various deep learning (DL) methods may be employed to determine the signatures. Signatures within electronic signals that are indicative of an anchor occurrence may be referred to as anchor signatures and signatures that indicate contextual features of an anchor occurrence may be referred to as context signatures.

The training of an anchor model may be performed to be specific to the electronic device and/or the specific user. For example, the training of an anchor model may be performed using training data associated with a specific user and/or a specific electronic device, where the training data is not associated with other users and/or other devices. In some embodiments, the model may be trained using the computing resources (e.g., the processing power, storage, and/or memory) of the electronic device. The training data (e.g., electronic signals) may be generated by the specific device and from device-interactions associated with the specific user. The training data and/or the trained anchor model need not be provided to other devices. Thus, the training of the model need not employ computing resources, separate from the computing resources of the particular user's particular

electronic device. Furthermore, the user's privacy may be protected because the training data and/or the trained model need not be provided to another electronic other than the user's device.

In other embodiments, at least portions of the training of a model may be off-loaded to computing resources associated with other devices. In such embodiments, the user's privacy may still be guarded via encryption of the training data and/or trained model. Before being off-loaded, the training data generated by the user's electronic device may be sufficiently encrypted. For example, an encrypted-version of the trained model and/or training data may be stored via various user-accessible cloud-computing resources. The encryption key may be stored on the user's device, such that the trained model and/or the training data may be inaccessible to other users and/or other devices. Thus, even though the training data and/or the trained model may be "backed-up" via cloud services, while the user's privacy is still protected.

In the various embodiments, each type of anchor (e.g., waking up, entering a LOI, end of last calendar event, finishing a workout, and the like) may be trained separately. In some embodiments, an anchor model may be trained for multiple anchor types in parallel. In still other embodiments, multiple anchor types may be trained in series. A combination of parallel and serial training periods may be employed for multiple anchor types. For each anchor type, the training period of an anchor model may include a training data acquisition phase, as well as three learning phases. The training data acquisition phase includes acquiring a sufficient amount of training data for the anchor type. As explained below, the first learning phase of the training may be referred to as a filtering phase, the second learning phase may be referred to as a selecting and/or ranking phase, and the third learning phase may be referred to as a temporal offsetting phase.

Acquiring training data includes acquiring and/or monitoring training signals generated by the electronic device. The training signals include signatures indicating a statistically significant number of anchor occurrences and a statistically significant number of training actions, e.g., actions executed by the device during the data acquisition phase. The signals may also include signatures indicating contextual features, contextual conditions, or other contextual information associated with anchor occurrences. The contextual information for each anchor occurrence may be encoded in metadata and associated with and/or included in the data encoding the anchor occurrence. The actions may include user-initiated actions that are executed by the device, via the user interacting with the device during the training data acquisition phase. Each action in the training data may be classified as one or more actions types in a set of enabled action types that are executable by the device. Thus, the set of enabled action types may be a set of possible action types that are enabled and/or executable by the electronic device. For example, the set of enabled action types may include playing a podcast episode via a podcasting application, starting a workout provided by workout application, sending a message via a messenger application, requesting rideshare service via a ridesharing application, or the like.

After a sufficient amount of training data associated with an anchor type has been acquired via the electronic device during the data acquisition phase, the training of the anchor model for the anchor type may begin with the filtering phase. During the filtering phase, a set of candidate action types may be determined and/or generated. The set of candidate action types may be determined by filtering the set of

enabled action types. Because the filtering phase includes filtering the set of enabled action types, the set of candidate action types may be a subset of the set of enabled action types. The filtering of the set of enabled actions may include determining or generating statistical correlations, correspondences, and/or associations between the training events and the anchor occurrences of the training data. The filtering may be performed via the statistical correlations. For example, the correlations may be employed to determine which action types are most highly associated (or correlated) with the anchor type being trained. As described throughout, various methods may be employed to generate the correlations and filtering criteria, such as but not limited to determining class conditional probabilities or posterior probabilities conditioned on the anchor type. In some embodiments, the filtering criteria may include entropy-based filtering criteria, local maxima or minima-based filtering criteria, or the like.

During the ranking phase, the set of candidate actions types may be ranked, and at least portions of the candidate action types may be selected based on the rankings. The rankings and selection of the candidate action types may be based the contextual conditions or contextual features associated with the anchor occurrences. That is, the action types that the user is most likely to initiate or execute, in response to (or in expectation of) the occurrence of an anchor, conditioned on particular contextual conditions or contextual features are highly ranked and selected for. In some embodiments, a classification decision tree is constructed for each action type of the set of candidate action types. The nodes of a classification decision tree for an action type may indicate one or more possible contextual conditions or contextual features of the anchor occurrences that are associated with or conditioned on the action type and including the corresponding contextual conditions or features. The nodes may additionally and/or alternatively be conditioned on the context of the action type of the tree. For example, a node may be conditioned on the number of times, or frequency, that the user has initiated the action type in a given amount of time (e.g., the previous hour, 12 hours, day, week, or the like). To determine a probability or likelihood that the user initiates a particular action type, conditioned on an anchor occurrence types and on one or more contextual conditions, the nodes of the corresponding classification decision tree may be traversed and the probability may be calculated based as described throughout. The context-dependent classification decision trees may be employed to map multiple action types to a single anchor type based on the context of anchor occurrences. The multiple action types may be differentiated for a single anchor type via the different contexts of anchor occurrences. For instance, each of a first action type (e.g., playing an episode of a podcast) and a second action type (e.g., starting a workout) may be mapped to the single anchor type of waking up. Thus, the ranking and selecting of action types may be based on the classification decision trees.

During the temporal offset phase of training for an anchor type, a temporal offset may be determined for one or more action types that may be suggested in response to (or in expectation of) an occurrence of the anchor type. To determine the temporal offset, one or more temporal distributions may be generated from the training data. The temporal distribution may include a histogram of the temporal difference between anchor occurrences and user-initiated actions. In some embodiments, a separate temporal distribution may be generated for an action type. The temporal offset may be determined via one or more statistical characteristics of the

distribution, e.g., the mean of the distribution, the median of the distribution, the mode of the distribution, a variance of the distribution, or the like. For example, when training the model for the anchor type of waking up, the training data may be employed to generate a temporal histogram for the action type of playing a podcast. The x-axis of the histogram may include bins for the temporal value between the occurrence of the user waking up and the user initiating playing an episode of one or more podcasts. The y-axis of the histogram may indicate the number of times (or frequency) that the user initiates playing a podcast at the temporal value and after the occurrence of the user waking up. In at least some embodiments, a separate temporal distribution may be generated for each action type and for each contextual condition, or combination of contextual conditions. For instance, separate histograms (similar to the one described above) may be generated for wake up anchors that occurred in the morning on a weekday (or work day for the user) and wake up anchors that occurred in the morning on a weekend day.

Once an anchor model is adequately trained for one or more anchor types, the trained model may be deployed by the electronic device. Once the trained model is deployed, signals generated by the device may be monitored and/or analyzed. An occurrence of an anchor type that the model was trained for may be detected and/or identified. As noted above, in addition to indicating the anchor occurrence, the signals may indicate one or more contextual conditions of the anchor occurrence. The contextual conditions may be encoded in metadata associated with the anchor occurrence. In response to and/or in accordance with the detected anchor occurrence, the model may be employed to determine and provide one or more suggested actions and a temporal offset for each of the one or more suggested actions. The one or more suggested actions, as well as the associated temporal offsets, may be based on the contextual conditions of the anchor occurrence. At the temporal offset for suggested action, as measured from the anchor occurrence, the electronic device may provide the user with a notification indicating the suggested action. The user may be enabled to select the notification, and upon selection by the user, the electronic device may execute the selected suggested action. As discussed throughout, the model may be employed to detect anchor occurrences that are expected to occur in the future (e.g., a flight later in the day that, for which, the user is scheduled to take). Via a negative temporal offset, a suggested action (e.g., requesting a rideshare to the airport) may be provided to the user prior to the scheduled flight.

In a non-limiting embodiment, an anchor type associated with the user may be trained after a sufficient amount of training data has been acquired via the user interacting the device upon waking up. The training data may include signals generated by the device. A portion of the signals may encode multiple occurrences of one or more anchor types and another portion of the signals may encode actions initiated by the user. Anchor occurrences encoded in the training data may be referred to as training anchor occurrences and actions encoded in the training data may be referred to as training actions. The user may have initiated at least a portion of the training actions in response to training anchor occurrences. The signals may additionally encode metadata indicating contextual conditions of at least some of the anchor occurrences. Each of the training actions may be classified as one or more of the action types that are enabled via the electronic device. More specifically, each training action may be classified as one of the action types in the set of enabled action types.

In some embodiments, an anchor model may be trained for multiple anchor types in parallel. In other embodiments, the model may be trained for anchor types in a serial progression. After a sufficient amount of training data associated with a particular anchor type has been collected, the filtering phase for that particular anchor type may begin. In the various embodiments, a sufficient amount of training data for a particular anchor type may include an amount of training data that encodes a statistically significant number of anchor occurrences of the particular anchor type and a statistically significant number co-occurring actions of one or more action types. As used herein, a statistically significant number of anchor occurrences and/or actions may refer to a sufficient number of anchor occurrences and/or actions, such that one or more statistical metrics (e.g., error bars, confidence levels, or the like) may be quantified to within one or more thresholds. Note that when associating multiple actions to a single anchor type, where the multiple actions are differentiated via the contextual conditions of the anchor occurrence, a larger volume of training data may be required.

During the filtering phase, various candidate action types are statistically correlated with the particular anchor type being trained, based on anchor occurrences and user-initiated actions encoded in the training data (e.g., co-occurrences of anchors and actions within the training data). More specifically, a large, strong, or high correlation between an anchor type and a particular action type indicate frequent co-occurrences, within the training data, of anchors and actions that are classified as the particular action type. A small, low, or weak correlation between an anchor type and a particular action type indicate infrequent, or non-existent, co-occurrences, within the training data, of anchor and actions that are classified as the particular action type. As discussed throughout, the co-occurrences in the training data need not be temporally simultaneous, but may be separated by temporal ranges, characterized via one or more temporal offsets. The correlations may be conditioned on the contextual conditions or features of anchor occurrences. The correlations are employed to filter the electronic device's set of enabled actions types to generate or determine a set of candidate action types. Various methodologies may be employed to determine the correlations between the anchor types and action types. For example, various supervised or unsupervised machine learning (ML) methods may be employed to identify patterns (e.g., co-occurrences of anchors and actions) within the signals of the training data.

In the following discussion, the exemplary but non-limiting anchor type of waking up, is employed to discuss the training of an anchor type. However, this discussion is non-limiting, and the discussion may be applicable to any number of other anchor types. Once a sufficient amount of training data associated with the waking up anchor type, which encodes a statistically significant number of occurrences of the user waking up and a statistically significant number of actions initiated upon the user waking up, the filtering phase of training the model for the waking up anchor type may be performed. The filtering phase includes generating, determining, and/or identifying a set of candidate action types that is a subset of the set of enabled actions. In various embodiments, correlations between the anchor occurrences and the action types are determined via the training data. More specifically, a statistical correlation between the anchor type (e.g., waking up) and each enabled action type may be determined via the training data. Such a statistical correlation between the anchor type and a particular action type may indicate a frequency of co-occur-

rence of an occurrence of the anchor and the user initiating the particular action type. The correlations may be determined via one or more probability distributions extractable from the training data. For example, one or more class conditional probability distributions and/or one or more posterior probability distributions may be extracted from the training data. A probability distributions may be conditions on the co-occurrence of the anchor and one or more actions. The correlations may be determined via the one or more probability distributions.

Note that the co-occurrence of anchors and actions need not be simultaneous, but may be separated by a range of temporal values. As discussed below, the third learning phase (i.e., the temporal offset phase) of the training is directed towards determining a temporal offset based on the range of temporal values between an anchor occurrence and the initiation and/or execution of an action. For example, if the user frequently plays a particular podcast shortly after waking up (e.g., typically between 10 minutes and 20 minutes), a relatively large or strong correlation between the anchor and the action type (e.g., playing the particular podcast) is determined. If the user rarely (or never) views a bus schedule shortly after waking up, a relatively small or weak (or zero) correlation between the anchor and the action type of launching a bus schedule application is determined. Enabled action types that are more strongly correlated with the anchor type (e.g., playing a podcast) are included in the set of candidate action types. Enabled action types that are less correlated with the anchor type (e.g., launching the bus schedule application) are not included in the set of candidate action types. In some embodiments, the set of enabled action types is filtered to determine the set of candidate actions types. Enabled action types that "pass" through the filter are included in the set of candidate action types.

As noted above, various supervised or unsupervised ML methods may be employed to determine the correlations. In general, ML methods may be employed to identify patterns of anchor and event co-occurrences in the training data that is associated with the anchor type. The correlations that are employed to identify the set of candidate action types may be determined based on the patterns (e.g., co-occurrences) of anchors and events in the training data. One non-limiting example of an unsupervised ML method applicable to the various embodiments is associated rule learning. In some embodiments, the correlations may be generated by applying one or more association rule learning methods and/or algorithms, such as but not limited to apriori learning algorithm, on the training data. In general, an association rule learning method is employed to analyze co-occurrences of the anchor and action types within the training data. Apriori rule learning, which is directed towards learning association rules from co-occurring events (e.g., anchors and actions) is one such algorithm that may be applied to the training data to determine the correlation between anchor occurrences and action types.

In other embodiments, a neural network (NN) architecture that implements an attention mechanism may be employed to determine the correlations between anchor occurrences and action types, within the training data. When implemented within a NN, attention mechanism increase the ability of the NN to discover and/or learn patterns within training data. In general, the analysis implemented by the NN is directed to portions of training data, where the patterns are more likely to be discovered. That is, the "attention" of the NN is directed towards more appropriate portions of the training data, rather than applying the same amount of attention to all portions of the training data. For

example, with attention applied towards “long term patterns”, patterns associated with “long term” dependencies within the training data may be more readily discovered than with NN that don’t implement attention mechanisms. Accordingly, when attention mechanisms are employed, the co-occurrences of anchors and actions may be more readily discovered. In still other embodiments, other methods that analyze patterns within data may be employed to determine the correlations between anchor occurrences and user actions. In still other embodiments, a SparseMax algorithm may be employed applied to the training data. That is, a deep learning NN may be employed to identify patterns in the training data, where an activation function of at least one of the layers of the NN is a SparseMax function. A SparseMax activation function has the general property that the output vector is a relatively sparse vector. Thus, the strongest correlations are identified in the training data. In some embodiments, a canonical correlation analysis (CCA) may be employed to determine the correlations in the training data.

The filtering of the set of enabled actions types to generate the set of candidate action types may be based on a threshold applied to the correlation determined for each action type. In other embodiments, the filtering may be based on a relative and/or absolute number of action types to be included in the set of candidate action types. In one non-limiting example, the filtering phase of the training of the model for the waking up anchor type may filter the set of enabled action types, such that the set of candidate action types includes five action types: playing a podcast, playing a song in the user’s music library, viewing recent photos via a photo application, checking the weather forecast via a weather application, or playing a game installed on the device. Upon the identification of the set of candidate actions types, via the filtering phase, the ranking phase of the training period may begin.

During the ranking phase, the set of candidate actions types may be ranked, and at least portions of the candidate action types may be selected based on the rankings. The rankings and selection of the candidate action types may be based on the contextual conditions or contextual features associated with the anchor occurrences encoded in the training data. That is, the action types, of the action types in the set of candidate action types, that the user is most likely to initiate or execute, in response to (or in expectation of) the occurrence of an anchor are ranked highly and selected for in the ranking phase. The selected action types may be conditioned on particular contextual conditions or contextual features of the anchor occurrences. In some embodiments, a ranking for a particular action type of the set of candidate action types may be determined based on training actions that are classified as the particular action type. The ranking of the particular action type may be further based on the contextual condition indicated by metadata associated with the training anchor occurrences that are correlated with the training actions that are classified as the particular action type.

In the ranking phase, a decision tree learning method may be employed to learn to distinguish amongst the contextual conditions of anchor occurrences and determine a suggested action based on the contextual conditions. For instance, a classification decision tree algorithm may be employed, such as but not limited to iterative dichotomiser 3 (ID3) algorithm may be applied to the relevant portion of the training data to generate a classification decision tree for each action type (or each of the highest ranked action types) of the set of candidate action types. A classification decision tree for a particular action type indicates which contextual

conditions of the anchor occurrence to associate with the action type, and which contextual conditions to not form an association with the anchor occurrence. In this way, the anchor model is trained to distinguish amongst the contextual conditions of the anchor occurrence and tailor the one or more suggested actions to the anchor occurrence based on the contextual conditions. The nodes indicate one or more contextual conditions and whether the one or more contextual conditions “classify” the anchor occurrence (with the contextual conditions) as being associated with the action type.

FIG. 8A illustrates a classification decision tree 800 for the action type of playing a particular podcast and the anchor type of the user waking up, according to various examples. Classification decision tree 800 was generated via an application of a classification decision tree algorithm on the training data. It is understood that classification decision tree 800 is an exemplary embodiment only, and the embodiments may vary. A classification decision tree generally subdivides the training data into smaller and smaller subsets of training data. Each subset is conditioned on the contextual conditions associated with the nodes. Each node indicates whether the node (and thus the associated subset of training data) is in accordance with the contextual conditions encountered while traversing the tree. The upper node 802 indicates that the classification decision tree 800 is for the classification of “play a particular podcast.” Tree 800 is traversed, starting root node 804, by traversing the contextual conditions associated with an anchor occurrence. In this discussion, node 804 is referred to as the root node because it indicates the first “decision” in the tree.

In a classification tree (e.g., tree 800), the root node (e.g., node 804) is typically conditioned on the contextual condition that is most strongly predictive of the class (e.g., playing the particular podcast). In tree 800, root node 804 is conditioned on whether the anchor (e.g., waking up) occurred on a weekday or weekend. Root node indicates that the training data included 44 samples of the anchor occurrences, and that 23 of the 44 samples occurred on a weekday and the remaining 13 samples occurred on a weekend (e.g., values=[31, 13]). Nodes in decision trees may be characterized by a Gini metric, Gini coefficient, Gini index, which indicates the inequality of the underlying distribution, as conditioned on the contextual condition of the node. In some embodiments, the Gini metric may vary between 0.0 and 1.0 (or some otherwise normalized range). Root node 804 is associated with a Gini metric of 0.499. If all the anchor occurrences happened on a weekday, the Gini metric may be closer to 1.0, while if all the anchor occurrences happened on a weekend, the Gini metric would be closer to 0.0.

Starting from the root node 804, if the anchor condition occurred on a weekday, then traversing flows to node 806. If the anchor occurrence occurred on a weekend, then traversing flows to node 808. Node 806 is conditioned on the contextual condition of playing the podcast at least 8 times in the previous 10 days of the anchor occurrence. Node 808 is conditioned on the contextual condition of the anchor occurrence occurring on a Saturday, rather than a Sunday. Nodes 806 and 808 provide the sample numbers, values of the sample numbers, and Gini metrics associated with the distribution. Nodes 810, 812, and 814 are leaf nodes and indicate the various distributions, sample numbers of the distribution, and associated Gini metrics. The class indicator at each node indicates whether an anchor occurrence with the associated contextual conditions should be classified as the class (e.g., play the particular podcast). Tree 800 indicates that the action type of playing the particular podcast

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should be suggested when the user wakes up on a weekday and if the user has been playing the podcast frequently in the last 10 days (e.g., greater than 7 times in the last 10 days).

FIG. 8B illustrates a classification decision tree **820** for the action type of playing a particular song and the anchor type of the user waking up, according to various examples. Tree **820** is similar to tree **800** of FIG. 8A, except the tree **820** was generated for the action type of playing a particular song from the user's music library upon waking up. Tree **820** indicates that a suggested action of playing the particular song when the user has listened to the song (upon waking up) greater than 6 times in the last 10 days, but less than 23 times in the last 30 days. Thus, the embodiments may provide the suggested action of playing the particular song when the user has listened to the song relatively frequently in the last 10 days, and the user has not listened to it too frequently in the last 30 days. The classification decision tree algorithm may be employed to generate such a classification decision tree for each action type in the set of candidate action types. In some embodiments, as indicated by tree **800** and tree **820**, if the contextual conditions of the anchor occurrence are such that the anchor occurrence may be classified as both playing the particular podcast and playing the particular song, both action types may be suggested to the user. Note that each action type may have a separate temporal offset. Thus, the suggested action types may be provided at separate temporal offsets. For example, if the temporal offset for playing the podcast is 30 minutes and the temporal offset for playing the particular song is 15 minutes, the embodiments may suggest playing the song 15 minutes and waking up and suggest playing the podcast 30 minutes after waking up.

As indicated above, the temporal offset phase of the training is directed towards determining a temporal offset for the suggested action types. In some embodiments, one or more of the candidate action types may be selected for determining the temporal offset for the candidate action type. In some embodiments, selecting the action types may be based on the ranking of the candidate action types and/or the classification decision trees generated for the candidate action types. The portion of the training data that is associated with the selected action type is employed to generate a temporal distribution of the time when the user initiated the selected action type, in response to the anchor occurrence. The temporal distribution for that action type may be employed to determine the temporal offset for that action type. As noted throughout, the temporal offset for an action type may be employed to determine the point of time (after or before) the occurrence of an anchor that the suggested action is provided to the user.

In various embodiments, a histogram may be generated of the temporal distribution generated for the selected candidate action type. The temporal offset may be determined via the histogram. FIG. 9 illustrates a histogram **900** for the temporal distribution of the action type of playing a particular podcast, in conjunction with a waking up anchor, according to various examples. Histogram **900** is provided only as an example, and the embodiments may vary in the construction of histograms for the various action types. Histogram **900** was generated from the training data associated with the anchor type waking up and training actions (e.g., the user initiating the playing of the podcast) that the user initiated in response to the anchor occurrence. The x-axis indicates the time (in seconds) between the anchor occurrence and the user initiating playing the particular podcast. In various embodiments, the x-axis is subdivided into a plurality of bins. The y-axis indicates the number of

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times (within the training data) that the action type was initiated by the user within the time indicated by the x-axis value of the bin. Thus, the y-axis value of the various bins indicate the temporal distribution for the action type and anchor occurrence. The temporal offset may be determined via a statistical metric of the temporal distribution. For example, in some non-limiting embodiments, the temporal offset for playing the particular podcast may be determined via at least one of the mean or median of the temporal distribution shown in histogram **900**. As seen in FIG. 9, most of the times that the user initiated playing the podcast, upon waking up, were within 700 seconds. Thus, in this non-limiting example, the temporal offset for the action type of playing the particular podcast, upon waking up, may be set to approximately 10 minutes.

Although not shown, another temporal distribution (and histogram) may be generated for the action type of playing a particular song may be generated and a separate temporal offset may be determined. In some embodiments, if enough statistics are included in the training data, a separate temporal distribution and/or histogram may be generated for each of the contextual conditions of the anchor type. For example, separate histograms may be generated for anchor occurrences where the user wakes up on a weekday and anchor occurrences where the user wakes up on a weekend.

Upon completion of the filtering, ranking, and temporal offset phases of the training, the anchor model may be updated to provide suggested actions, as learned via the training. Once updated, the anchor model may be deployed to the device to begin providing suggested actions, in response (or anticipation) of anchor occurrences. In various embodiments, once deployed at the electronic device, the model may be iteratively updated, periodically or from time to time, based on user feedback. That is, the model may be iteratively updated, based on the user's engagement with the suggested actions. When the user positively engages with a particular suggested action (e.g., the user selects the particular suggested action to initiate its execution), the model is updated to reinforce or increase the probability that the model provides the suggested action in response to future anchor occurrences. When the user does not engage (or negatively engages) with the particular suggested action (e.g., the user does not select the particular suggested action to initiate its execution), the model is updated to decrease the probability that the model provides the particular suggested action in response to future anchor occurrences. In some embodiments, one or more reinforcement learning methods may be employed to iteratively update the anchor models, based on user engagement with the suggested actions or other user feedback. The reward function employed in such reinforcement learning embodiments may be greater when the user selects suggested actions, than when the user does not select suggested actions. Thus, the anchor models may be iteratively updated based on the user's response to the suggested actions.

FIG. 10 illustrates process **1000** for training a predictive model that enables providing suggested actions in response to an occurrence of an anchor event, according to various examples. Process **1000** is performed, for example, using one or more electronic devices implementing a digital assistant. In some examples, process **1000** is performed using a client-server system (e.g., system **100**), and the blocks of process **1000** are divided up in any manner between the server (e.g., DA server **106**) and a client device. In other examples, the blocks of process **1000** are divided up between the server and multiple client devices (e.g., a mobile phone and a smart watch). Thus, while portions of

process **1000** are described herein as being performed by particular devices of a client-server system, it will be appreciated that process **1000** is not so limited. In other examples, process **1000** is performed using only a client device (e.g., user device **104**) or only multiple client devices. In process **1000**, some blocks are, optionally, combined, the order of some blocks is, optionally, changed, and some blocks are, optionally, omitted. In some examples, additional steps may be performed in combination with the process **1000**. The operations described above with reference to FIG. **10** are optionally implemented by components depicted in FIGS. **1-4**, **6A-6B**, and **7A-7C**. It would be clear to a person having ordinary skill in the art how other processes are implemented based on the components depicted in FIGS. **1-4**, **6A-6B**, and **7A-7C**.

Process **1000** begins at block **1002**, where training data is acquired for training a predictive model, such as but not limited to an anchor model. Once trained, the anchor model is employed to provide suggested actions, in response to (or in anticipation of) the occurrence of an anchor and/or anchor event. The anchor may be an event that indicates one or more behaviors of a user of an electronic device. Accordingly, in the foregoing discussion, the anchor may be referred to as an event, and an occurrence of the anchor may be referred to as an event occurrence. The training data includes signals generated by the electronic device that enables a set of enabled actions. The signals may be referred to as training signals. At block **1004**, the signals may be analyzed to identify event occurrences, associated contextual conditions, and actions, initiated by the user of the device, in response to (or in anticipation of) the event occurrences. The event occurrences, the associated contextual conditions, and the training actions may be encoded in the signals. The contextual conditions may be included in metadata associated with the event occurrences. Each of the training actions may be classified as one or more of the action types of the set of enabled actions.

At block **1006**, statistical correlations between the event occurrences and the action types based on the training signals. In some embodiments, the correlations may be determined between the event occurrences and at least a portion of the training actions. The statistical correlations may be determined via any of the various embodiments discussed herein. For example, the correlations may be determined via one or more supervised or unsupervised machine learning (ML) methods and/or algorithms. At block **1008**, a set of candidate action types may be determined based on the correlations between the event occurrences and the action types. The set of candidate action types may be a subset of the set of enabled actions. In some embodiments, the set of enabled action types may be filtered to generate and/or identify the set of candidate actions. Any of the various filters and/or filtering methods discussed herein may be employed to filter the set of enabled action types. In some embodiments, blocks **1006** and **1008** may include the filtering phase of training the predictive model.

At block **1010**, a ranking for each of the action types of the set of candidate action types may be determined and/or generated. The rankings may be based on a portion of the training actions that are classified as the action type being ranked, the contextual conditions of the event occurrences correlated with the portion of the training actions, and the correlations between the portion of the training actions and the associated event occurrences. At block **1012**, one or more actions types may be selected from the set of candidate action types. The selection of the action types may be based on the rankings of the candidate action types. In various

embodiments, blocks **1010** and **1012** may include the ranking phase of training the predictive model. Thus, at blocks **1010** and/or **1012**, one or more classification decision trees, as discussed throughout, may be generated for each of the ranked action types. The classification decision trees may be based on the contextual conditions of the associated event occurrences. The selection of the action types may be based on the classification decision trees. Non-limiting examples of such classification decision trees are discussed in conjunction with at least FIGS. **8A-8B**.

At block **1014**, one or more temporal distributions may be generated for each action type selected at block **1012**. The temporal distribution for a particular action type may indicate a temporal difference between event occurrences and associated training actions classified as the selected action type. A separate temporal distribution may be generated for each action type selected at block **1012**. A separate temporal distribution may be generated for each of the different contextual conditions of the event occurrences. A histogram may be generated for each of the temporal distributions. A non-limiting example of a histogram is discussed in conjunction with at least FIG. **9**. At block **1016**, a temporal offset is determined for each of the one or more action types selected at block **1012**. The temporal offset may be determined based on the temporal distribution and/or histogram generated for the selected action type. In various embodiments, blocks **1014** and **1016** may include the temporal offset phase of training the predictive model.

At block **1018**, the predictive model may be updated, in accordance to the training of process **1000**. In some embodiments, the predictive model may be updated to provide the user, at the electronic device and in response to the occurrence of another event, one or more suggested actions in accordance to the event type of the other event occurrence, one or more of the selected action types, the one or more temporal offsets for the one or more selected action types, and the contextual conditions of the other event occurrence. That is, the trained event or anchor model may be deployed at the electronic device. Various embodiments for deploying a trained predictive model on an electronic device are described in conjunction with at least FIG. **11**.

FIG. **11** illustrates process **1100** for deploying a trained predictive model at an electronic device, according to various examples. Similar to process **1000**, process **1100** may be performed, for example, using one or more electronic devices implementing a digital assistant. Process **1100** may be performed by any device, and/or combination of devices, similar to the devices and/or combination of devices that process **1000** may be performed by.

Process **1100** starts, at block **1102**, where an event (or anchor) occurrence, as well as contextual conditions, are detected via monitoring and analysis of signals generated by the electronic device. At block **1104**, a trained predictive model is employed to determine one or more suggested actions and one or more temporal offsets based on the detected event occurrence and the associated contextual conditions. In some embodiments, the one or more suggested actions and the one or more associated temporal offsets are received, via the predictive model, at the electronic device. In some embodiments, the predictive model may be trained by one or more embodiments described at least in conjunction with FIG. **10**. At block **1106**, the electronic device is employed to provide, to the user, the one or more suggested actions, in accordance with the associated one or more temporal offsets. At block **1108**, an in response to receiving an indication and/or selection of a provided

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suggested action, initiate the execution of the selected suggested action, via the electronic device.

In various embodiments, once deployed at the electronic device, the model may be iteratively updated, periodically or from time to time, based on user feedback. That is, the model may be iteratively updated, based on the user's engagement with the suggested actions. Such iteratively updating is not shown in FIG. 11. However, it is understood that process 1100 may be modified to include iteratively updating the predictive model based on user engagement and/or feedback in response to the suggested actions. When the user positively engages with a particular suggested action (e.g., the user selects the particular suggested action to initiate its execution), the model is updated to reinforce or increase the probability that the model provides the suggested action in response to future anchor occurrences. When the user does not engage (or negatively engages) with the particular suggested action (e.g., the user does not select the particular suggested action to initiate its execution), the model is updated to decrease the probability that the model provides the particular suggested action in response to future anchor occurrences. In some embodiments, one or more reinforcement learning methods may be employed to iteratively update the anchor models, based on user engagement with the suggested actions or other user feedback. The reward function employed in such reinforcement learning embodiments may be greater when the user selects suggested actions, than when the user does not select suggested actions. Thus, the anchor models may be iteratively updated based on the user's response to the suggested actions.

In accordance with some implementations, a computer-readable storage medium (e.g., a non-transitory computer readable storage medium) is provided, the computer-readable storage medium storing one or more programs for execution by one or more processors of an electronic device, the one or more programs including instructions for performing any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises means for performing any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises a processing unit configured to perform any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises one or more processors and memory storing one or more programs for execution by the one or more processors, the one or more programs including instructions for performing any of the methods or processes described herein.

Example methods are disclosed herein. One example method may be for training a predictive model (e.g., and anchor model) for an anchor (or event) that indicates a behavior user of an electronic device enabled to execute each action type of a set of enabled action types. The electronic device may have one or more processors and a memory. The electronic device may perform one or more actions and or operations. The method may include, at the electronic device and based on a plurality of signals generated by the electronic device, detecting a plurality of training actions and a plurality of event occurrences of the event. Each of the plurality of event occurrences may be associated with metadata indicating a contextual condition of the event occurrence. Each of the plurality of training actions may be initiated via the user interacting with the electronic device

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and is classified as an action type of the set of enabled action types. A set of candidate action types may be determined. Determining the set of candidate action types may be based on a plurality of correlations between each of the plurality of training actions and each of the plurality of event occurrences. The set of candidate action types may be a subset of the set of enabled action types. A ranking for each action type of the set of candidate action types may be determined. Determining the ranking for an action type may be based on a portion of the plurality of training actions that are classified as the action type. Determining the ranking of the action type may be further based on the one or more contextual conditions indicated by the metadata associated with a portion of the plurality of events occurrences that are correlated, via the plurality of correlations, with the portion of the plurality of training actions. In some embodiments, a first action type of the set of candidate action types may be selected based on the determined ranking for each action type of the set of candidate action types. A first portion of the plurality of training actions and a first portion of the plurality of event occurrences may be selected. Each of the first portion of the plurality of training actions may be classified as the first action type. Each of the first portion of the plurality of event occurrences may be correlated, via the plurality of correlations, with at least one of the first portion of the plurality of training actions. A temporal offset for the first action type may be determined based on a temporal distribution of the first of the plurality of training actions, with respect to the first portion of the plurality of event occurrences. In some embodiments, the predictive model may be updated to generate, in response to another occurrence of the event, a suggested action. The provided suggested action is in accordance with the first action type and the temporal offset of the first action type.

The example method may include employing an association rule learning algorithm to determine the plurality of correlations between each of the plurality of training actions and each of the plurality of event occurrences. In some embodiments, a canonical correlation analysis may be employed to determine the plurality of correlations between each of the plurality of training actions and each of the plurality of event occurrences. In still other embodiments, attention within a neural network may be employed to determine the plurality of correlations between each of the plurality of training actions and each of the plurality of event occurrences. In or more other embodiments, a SparseMax algorithm may be employed to determine the plurality of correlations between each of the plurality of training actions and each of the plurality of event occurrences.

Determining the set of candidate action types may include applying a filter on the set of enabled action types and the filter is based on the plurality of correlations between each of the plurality of training actions and each of the plurality of event occurrences. The filter may be an entropy filter. In other embodiments, the filter is a class-conditional probability filter. In still other embodiments, the filter is one of a local maxima filter or a local minima filter. In some embodiments, the filter may include filtering on the number of times that the candidate action type has been initiated in response to (or in expectation of) the occurrence of the anchor in a previous amount of time (e.g., how many times the user has initiated the candidate action in the previous 28 days). In other embodiments, the filter may filter on the posterior probability associated with the candidate action type. In still other embodiments, the filter may filter on the class conditional probability associated with the candidate action type (e.g., probability of the anchor conditioned on the candidate action

type). The filter may be a combination of multiple filters. For instance, the filter may be a combination of a filter on the number of times (or frequency) that the user has initiated the candidate action type in response to the anchor, a filter on the posterior probability of the candidate action type, and a filter on the class conditional probability of the anchor conditioned on the candidate action type.

The example method may further include, for each action type of the set of candidate action types, generating a classification decision tree. The nodes of the classification decision tree may be conditioned on contextual conditions associated with the portion of the plurality of events occurrences that are correlated with the portion of the plurality of training actions that are classified as the action type. The classification decision tree may classify other event occurrences with the contextual conditions as associated with the action type. In some embodiments, selecting the first action type of the set of candidate action types may be further based on the classification decision tree for the first action type. The classification decision tree for each action type of the set of candidate action types may be employed to associate a plurality of action types of the set of candidate action types with the event. The plurality of action types may be differentiated by the contextual conditions of the plurality of event occurrences.

The example method may further include generating a first histogram for the first action type based on the temporal distribution of the first of the plurality of training actions. The temporal offset for the first action type may be determined based on the first histogram. The method may further include generating a second histogram for the first action type based on a second temporal distribution of the plurality of training actions. The temporal distribution may be conditioned on a first set of contextual conditions and the second temporal distribution may be conditioned on a second set of contextual conditions. A second temporal offset for the first action type may be determined based on the second temporal distribution of the plurality of training actions. The predictive model may be updated to generate, in response to a first occurrence of the event that is condition on the first set of contextual conditions, the suggested action, in accordance with the first action type and the temporal offset of the first action type. The predictive model may be further updated to generate, in response to a second occurrence of the event that is condition on the second set of contextual conditions, a second suggested action, in accordance with the first action type and the second temporal offset of the first action type. The temporal offset for the first action type may be further based on at least one of a mean or a median of the temporal distribution of the plurality of training actions. The predictive model may be stored and/or encrypted on the electronic device. The temporal offset may be less than zero or greater than zero. Generating the suggested action may include providing a notification, at the electronic device, at a time that is separated from the other event occurrence by the temporal offset.

Another example method may be for employing a predictive model for an event that indicates a behavior of a user of an electronic device. The other method may include, based on one or more signals generated by the electronic device, detecting an event occurrence of the event. The event occurrence may be associated with metadata indicating a contextual condition of the event occurrence. In accordance with the event occurrence, a suggested action and a temporal offset may be received from the predictive model. The suggested action may be provided to the user within the temporal offset from the event occurrence.

In the method, the event may include a termination of a do not disturb mode of the electronic device and the behavior of the user includes the user waking up. Detecting the event occurrence may include determining, based on a positioning signal of the one or more signals, that the electronic device is positioned at a particular location. The particular location may include at least one of an airport, a bus terminal, or a train terminal. In other embodiments, the particular location includes at least one of a residence of the user, a workplace of the user, a recreational location, or a fitness center. The event occurrence may include an occurrence of a calendar event of an electronic calendar associated with the user. Providing the suggested action may include providing the suggested action after the event occurrence, and in accordance to the temporal offset. In other embodiments, providing the suggested action includes providing the suggested action before the event occurrence, and in accordance to the temporal offset. In still other embodiments, providing the suggested action includes employing a display of the electronic device to providing a pop-up notification indicating the suggested action.

The method may further include receiving a user selection, at the electronic device, of the suggested action. The electronic device may be employed to automatically execute the suggested action. The predictive model may be trained based on a plurality of training signals generated by the electronic device. One or more machine learning (ML) methods may be employed to train the predictive model, via the training signals. In some embodiments, the method further includes, in accordance with the predictive model, providing a second suggested action to the user within a second temporal offset from the event occurrence. The suggested action may be based on at least an event type of the event occurrence. The suggested action may be further based on at least one or more contextual conditions of the event occurrence. The one or more contextual conditions of the event occurrence may include at least one of a day of a week, a time of a day, or a location of the electronic device. The suggested action may include at least one of playing content on the electronic device, launching an application installed on the electronic device, or sending a message to a first contact within a plurality of contacts for the user. The application installed on the electronic device may include at least one of a rideshare application, a physical fitness application, or a food delivery application.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in the art are thereby enabled to best utilize the techniques and various embodiments with various modifications as are suited to the particular use contemplated.

Although the disclosure and examples have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims.

As described above, one aspect of the present technology is the gathering and use of data available from various sources provide suggested actions in response detecting events and/or anchors. The present disclosure contemplates

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that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, twitter IDs, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other identifying or personal information.

The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, the personal information data can be used to provide suggested actions in response to detected events and/or anchors to the user. Accordingly, use of such personal information data enables users to calculated control of the delivered content. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.

The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, in the case of providing suggested actions, the present technology can be configured to allow users to select to "opt in" or "opt out" of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide training

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data and/or trained models to other devices and/or other users. In yet another example, users can select to limit the length of time mood-associated data is maintained or entirely prohibit the development of a baseline mood profile. In addition to providing "opt in" and "opt out" options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data at a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data.

What is claimed is:

1. An electronic device, comprising:

one or more processors;

a memory storing one or more programs, the one or more programs including instructions, which when executed by the one or more processors of the electronic device, cause the electronic device to:

based on one or more signals generated by the electronic device, detect an event occurrence of the event, wherein the event occurrence is associated with meta-data indicating a contextual condition of the event occurrence, wherein detecting the event occurrence includes detecting a launching of an application;

in accordance with the event occurrence, receive a first suggested action, a second suggested action; and

a first temporal offset, a second temporal offset, and a third temporal offset from the predictive model, wherein the second temporal offset is different from the third temporal offset; and

provide the first suggested action to the user within the first temporal offset from the event occurrence and detecting an input from a user;

in response to detecting the input from the user:

in accordance with a determination that the input from the user corresponds to the first suggested action, provide the second suggested action to the user within the second temporal offset; and

in accordance with a determination that the input from the user does not correspond to the first suggested action, provide the second suggested action to the user within the third temporal offset.

2. The electronic device of claim 1, wherein the event includes a termination of a do not disturb mode of the electronic device and a behavior of the user includes the user waking up.

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3. The electronic device of claim 1, wherein detecting the event occurrence includes determining, based on a positioning signal of the one or more signals, that the electronic device is positioned at a particular location.

4. The electronic device of claim 3, wherein the particular location includes at least one of an airport, a bus terminal, or a train terminal.

5. The electronic device of claim 3, wherein the particular location includes at least one of a residence of the user, a workplace of the user, a recreational location, or a fitness center.

6. The electronic device of claim 1, wherein the event occurrence includes an occurrence of a calendar event of an electronic calendar associated with the user.

7. The electronic device of claim 1, wherein providing the first suggested action includes providing the first suggested action after the event occurrence, and in accordance to the temporal offset.

8. The electronic device of claim 1, wherein providing the first suggested action includes providing the first suggested action before the event occurrence, and in accordance to the temporal offset.

9. The electronic device of claim 1, wherein providing the first suggested action includes employing a display of the electronic device to providing a pop-up notification indicating the first suggested action.

10. The electronic device of claim 1, wherein the one or more programs further including instructions, which when executed by the one or more processors of the electronic device, cause the electronic device to:

employ the electronic device to execute the first suggested action.

11. The electronic device of claim 1, wherein the predictive model was trained based on a plurality of training signals generated by the electronic device.

12. The electronic device of claim 1, wherein one or more machine learning (ML) methods were employed to train the predictive model.

13. The electronic device of claim 1, wherein the first suggested action is based on at least an event type of the event occurrence.

14. The electronic device of claim 1, wherein the first suggested action is based on at least one or more contextual conditions of the event occurrence.

15. The electronic device of claim 14, wherein the one or more contextual conditions of the event occurrence include at least one of a day of a week, a time of a day, or a location of the electronic device.

16. The electronic device of claim 1, wherein the first suggested action includes at least one of playing content on the electronic device, launching the application installed on the electronic device, or sending a message to a first contact within a plurality of contacts for the user.

17. The electronic device of claim 16, wherein the application installed on the electronic device includes at least one of a rideshare application, a physical fitness application, or a food delivery application.

18. A non-transitory computer-readable storage medium comprising one or more programs for execution by one or more processors of an electronic device, the one or more programs including instructions which, when executed by the one or more processors, cause the electronic device to: based on one or more signals generated by the electronic device, detect an event occurrence of the event, wherein the event occurrence is associated with meta-data indicating a contextual condition of the event

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occurrence, wherein detecting the event occurrence includes detecting a launching of an application;

in accordance with the event occurrence, receive a first suggested action, a second suggested action; and a first temporal offset, a second temporal offset, and a third temporal offset from the predictive model, wherein the second temporal offset is different from the third temporal offset; and

provide the first suggested action to the user within the first temporal offset from the event occurrence and detecting an input from a user;

in response to detecting the input from the user:

in accordance with a determination that the input from the user corresponds to the first suggested action, provide the second suggested action to the user within the second temporal offset; and

in accordance with a determination that the input from the user does not correspond to the first suggested action, provide the second suggested action to the user within the third temporal offset.

19. The non-transitory computer-readable storage medium of claim 18, wherein detecting the event occurrence includes determining, based on a positioning signal of the one or more signals, that the electronic device is positioned at a particular location.

20. The non-transitory computer-readable storage medium of claim 18, wherein the event occurrence includes an occurrence of a calendar event of an electronic calendar associated with the user.

21. The non-transitory computer-readable storage medium of claim 18, wherein providing the first suggested action includes providing the first suggested action after the event occurrence, and in accordance to the temporal offset.

22. The non-transitory computer-readable storage medium of claim 18, wherein providing the first suggested action includes providing the first suggested action before the event occurrence, and in accordance to the temporal offset.

23. The non-transitory computer-readable storage medium of claim 22, wherein the one or more contextual conditions of the event occurrence include at least one of a day of a week, a time of a day, or a location of the electronic device.

24. The non-transitory computer-readable storage medium of claim 18, wherein providing the first suggested action includes employing a display of the electronic device to providing a pop-up notification indicating the first suggested action.

25. The non-transitory computer-readable storage medium of claim 18, wherein the first suggested action is based on at least one or more contextual conditions of the event occurrence.

26. A method for employing a predictive model for an event that indicates a behavior of a user of an electronic device, the method comprising:

employing one or more processors and a memory of the electronic device to perform operations comprising:

based on one or more signals generated by the electronic device, detecting an event occurrence of the event, wherein the event occurrence is associated with meta-data indicating a contextual condition of the event occurrence, wherein detecting the event occurrence includes detecting a launching of an application;

in accordance with the event occurrence, receiving a first suggested action, a second suggested action; and a first temporal offset, a second temporal offset, and a third

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temporal offset from the predictive model, wherein the second temporal offset is different from the third temporal offset; and
 providing the first suggested action to the user within the first temporal offset from the event occurrence and detecting an input from a user;
 in response to detecting the input from the user:
 in accordance with a determination that the input from the user corresponds to the first suggested action, providing the second suggested action to the user within the second temporal offset; and
 in accordance with a determination that the input from the user does not correspond to the first suggested action, providing the second suggested action to the user within the third temporal offset.

27. The method of claim 26, wherein detecting the event occurrence includes determining, based on a positioning signal of the one or more signals, that the electronic device is positioned at a particular location.

28. The method of claim 26, wherein the event occurrence includes an occurrence of a calendar event of an electronic calendar associated with the user.

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29. The method of claim 26, wherein providing the first suggested action includes providing the first suggested action after the event occurrence, and in accordance to the temporal offset.

30. The method of claim 26, wherein providing the first suggested action includes providing the first suggested action before the event occurrence, and in accordance to the temporal offset.

31. The method of claim 26, wherein providing the first suggested action includes employing a display of the electronic device to providing a pop-up notification indicating the first suggested action.

32. The method of claim 26, wherein the first suggested action is based on at least one or more contextual conditions of the event occurrence.

33. The method of claim 32, wherein the one or more contextual conditions of the event occurrence include at least one of a day of a week, a time of a day, or a location of the electronic device.

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